Auditing a quantum gravity calculation with an AI model: corrected scalar self-mass and unchanged late-time mode functions during inflation

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We present an independent, step-by-step recomputation of the one-loop quantum-gravitational self-mass-squared of a massless, minimally coupled scalar on de Sitter background in a non-de Sitter-invariant gauge [E. O. Kahya and R. P. Woodard, Phys. Rev. D **76**, 124005 (2007)], and of the Schwinger–Keldysh analysis of the late-time scalar mode functions based on it [Phys. Rev. D **77**, 084012 (2008)]. Each main-text step was re-derived from its inputs, in general D where the papers keep it general, and cross-checked by independent code; the appendix of the first paper on extracting derivatives was audited (worked examples and 30-digit tests of its totals) rather than re-derived. Most results are confirmed, but (i) the printed self-mass-squared is -1 times the one defined by the papers' own effective field equation, with both the 4-point and 3-point parts reversed; (ii) the final table of nonlocal terms needs two corrections (a missing non-analytic term, and a constant that drops out of the retarded kernel), established by a 30-digit global pointwise test; (iii) two finite errors shift counterterm coefficients by absorbable amounts. We reproduce Tables 2–5 of the second paper exactly (Table 5 in its own sign convention; in the definition sign every entry flips, the total remaining zero) and, with the corrected self-mass, redo its late-time analysis through order k^{10} without its simplifying approximation, whose justification fails term by term. The non-falling part of the source cancels order by order, and the mode function tends to its tree-order value times a constant: one-loop quantum gravity gives no significant late-time correction to the scalar mode functions. Limitations: the global tests are numerical, the cancellation is not proven beyond k^{10}, and a possible extra local term is only partially excluded. The work used an AI model (Claude Opus 5.5); notes and code are public.

I. INTRODUCTION

A. Physical motivation

Massless, minimally coupled scalars and gravitons are unique in possessing zero mass without classical conformal invariance. Simple arguments based on the energy-time uncertainty principle suggest that quantum effects from these particles are strongly enhanced during primordial inflation [1]. At lowest order inflation rips long-wavelength virtual quanta of these fields out of the vacuum; this is the origin of the primordial scalar [2] and tensor [3] perturbations. At higher orders the sea of long-wavelength scalars and gravitons interacts with itself and with other fields, and loop corrections can be enhanced by powers of the logarithm of the scale factor [4]. Gravitons couple through derivatives, which weakens the enhancement, but their effects need not be small: for massless fermions, one-loop quantum gravity induces a time-dependent field-strength renormalization that eventually becomes nonperturbatively large [5].

It is therefore natural to ask whether the inflationary production of gravitons has a significant effect on the propagation of a massless, minimally coupled scalar. The question has phenomenological interest. The inflaton potential of scalar-driven inflation is so flat that the inflaton mode functions are effectively those of a massless, minimally coupled scalar, and the scalar power spectrum is routinely computed by setting the scalar-scalar correlator equal to its value for such a scalar on de Sitter background [6, 7]. A significant one-loop correction to the scalar mode functions, in particular a secularly growing field-strength renormalization of the kind found for fermions, would invalidate this procedure.

B. What Papers I and II did

The question was addressed in two papers by R. P. Woodard and the present author. In Paper I [8] the one-loop self-mass-squared $-iM^2(x;x')$ of a massless, minimally coupled scalar was computed on the conformal coordinate patch of D-dimensional de Sitter space, $ds^2 = a^2(\eta)(-d\eta^2 + d\vec{x}\cdot d\vec{x})$ with $a = -1/(H\eta)$. The graviton propagator

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was taken in the non-de Sitter-invariant gauge (I.15), in which it is a sum of constant tensor factors times three scalar propagators [9, 10]. The calculation used dimensional regularization. The 4-point vertices give a purely local contribution. The products of two 3-point vertices were split into local, logarithm and "normal" parts; the normal parts were made integrable by extracting derivatives (the Appendix of Paper I), and their remaining divergences were separated into local terms. These were removed by fourth-order BPHZ counterterms [11]: two de Sitter-invariant ones and one noninvariant one, which is needed because the gauge breaks de Sitter invariance. The final result (I.167) consists of arbitrary finite counterterm contributions plus a table of thirteen finite nonlocal terms, Table X of Paper I.

In Paper II [12] this self-mass-squared was used to correct the linearized scalar field equation. To obtain a real and causal equation the Schwinger–Keldysh formalism [13–16] was used. Writing $\Phi = u + \kappa^2 \Phi_1$, with $u(\eta, k)$ the tree-order mode function, the one-loop correction Φ_1 obeys a sourced mode equation, (II.29). Its source was evaluated at late times under a simplification, (II.44): $u(\eta', k) \to u(0, k)$ and $e^{i\vec{k}\cdot(\vec{x}\,'-\vec{x})} \to 1$, which removes every term of the table containing a spatial Laplacian. Several of the nineteen remaining contributions (Table 5 of Paper II) grow individually like $\ln^2 a$ or $\ln a$, but their sum vanishes. Paper II concluded that the plane-wave mode functions receive no significant one-loop corrections at late times.

C. Why a recomputation

Both calculations are long. Paper I runs through about one hundred and eighty numbered equations in the main text, thirty-eight in its appendix, and twenty tables, and the coefficients of its final table emerge from many cancellations between separately computed pieces. The late-time conclusion of Paper II rests on an exact cancellation among individually growing terms, obtained under an approximation whose validity was argued rather than demonstrated. An independent, step-by-step recomputation is the most reliable way to establish which parts of these results are correct, and to quantify the effect of any error on the physical conclusion. The present work provides such a recomputation, with the exceptions and limitations stated in Secs. I D and VIII D, and presents the corrected result in one place.

D. Method

The recomputation was organized as eighteen consecutive tasks, each covering a block of equations of Paper I or Paper II (our Table 1). The following rules were applied throughout.

- Every equation of the main text of Paper I and of Paper II was re-derived from its inputs: the Lagrangian, the propagators, and results established in earlier tasks. The printed target was not used as input; it was compared with only at the end of each step. The stated exceptions are the following. (a) The gauge-fixed graviton propagator (I.15)–(I.25) was taken from Paper I and Ref. [9] as input and was not re-derived. (b) The appendix "Extracting derivatives" of Paper I, (I.A1)–(I.A38) and Tables XI–XX, was audited rather than redone: worked examples were checked by hand, Tables XII and XIV were transcribed, and the totals of all ten tables were verified by global pointwise tests (Sec. V E). (c) Tables IV and V of Paper I were computed from these printed appendix totals. In the assembly of Table X, Table V enters as printed (apart from one analytic constant corrected after the global test, Sec. V G), in particular its polynomial terms, after its non-analytic terms had been confirmed. Tables 2–5 of Paper II were never used as input.

- The spacetime dimension D was kept general wherever Paper I keeps it general; $D = 4$ was set only in terms that are not multiplied by a pole.

- Every analytic step was cross-checked by independent code where possible (Python with `sympy` and `mpmath`), kept separate from the derivation: symbolic identities in general D where possible, and numerical tests at random points in 30-digit arithmetic otherwise. The few steps done by hand only are listed in Sec. VIII D.

- Every discrepancy with the printed papers was classified as typographical, finite and absorbable into the counterterms, substantive, or a matter of convention.

The step-by-step re-derivations and the code checks were carried out with an AI model, Claude Opus 5.5 (Anthropic), used as a calculation and checking tool. The original working reports were written in Turkish; they were translated into English derivation notes (nine chapters and a summary chapter of findings), which, together with every script and its expected output, are publicly available [17]. The present paper gives the derivation in condensed form; the notes give the intermediate steps in full.

E. Conventions

Equation (n) of Paper I is cited as (I.n), equation (n) of Paper II as (II.n). Tables of Paper I carry Roman numerals (Table X), tables of Paper II Arabic numerals (Table 5). Equations of the appendix of Paper I are cited as (I.A1), etc.; equation and table numbers of Paper II refer to its arXiv version, arXiv:0710.5282v1. Tables of the present paper are also numbered with Arabic numerals, but they are always cited as "our Table"; an unqualified Roman table number refers to Paper I. We follow the notation of Paper I: $g_{\mu\nu} = a^2(\eta_{\mu\nu} + \kappa h_{\mu\nu})$ with $\kappa^2 = 16\pi G$, and the de Sitter length function

$$y(x;x') = aa'H^2\Big[\|\vec{x}-\vec{x}\,'\|^2 - \big(|\eta-\eta'|-i\delta\big)^2\Big], \qquad a \equiv a(\eta), \quad a' \equiv a(\eta'), \qquad x \equiv \frac{y}{4}, \tag{1}$$

where the variable $x = y/4$ appears only as the argument of the coefficient functions of the tables. Paper II uses the same y with $H = 1$.

Two sign conventions. Paper I defines the self-mass-squared through the linearized effective field equation (I.168),

$$\partial_\mu\big(\sqrt{-\hat{g}}\,\hat{g}^{\mu\nu}\partial_\nu\Phi\big)(x) - \int d^4x'\, M^2(x;x')\,\Phi(x') = 0, \tag{2}$$

where $\hat{g}_{\mu\nu} = a^2\eta_{\mu\nu}$ is the background metric. Paper I prints only this field equation. Writing the one-particle-irreducible effective action as $\Gamma = S_0 + \Gamma_1$, with S_0 the tree action of the background field and Γ_1 its quantum part, Eq. (2) is $\delta\Gamma/\delta\Phi = 0$ with

$$-iM^2(x;x') = \frac{\delta^2\,(i\Gamma_1[\Phi])}{\delta\Phi(x)\,\delta\Phi(x')}\Big|_{\Phi=0} \tag{3}$$

(Sec. II F). We call the sign fixed by Eqs. (2)–(3) the *definition sign*. The sign of the self-mass contributions as printed in Papers I and II is called the *paper convention* (or Table-X convention); our Table 3 lists which printed equations are in which convention. One of our results is

$$\text{definition sign} = -1 \times \text{paper convention}. \tag{4}$$

Corrected tables are quoted in the paper convention, so that they can be compared entry by entry with the printed ones, and the conversion is stated explicitly. Physical statements are made in the definition sign.

Units. The entries of Table X of Paper I and of Table 1 of Paper II are coefficients of $\kappa^2H^4/(4\pi)^4$. The counterterm coefficients α_i and β_i are quoted in units of $H^{D-4}/(4\pi)^{D/2}$. We write $\hat{k} \equiv k/H$, and Euler's constant as $\gamma_{\rm E}$ (it is γ in Papers I and II), to avoid a clash with the operators γ_i of Paper I.

Counting of κ^2. As in (II.23) we write $\Phi = u + \kappa^2\Phi_1$, where Φ_1 is sourced by the one-loop self-mass-squared M_1^2 with the factor κ^2 stripped off,

$$\Big[\partial_t^2 + 3H\partial_t + \frac{k^2}{a^2}\Big]\Phi_1 = \mathcal{R}_1 \equiv -\frac{1}{a^4}\int_{\eta_i}^0 d\eta'\,u(\eta',k)\int d^3x'\,\big\{M^2_{++}(x;x') + M^2_{+-}(x;x')\big\}_1\,e^{i\vec{k}\cdot(\vec{x}\,'-\vec{x})}. \tag{5}$$

The factor κ^2 is thus counted once. The source $\mathcal{R}_1$ is quoted in units of $H^4u(0,k)/16\pi^2$.

F. Organization of the paper

The paper follows the order of Papers I and II, so that it can be read side by side with them. Our Table 1 maps each of our sections to the corresponding sections, equations and tables of the printed papers, and to the chapters of the public derivation notes [17]. Within each section we present our derivation as the main line, with explicit pointers ("cf. (I.n)") to the printed equations, and compare with the printed values in small tables or footnotes.

G. Summary of results

Most of the calculation of Paper I is confirmed: the Feynman rules (apart from misprints), the coincidence limits (apart from the finite part of (I.40)), the split of the 3-point contribution, Table III, Table IV (apart from one sign misprint that the paper did not use in its calculation), all non-analytic terms of Table V, Tables VI–IX, and the counterterm analysis, with two invariant counterterms and the single noninvariant one (I.145). The coefficient β_1

TABLE 1. Correspondence between the sections of this paper, the printed Papers I [8] and II [12], and the chapters of the derivation notes [17].

Sec.	Content	Paper I / Paper II	Notes (tasks)
II	Lagrangian, vertices, propagators, definition of M^2, derivative rules	I: Sec. II, (I.1)–(I.30), Tables I–II; derivative rules (text after (I.31)); definition (I.168); calibration with (I.130)–(I.131)	Chs. 1–3 (1–2; parts of 3 and 6–8)
III	4-point contribution, coincidence limits	I: Sec. III A, (I.31)–(I.44)	Ch. 2 (3–5)
IV	3-point contribution: setup, local and logarithm terms	I: Sec. III B (introduction, B.1, B.2), (I.45)–(I.67)	Chs. 3–4 (6–10)
V	3-point normal terms: Table III, extraction of derivatives, Tables IV–V	I: Sec. III B.3, (I.68)–(I.125), Tables III–V; Appendix, (I.A1)–(I.A38), Tables XI–XX	Chs. 5–6 (11–13)
VI	Counterterms, residual symmetries, divergences, Table X, renormalized result	I: Sec. IV, (I.126)–(I.167), Tables VI–X	Ch. 7–8 (14–17)
VII	Schwinger–Keldysh equation, Tables 2–5, late times beyond (II.44), mode function	II: Secs. 2–3, (II.1)–(II.45), Tables 1–5	Ch. 9 (18)
VIII	Summary of differences, errata, conclusions, limitations	I: Sec. V; II: Sec. 4	all
A	Verification methods	—	code

vanishes, and β_2 is finite; the pole $-4/(D-4)$ of β_3 (paper convention) is confirmed within Paper I's treatment of the outer operators of the normal terms (Secs. V H and VI D). Tables 2–5 of Paper II are reproduced exactly (Table 5 in its own sign convention). Three things change in the self-mass-squared.

The overall sign (S1). The printed $-iM^2$ is -1 times the quantity defined by Paper I's own effective field equation (I.168) [our Eqs. (2)–(3)]. The 4-point part acquires the reversed sign through the reading of the derivative rule in (I.32), the 3-point part through an overall sign slip in the step from (I.46) to (I.47)–(I.49). Since both parts are reversed, their relative sign, and the local cancellation between (I.44) and (I.67), are consistent. Several independent arguments support this sign (Sec. III). The sign of the local part is unphysical, because the finite counterterms $\Delta\alpha_i$ are arbitrary; the nonlocal part, however, has no counterterm freedom, and in the definition sign it is $-$(corrected Table X).

Two corrections of Table X (S2, S3). In the $(aa')^2\nabla^4$ row the contribution $8\ln x - 16x\ln x$ of Table III is missing; the corrected row is $\frac{34}{3}\ln x - 40x\ln x + 24x^2\ln x - 36x^2$. This term is non-analytic and changes the physical nonlocal self-mass. In the $(aa')^3(a^2+a'^2)H^4$ row the constant is $+18$ instead of -54; this term is analytic and drops out of the retarded kernel identically. Only with both corrections does Table X agree, pointwise to 30 digits at random points, with the direct sum of the normal contributions and the nonlocal logarithm term of (I.67).

Absorbable finite shifts (B1–B6). Finite errors in (I.40) and (I.59) shift $\beta_2 : -\frac{61}{3} \to -\frac{49}{3}$ and $\beta_3 : +\frac{58}{3} \to +\frac{52}{3}$ (paper convention). These shifts are absorbed into the arbitrary finite parts of the counterterms; the divergence is unchanged.

The renormalized self-mass-squared in the definition sign is therefore

$$-iM^2_{\rm ren}(x;x') = i\kappa^2 a^4\Big(\Delta\alpha_1\Box^2 + \Delta\alpha_2 H^2\Box + \Delta\alpha_3 H^2\frac{\nabla^2}{a^2}\Big)\delta^4(x-x') - \big[\text{corrected Table X}\big], \tag{6}$$

where the factors H^2, missing in the printed (I.167) (erratum C18), are required by dimensions. The corrected Table X is given in our Table 31.

Late times. None of these corrections changes the late-time physics. We redid the analysis of Paper II with the corrected self-mass-squared in the definition sign and without the simplification (II.44), keeping the exact $u(\eta',k)$, the phase $e^{i\vec{k}\cdot(\vec{x}'-\vec{x})}$, the k^2/a^2 part of $\Box$ and every row containing ∇^2. The justification given for (II.44) fails term by term (S4): two rows of the table separately grow like $\mp\frac{4}{15}\hat{k}^2 a$. Nevertheless the non-falling part of the source cancels separately at each order $k^0,\dots,k^{10}$, for the printed as well as for the corrected table (verified exactly through k^{10}, not proven to all orders; see Sec. VIII D). The correction of the $(aa')^2\nabla^4$ row changes only falling terms: it removes the $\hat{k}^4/a$, $\hat{k}^6/a,\dots$ tails, so that with the corrected table the source falls like $\ln a/a^2$. The mode function behaves as

$$\Phi(\eta,k) = u(\eta,k) + \kappa^2\Phi_1(\eta,k) \xrightarrow{a\to\infty} u(0,k)\Big\{1 + \frac{\kappa^2 H^2}{16\pi^2}\Big[\tilde{C} + O\Big(\frac{\ln a}{a^2}\Big)\Big]\Big\}, \tag{7}$$

where $\tilde{C}$ is a constant field-strength renormalization and the tree-order approach $u(\eta,k) = u(0,k)[1+\hat{k}^2/(2a^2)+\dots]$

is not displayed. Through order k^{10}, and given the kernel and local terms as derived, the main conclusion of Paper II stands: one-loop quantum gravity gives no significant correction to the scalar mode functions at late times.

In the classification of Sec. VIII there are four substantive findings: the overall sign (S1), the two corrections of Table X (S2, S3) and the failure of the justification of (II.44) (S4, an argument only, with no effect on the result). The finite errors and their consequences for $\beta_{2,3}$ and $\alpha_{2,3}$ are the six absorbable items B1–B6. In addition we list 23 typographical errors, C1–C23, that have no effect on any result. All differences are collected in Sec. VIII, together with the limitations of this work.

II. THE MODEL, PROPAGATORS AND FEYNMAN RULES

This section collects the ingredients of the computation: the Lagrangian and its expansion to second order in the graviton field (Sec. II of Paper I, Eqs. (I.1)–(I.8) and Tables I–II), the gauge-fixed graviton propagator and the scalar propagators [(I.9)–(I.30)], the definition of the self-mass-squared [(I.168)], and the rules by which the vertex operators are turned into kernels. Paper I states these rules only in words, after (I.31) and before (I.45). The overall sign of the printed self-mass can be traced back to them, so we state them here as a lemma (Sec. II G). The spacetime dimension D is general throughout.

Apart from the sign of the self-mass, every difference from the printed equations in this part of the calculation is a misprint that did not enter the paper's computation. The differences are listed in our Table 5 at the end of the section. Full derivations are in Chs. 1–3 of the public derivation notes, and the scripts are in `code/01_feynman_rules`, `code/02_four_point` and `code/03_three_point_setup` of the repository [17]. All scripts quoted below were run and reproduce their logged output.

A. Lagrangian, background and notation

The theory is a massless, minimally coupled scalar coupled to gravity with a positive cosmological constant [cf. (I.1)]:

$$\mathcal{L} = -\tfrac{1}{2}\,\partial_\mu\phi\,\partial_\nu\phi\,g^{\mu\nu}\sqrt{-g} + \frac{1}{16\pi G}\big(R-(D-2)\Lambda\big)\sqrt{-g}, \qquad \Lambda \equiv (D-1)H^2. \tag{8}$$

Paper I prints the first term as $-\frac{1}{2}\partial_\mu\phi\,\partial_\mu\phi\,g^{\mu\nu}\sqrt{-g}$. The only reading without a free index is the one in (8) (item C1 of our Table 5). The scalar is a spectator with zero background value. The background is the conformal patch of D-dimensional de Sitter space, and the graviton field is defined by conformally rescaling the metric [cf. (I.2), (I.3)]:

$$ds^2 = a^2\big(-d\eta^2 + d\vec{x}\cdot d\vec{x}\big), \qquad a(\eta) = -\frac{1}{H\eta}, \qquad g_{\mu\nu} \equiv a^2\big(\eta_{\mu\nu}+\kappa h_{\mu\nu}\big), \qquad \kappa^2 \equiv 16\pi G. \tag{9}$$

We use the signature $(-,+,\ldots,+)$, so $a>0$ on $-\infty<\eta<0$. The indices of $h_{\mu\nu}$ are raised and lowered with $\eta_{\mu\nu}$ only, so $h\equiv\eta^{\mu\nu}h_{\mu\nu}$ and $h^{\mu\nu}\equiv\eta^{\mu\alpha}\eta^{\nu\beta}h_{\alpha\beta}$. Likewise $\partial^\mu\equiv\eta^{\mu\nu}\partial_\nu$ and $\partial\phi\cdot\partial\phi\equiv\eta^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$. This is not merely notation: the coefficients of (I.6) below hold only in this convention, since raising with the full metric would change the $O(\kappa)$ terms. The background quantities carry a hat [cf. (I.9)],

$$\hat{g}_{\mu\nu} = a^2\eta_{\mu\nu}, \qquad \hat{g}^{\mu\nu} = \frac{\eta^{\mu\nu}}{a^2}, \qquad \sqrt{-\hat{g}} = a^D, \qquad \hat{R} = D(D-1)H^2. \tag{10}$$

The gauge used below treats time and space differently, so we also need the spatial parts of the metric and of the Kronecker delta [cf. (I.10)]:

$$\bar{\eta}_{\mu\nu} \equiv \eta_{\mu\nu} + \delta^0_\mu\delta^0_\nu, \qquad \bar{\delta}^\mu_\nu \equiv \delta^\mu_\nu - \delta^\mu_0\delta^0_\nu. \tag{11}$$

Finally, the scalar d'Alembertian of the background [cf. (I.42)],

$$\Box \equiv \frac{1}{\sqrt{-\hat{g}}}\,\partial_\mu\big(\sqrt{-\hat{g}}\,\hat{g}^{\mu\nu}\partial_\nu\big) = \frac{1}{a^D}\,\partial_\mu\big(a^{D-2}\partial^\mu\big), \tag{12}$$

is used throughout the rest of the paper.

B. Expansion of the scalar kinetic term

Write the metric as the matrix product $g_{\mu\nu} = a^2\eta_{\mu\rho}(\delta^\rho{}_\nu + \kappa h^\rho{}_\nu)$. The inverse metric then follows from the Neumann series of $(\mathbf{1}+\kappa\mathsf{h})^{-1}$, with $(\mathsf{h})^\mu{}_\nu \equiv h^\mu{}_\nu$. We also obtained it independently by matching coefficients in $g_{\mu\rho}g^{\rho\nu} = \delta_\mu{}^\nu$. The volume element follows from $\det(\mathbf{1}+\kappa\mathsf{h}) = \exp\mathrm{Tr}\ln(\mathbf{1}+\kappa\mathsf{h})$, which we cross-checked by a direct Leibniz expansion of the determinant. The results are [cf. (I.4), (I.5)]

$$g^{\mu\nu} = \frac{1}{a^2}\Big(\eta^{\mu\nu} - \kappa h^{\mu\nu} + \kappa^2 h^\mu{}_\rho h^{\rho\nu} - \kappa^3 h^\mu{}_\rho h^\rho{}_\sigma h^{\sigma\nu} + O(\kappa^4)\Big), \tag{13}$$

$$\sqrt{-g} = a^D\Big(1 + \tfrac{1}{2}\kappa h + \tfrac{1}{8}\kappa^2 h^2 - \tfrac{1}{4}\kappa^2 h^{\rho\sigma}h_{\rho\sigma} + O(\kappa^3)\Big). \tag{14}$$

The structure h^2 in (14) does not come from the logarithm. It comes only from the term $\frac{1}{2}X^2$ in the expansion of $\sqrt{\det(\mathbf{1}+\kappa\mathsf{h})} = e^X$, where $X \equiv \frac{1}{2}\mathrm{Tr}\ln(\mathbf{1}+\kappa\mathsf{h}) = \frac{1}{2}\kappa h - \frac{1}{4}\kappa^2 h^{\rho\sigma}h_{\rho\sigma} + O(\kappa^3)$. In the product, the only second-order cross term is $(\frac{1}{2}h)(-h^{\mu\nu})$. Since $\partial_\mu\phi$ does not depend on the metric, the kinetic term becomes [cf. (I.6)]

$$\begin{aligned}-\tfrac{1}{2}\partial_\mu\phi\,\partial_\nu\phi\,g^{\mu\nu}\sqrt{-g} = -\tfrac{1}{2}\partial_\mu\phi\,\partial_\nu\phi\,a^{D-2}\Big\{&\eta^{\mu\nu} - \kappa h^{\mu\nu} + \tfrac{1}{2}\eta^{\mu\nu}\kappa h + \tfrac{1}{8}\eta^{\mu\nu}\kappa^2 h^2 - \tfrac{1}{4}\eta^{\mu\nu}\kappa^2 h^{\rho\sigma}h_{\rho\sigma}\\ &- \tfrac{1}{2}\kappa^2 h h^{\mu\nu} + \kappa^2 h^{\mu\rho}h_\rho{}^\nu + O(\kappa^3)\Big\}.\end{aligned} \tag{15}$$

The common factor a^{D-2} is a^{-2} from the inverse metric times a^D from the determinant. It equals 1 at $D=2$, which reflects the Weyl invariance of the two-dimensional kinetic term. The parts linear and quadratic in h are

$$\kappa\mathcal{L}^{(1)} = \kappa a^{D-2}\Big[\tfrac{1}{2}h^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - \tfrac{1}{4}h\,\partial\phi\cdot\partial\phi\Big], \tag{16}$$

$$\kappa^2\mathcal{L}^{(2)} = \kappa^2 a^{D-2}\Big[-\tfrac{1}{16}h^2\,\partial\phi\cdot\partial\phi + \tfrac{1}{8}h^{\rho\sigma}h_{\rho\sigma}\,\partial\phi\cdot\partial\phi + \tfrac{1}{4}hh^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - \tfrac{1}{2}h^{\mu\rho}h_\rho{}^\nu\partial_\mu\phi\,\partial_\nu\phi\Big]. \tag{17}$$

Checks. Two special cases can be computed exactly from the full metric, without using the expansion. For $h_{\mu\nu} = c\,\eta_{\mu\nu}$ one has $\sqrt{-g}\,g^{\mu\nu} = a^{D-2}(1+\kappa c)^{(D-2)/2}\eta^{\mu\nu}$, and its κ^2 coefficient is $\frac{1}{8}(D-2)(D-4)c^2$. Write the general second-order bracket as $A\,h^{\mu\rho}h_\rho{}^\nu + B\,hh^{\mu\nu} + C\,\eta^{\mu\nu}h^2 + E\,\eta^{\mu\nu}h^{\rho\sigma}h_{\rho\sigma}$. Matching powers of D fixes $A=1$, $C=\frac{1}{8}$ and $B+E=-\frac{3}{4}$. For $h_{\mu\nu} = b\,\delta^0_\mu\delta^0_\nu$ one has $\sqrt{-g}\,g^{00} = -a^{D-2}(1-\kappa b)^{-1/2}$ and $\sqrt{-g}\,g^{ij} = a^{D-2}\delta^{ij}(1-\kappa b)^{1/2}$, which gives $C+E=-\frac{1}{8}$. Together the two cases fix all four coefficients, $(A,B,C,E) = (1,-\frac{1}{2},\frac{1}{8},-\frac{1}{4})$, in agreement with (15). The script `code/01_feynman_rules/check_eq6.py` builds the inverse metric and $\sqrt{-g}$ from the exact adjugate and determinant, expands them, and compares with the bracket of (15). The comparison is fully symbolic for $D=2,3$ and uses random rational $h_{\mu\nu}$ for $D=4,5,6$. The difference vanishes in all cases.

Result: (I.4), (I.5) and the bracket of (I.6) are exact for general D, with no gauge or trace condition. The printed left-hand side of (I.6) reads $-\frac{1}{2}\partial_\mu\phi\,\partial_\nu g^{\mu\nu}\sqrt{-g}$; the second ϕ is missing. This slip (C2) was noticed only while preparing the write-up.

C. Vertex operators: corrected Tables I and II

Paper I represents the interactions as vertex operators acting on the fields. The two scalar legs are numbered "1", "2" and the gravitons "3", "4". A factor $\frac{1}{2}$ is extracted for the two identical scalars, but not for the two identical gravitons of the 4-point vertices. With the standard rule that a vertex carries i times the corresponding coefficient in the action, this defines the normalization

$$\kappa\mathcal{L}^{(1)} = \frac{1}{2}\cdot\frac{1}{i}\sum_I\Big[V_I^{\alpha\beta}\phi_1\phi_2\Big]_{\phi_1=\phi_2=\phi}h_{\alpha\beta}, \qquad \kappa^2\mathcal{L}^{(2)} = \frac{1}{2}\cdot\frac{1}{i}\sum_I\Big[U_I^{\alpha\beta\rho\sigma}\phi_1\phi_2\Big]_{\phi_1=\phi_2=\phi}h_{\alpha\beta}h_{\rho\sigma}. \tag{18}$$

Consider a term $C\,T(\partial\phi,\partial\phi)\,h\cdots$, where C contains the number, κ^n and a^{D-2}. By (18) its vertex is $2iC\,T(\partial_1,\partial_2)$. This vertex is the full scalar functional derivative: for a symmetric $X^{\mu\nu}$, the part of $C\,X^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ bilinear in $\phi=\phi_1+\phi_2$ is $2C\,X^{\mu\nu}\partial_\mu\phi_1\partial_\nu\phi_2$. A 4-point operator U carries only one assignment of the two graviton legs, and the full vertex is $U[3,4]+U[4,3]$. When the two graviton legs are joined by the coincident propagator $X_{\alpha\beta\rho\sigma}$, which is pair symmetric (Sec. II D), the symmetry factor $\frac{1}{2}$ of the full vertex gives $\frac{1}{2}(U^{\alpha\beta\rho\sigma}+U^{\rho\sigma\alpha\beta})X_{\alpha\beta\rho\sigma} = U^{\alpha\beta\rho\sigma}X_{\alpha\beta\rho\sigma}$. This is why Paper I needs no symmetry factor. Combining $2iC$ with (15) gives a compact rule: every vertex equals $-i$ times the coefficient b_I of the corresponding structure in the bracket of (I.6), times $\kappa^n a^{D-2}$ and the contracted structure. The resulting vertices are listed in our Table 2.

TABLE 2. 3-point vertex operators $V_I^{\alpha\beta}$, contracted into $\phi_1\phi_2 h_{\alpha\beta}$, and 4-point vertex operators $U_I^{\alpha\beta\rho\sigma}$, contracted into $\phi_1\phi_2 h_{\alpha\beta}h_{\rho\sigma}$ (cf. Tables I and II of Paper I). The number b_I is the coefficient of the source structure in the bracket of (15), and every vertex equals $-ib_I\kappa^n a^{D-2}$ times its structure. Only U_2 differs from the print.

Vertex	Source structure in (15)	b_I	Derived (this work)	Paper I (printed)
V_1	$h^{\mu\nu}$	-1	$+i\kappa a^{D-2}\partial_1^\alpha\partial_2^\beta$	same
V_2	$\eta^{\mu\nu}h$	$+\frac{1}{2}$	$-\frac{i}{2}\kappa a^{D-2}\eta^{\alpha\beta}\,\partial_1\cdot\partial_2$	same
U_1	$\eta^{\mu\nu}h^2$	$+\frac{1}{8}$	$-\frac{i}{8}\kappa^2 a^{D-2}\eta^{\alpha\beta}\eta^{\rho\sigma}\,\partial_1\cdot\partial_2$	same
U_2	$\eta^{\mu\nu}h^{\rho\sigma}h_{\rho\sigma}$	$-\frac{1}{4}$	$+\frac{i}{4}\kappa^2 a^{D-2}\eta^{\alpha\rho}\eta^{\beta\sigma}\,\partial_1\cdot\partial_2$	$-\frac{i}{4}\kappa^2 a^{D-2}\eta^{\alpha\rho}\eta^{\beta\sigma}\,\partial_1\cdot\partial_2$
U_3	$hh^{\mu\nu}$	$-\frac{1}{2}$	$+\frac{i}{2}\kappa^2 a^{D-2}\eta^{\alpha\beta}\partial_1^\rho\partial_2^\sigma$	same
U_4	$h^{\mu\rho}h_\rho{}^\nu$	$+1$	$-i\kappa^2 a^{D-2}\partial_1^\alpha\eta^{\beta\rho}\partial_2^\sigma$	same

Eq. (I.7). Paper I prints $-\frac{1}{2}\kappa a^{D-2}\partial_\alpha\phi\,\partial_\beta\phi\,h^{\alpha\beta} \Rightarrow V_1^{\alpha\beta} = i\kappa a^{D-2}\partial_1^\alpha\partial_2^\beta$. The paper's own normalization, which can be read off from (I.8), is $U_1/C = 2i$. With this normalization a coefficient $-\frac{1}{2}\kappa$ would give $V_1 = -i\kappa a^{D-2}\partial_1^\alpha\partial_2^\beta$. The coefficient in (16) is $+\frac{1}{2}\kappa$, so the sign on the printed left-hand side is a misprint (C3). The printed V_1 is correct, and it is the vertex used in (I.46).

The sign of U_2. The printed $-\frac{i}{4}$ in row 2 of Table II is a misprint (C5). Two checks based on the full metric alone, without using (15), fix all coefficients uniquely. Take the ansatz $V = i\kappa a^{D-2}[v_1\,\partial_1^\alpha\partial_2^\beta + v_2\,\eta^{\alpha\beta}\partial_1\cdot\partial_2]$ and $U = i\kappa^2 a^{D-2}\sum_I u_I\,(\text{structure}_I)$ with the structures of our Table 2. For $h_{\mu\nu} = c\,\eta_{\mu\nu}$, (18) must reproduce $\kappa\mathcal{L}^{(1)} = -\frac{1}{4}(D-2)\kappa c\,a^{D-2}\partial\phi\cdot\partial\phi$ and $\kappa^2\mathcal{L}^{(2)} = -\frac{1}{16}(D-2)(D-4)\kappa^2c^2a^{D-2}\partial\phi\cdot\partial\phi$. This requires

$$\tfrac{1}{2}(v_1 + v_2 D) = -\tfrac{D-2}{4}, \qquad \tfrac{1}{2}\big[u_1 D^2 + (u_2+u_3)D + u_4\big] = -\tfrac{(D-2)(D-4)}{16}, \tag{19}$$

so $v_1 = 1$, $v_2 = -\frac{1}{2}$, $u_1 = -\frac{1}{8}$, $u_2+u_3 = \frac{3}{4}$ and $u_4 = -1$. For $h_{\mu\nu} = b\,\delta^0_\mu\delta^0_\nu$ one needs $u_1+u_2 = \frac{1}{8}$ and $u_1+u_2+u_3+u_4 = -\frac{3}{8}$. Hence $(u_1,u_2,u_3,u_4) = (-\frac{1}{8},+\frac{1}{4},+\frac{1}{2},-1)$ is the unique solution. The printed $u_2 = -\frac{1}{4}$ violates $u_1+u_2 = \frac{1}{8}$. With the printed Table II, the reconstructed Lagrangian differs from the exact one by $-\frac{1}{4}\kappa^2 a^{D-2}h^{\rho\sigma}h_{\rho\sigma}\,\partial\phi\cdot\partial\phi$. Similarly, the printed sign on the left-hand side of (I.7) would correspond to $v_1 = -1$ and fail the first condition in (19). The script `code/01_feynman_rules/check_vertices.py` confirms all of this in three ways:

(A) it obtains the full vertices by polarization with distinct legs (symbolic, $D = 3, 4$);

(B) it compares the tables with the exact metric ($D = 2,\dots,5$);

(C) it solves the linear system above.

The printed (I.33) is built with $+\frac{i}{4}$ (Sec. III A). The Table II entry is therefore a misprint only.

D. Gauge fixing and the graviton propagator

The graviton propagator is that of the noninvariant gauge of Refs. [8, 9]. The gauge-fixing term is [cf. (I.15)]

$$\mathcal{L}_{\rm GF} = -\tfrac{1}{2}a^{D-2}\eta^{\mu\nu}F_\mu F_\nu, \qquad F_\mu \equiv \eta^{\rho\sigma}\Big(h_{\mu\rho,\sigma} - \tfrac{1}{2}h_{\rho\sigma,\mu} + (D-2)Ha\,h_{\mu\rho}\delta^0_\sigma\Big). \tag{20}$$

After partial integration, the quadratic part of the gauge-fixed Lagrangian takes the form $\frac{1}{2}h^{\mu\nu}D_{\mu\nu}{}^{\rho\sigma}h_{\rho\sigma}$, with [cf. (I.16)–(I.19)]

$$D_{\mu\nu}{}^{\rho\sigma} \equiv \Big\{\tfrac{1}{2}\bar\delta_\mu{}^{(\rho}\bar\delta_\nu{}^{\sigma)} - \tfrac{1}{4}\eta_{\mu\nu}\eta^{\rho\sigma} - \frac{1}{2(D-3)}\delta^0_\mu\delta^0_\nu\delta^\rho_0\delta^\sigma_0\Big\}D_A + \delta^0_{(\mu}\bar\delta^{(\rho}_{\nu)}\delta^{\sigma)}_0\,D_B + \frac{1}{2}\Big(\frac{D-2}{D-3}\Big)\delta^0_\mu\delta^0_\nu\delta^\rho_0\delta^\sigma_0\,D_C, \tag{21}$$

$$D_A \equiv \partial_\mu\big(\sqrt{-\hat g}\,\hat g^{\mu\nu}\partial_\nu\big), \qquad D_B \equiv D_A - \frac{1}{D}\Big(\frac{D-2}{D-1}\Big)\hat R\sqrt{-\hat g}, \qquad D_C \equiv D_A - \frac{2}{D}\Big(\frac{D-3}{D-1}\Big)\hat R\sqrt{-\hat g}. \tag{22}$$

By (10), $D_B = D_A - (D-2)H^2a^D$ and $D_C = D_A - 2(D-3)H^2a^D$. The propagator is a sum of constant tensor factors times scalar propagators [cf. (I.20)–(I.25)]:

$$i\big[{}_{\mu\nu}\Delta_{\rho\sigma}\big](x;x') = \sum_{I=A,B,C} \big[{}_{\mu\nu}T^I_{\rho\sigma}\big]\, i\Delta_I(x;x'), \qquad D_I \times i\Delta_I(x;x') = i\delta^D(x-x'), \tag{23}$$

$$\big[{}_{\mu\nu}T^A_{\rho\sigma}\big] = 2\bar\eta_{\mu(\rho}\bar\eta_{\sigma)\nu} - \frac{2}{D-3}\bar\eta_{\mu\nu}\bar\eta_{\rho\sigma}, \qquad \big[{}_{\mu\nu}T^B_{\rho\sigma}\big] = -4\delta^0_{(\mu}\bar\eta_{\nu)(\rho}\delta^0_{\sigma)}, \tag{24}$$

$$\big[{}_{\mu\nu}T^C_{\rho\sigma}\big] = \frac{2}{(D-2)(D-3)}\big[(D-3)\delta^0_\mu\delta^0_\nu + \bar\eta_{\mu\nu}\big]\big[(D-3)\delta^0_\rho\delta^0_\sigma + \bar\eta_{\rho\sigma}\big], \tag{25}$$

$$D_{\mu\nu}{}^{\rho\sigma} \times i\big[{}_{\rho\sigma}\Delta^{\alpha\beta}\big](x;x') = \delta^{(\alpha}_\mu\delta^{\beta)}_\nu\, i\delta^D(x-x'). \tag{26}$$

We take (20)–(26) from Paper I and Ref. [9] as input; we did not re-derive the propagator. The gauge (20) breaks de Sitter invariance. Of the de Sitter transformations it breaks only the spatial special conformal ones. This property was checked in the counterterm analysis (Sec. VI B).

Properties used later. With $u_\mu \equiv \delta^0_\mu$, we have $u^\mu = -\delta^\mu_0$, $u\cdot u = -1$ and $\bar\eta_{\mu\nu} = \eta_{\mu\nu} + u_\mu u_\nu$. The tensor factors are symmetric within each index pair and under exchange of the pairs. Hence the graviton propagator obeys $i[{}_{\alpha\beta}\Delta_{\rho\sigma}](x;x') = i[{}_{\rho\sigma}\Delta_{\alpha\beta}](x';x)$, and its coincidence limit $X_{\alpha\beta\rho\sigma}(x) \equiv i[{}_{\alpha\beta}\Delta_{\rho\sigma}](x;x)$ satisfies $X_{\alpha\beta\rho\sigma} = X_{\rho\sigma\alpha\beta}$. In dimensional regularization, and for general D, X is a smooth function of x. The two symmetries together imply that the partial trace $N_{\lambda\kappa} \equiv X_{\lambda\rho}{}^\rho{}_\kappa$ is symmetric. The single and double traces needed for the contractions are

$$\begin{aligned}
&\eta^{\alpha\beta}T^A_{\alpha\beta\mu\nu} = -\frac{4}{D-3}\bar\eta_{\mu\nu}, \qquad \eta^{\alpha\beta}T^B_{\alpha\beta\mu\nu} = 0, \qquad \eta^{\alpha\beta}T^C_{\alpha\beta\mu\nu} = \frac{4\big[(D-3)u_\mu u_\nu + \bar\eta_{\mu\nu}\big]}{(D-2)(D-3)},\\
&\eta^{\alpha\beta}\eta^{\rho\sigma}T^A_{\alpha\beta\rho\sigma} = -\frac{4(D-1)}{D-3}, \qquad \eta^{\alpha\beta}\eta^{\rho\sigma}T^B_{\alpha\beta\rho\sigma} = 0, \qquad \eta^{\alpha\beta}\eta^{\rho\sigma}T^C_{\alpha\beta\rho\sigma} = \frac{8}{(D-2)(D-3)}.
\end{aligned} \tag{27}$$

The pair and within-pair symmetries were verified with explicit component arrays for $D = 4,\dots,7$, by the script `code/02_four_point/check_coincidence.py`. The traces enter the contractions checked there and those checked for $D = 4,5,6$ by `code/03_three_point_setup/check_47_49.py`.

E. Scalar propagators

All propagators are written in terms of the de Sitter length function $y(x;x')$ of Eq. (1) [cf. (I.11)]. The A-type propagator obeys the equation of a massless, minimally coupled scalar, $D_A\, i\Delta_A(x;x') = i\delta^D(x-x')$ [cf. (I.12)]. This equation has no de Sitter-invariant solution [18]. The solution relevant for cosmology is the one that preserves homogeneity and isotropy, the "E(3)" vacuum [19], and its minimal form is [20, 21] [cf. (I.13), (I.14)]

$$i\Delta_A(x;x') = A(y) + k\ln(aa'), \qquad k \equiv \mathcal{N}\frac{\Gamma(D-1)}{\Gamma(\frac{D}{2})}, \qquad \mathcal{N} \equiv \frac{H^{D-2}}{(4\pi)^{D/2}}, \tag{28}$$

$$\begin{aligned}
A(y) = \mathcal{N}\Bigg\{&\Gamma\Big(\frac{D}{2}-1\Big)\Big(\frac{4}{y}\Big)^{\frac{D}{2}-1} + \frac{\Gamma(\frac{D}{2}+1)}{\frac{D}{2}-2}\Big(\frac{4}{y}\Big)^{\frac{D}{2}-2} - \pi\cot\Big(\frac{\pi D}{2}\Big)\frac{\Gamma(D-1)}{\Gamma(\frac{D}{2})}\\
&+\sum_{n=1}^{\infty}\Big[\frac{1}{n}\frac{\Gamma(n+D-1)}{\Gamma(n+\frac{D}{2})}\Big(\frac{y}{4}\Big)^n - \frac{1}{n-\frac{D}{2}+2}\frac{\Gamma(n+\frac{D}{2}+1)}{\Gamma(n+2)}\Big(\frac{y}{4}\Big)^{n-\frac{D}{2}+2}\Big]\Bigg\}.
\end{aligned} \tag{29}$$

The first coefficient in (29) is $\Gamma(\frac{D}{2}-1)$. Paper I prints $\Gamma(\frac{D}{2}-1)/(\frac{D}{2}-1)$ (C4). The B-type and C-type propagators are de Sitter invariant. Their most singular part is the conformally coupled propagator [22] [cf. (I.26)–(I.28)]:

$$i\Delta_{\rm cf}(x;x') = \mathcal{N}\,\Gamma\Big(\frac{D}{2}-1\Big)\Big(\frac{4}{y}\Big)^{\frac{D}{2}-1}, \tag{30}$$

$$i\Delta_B = i\Delta_{\rm cf} - \mathcal{N}\sum_{n=0}^{\infty}\Bigg\{\frac{\Gamma(n+D-2)}{\Gamma(n+\frac{D}{2})}\Big(\frac{y}{4}\Big)^n - \frac{\Gamma(n+\frac{D}{2})}{\Gamma(n+2)}\Big(\frac{y}{4}\Big)^{n-\frac{D}{2}+2}\Bigg\}, \tag{31}$$

$$i\Delta_C = i\Delta_{\rm cf} + \mathcal{N}\sum_{n=0}^{\infty}\Bigg\{(n+1)\frac{\Gamma(n+D-3)}{\Gamma(n+\frac{D}{2})}\Big(\frac{y}{4}\Big)^n - \Big(n-\frac{D}{2}+3\Big)\frac{\Gamma(n+\frac{D}{2}-1)}{\Gamma(n+2)}\Big(\frac{y}{4}\Big)^{n-\frac{D}{2}+2}\Bigg\}. \tag{32}$$

Equivalently [23, 24] [cf. (I.29), (I.30)], $i\Delta_B = \mathcal{N}\,\frac{\Gamma(D-2)\Gamma(1)}{\Gamma(D/2)}\,{}_2F_1\big(D-2,1;\frac{D}{2};1-\frac{y}{4}\big)$ and $i\Delta_C = \mathcal{N}\,\frac{\Gamma(D-3)\Gamma(2)}{\Gamma(D/2)}\,{}_2F_1\big(D-3,2;\frac{D}{2};1-\frac{y}{4}\big)$. In (29)–(32) the infinite sums vanish at $D=4$, and each of their terms is a positive power of y. Away from coincidence,

$$\Box f(y) = H^2\Big[(4y-y^2)f''(y) + D(2-y)f'(y)\Big], \tag{33}$$

which we verified with explicit coordinates for $D=4,5$ (`code/03_three_point_setup/check_52.py`, check (g)). As a check that the definitions were read correctly, we compared the series (31)–(32) with the hypergeometric forms at $D=0.6$ and $y=0.5$. They agree to about 10^{-31} (`code/02_four_point/check_coincidence.py`).

The first coefficient of (I.14). Three independent arguments fix the coefficient c_0 of $\mathcal{N}(4/y)^{D/2-1}$ in $A(y)$ to $\Gamma(\frac{D}{2}-1)$.

1. The leading singularity of $A(y)$ is that of $i\Delta_{\rm cf}$, (30). Up to the conformal factor $(aa')^{1-D/2}$, this is the flat-space Feynman propagator $\Gamma(\frac{D}{2}-1)/(4\pi^{D/2})\,(\Delta x^2)^{1-D/2}$, with exactly this coefficient.

2. Away from coincidence, (33) and the propagator equation give $(4y-y^2)A''+D(2-y)A' = \text{const}$. The coefficient of $y^{1-D/2}$ must therefore cancel between the first two terms of (29). This requires $c_0 = (\frac{D}{2}-2)c_1/[(\frac{D}{2}-1)\frac{D}{2}]$. With the printed second coefficient $c_1 = \Gamma(\frac{D}{2}+1)/(\frac{D}{2}-2)$ this gives $c_0 = \Gamma(\frac{D}{2}-1)$.

3. The mixed second derivative of $i\Delta_A$ has a contact term. The inner derivatives annihilate $k\ln(aa')$, and [cf. (I.52)]

$$\partial_\rho\partial'_\sigma\, i\Delta_A(x;x') = \delta^0_\rho\delta^0_\sigma\,\frac{i}{a^{D-2}}\,\delta^D(x-x') + A''(y)\,\frac{\partial y}{\partial x^\rho}\,\frac{\partial y}{\partial x'^\sigma} + A'(y)\,\frac{\partial^2 y}{\partial x^\rho \partial x'^\sigma}. \tag{34}$$

   The local coefficient was obtained in three ways: directly from the series with the $i\delta$ prescription, from the propagator equation (I.12) alone, and from the canonical equal-time commutator. The direct computation gives $i\,c_0(\frac{D}{2}-1)a^{2-D}/\Gamma(\frac{D}{2})$. The other two routes do not use the normalization of $A(y)$, and both give $i\,a^{2-D}$.

With the printed c_0, the local coefficient in (34) would carry an extra factor $2/(D-2)$. This factor is 1 at $D=4$. Multiplied by the pole of $i\Delta_A(x;x)$, however, it would leave the finite error $2k/(D-2)$. The printed (I.52) has the correct coefficient, so the error is in the printing only. It does not affect the coincidence limits either, because the term vanishes as $y\to 0$ in dimensional regularization. The derivation of (34) and its use in the 3-point contribution are given in Sec. IV C (`code/03_three_point_setup/check_52.py`, checks (e) and (f)).

F. The self-mass-squared and the two sign conventions

Paper I defines the self-mass-squared through the effective field equation for the quantum-corrected mode functions. Paper I prints [(I.168)]

$$\partial_\mu\big(\sqrt{-\hat g}\,\hat g^{\mu\nu}\partial_\nu\Phi(x)\big) - \int d^4x'\,M^2_{\rm ren}(x;x')\,\Phi(x') = 0.$$

We adopt this equation as the definition of M^2. For the regulated kernel we use the same form in D dimensions. Write the effective action of the background Φ as

$$e^{i\Gamma[\Phi]} = \int [Dh][D\varphi]_{\rm 1PI}\,e^{iS[\Phi+\varphi,h]}, \qquad \Gamma[\Phi] = S_0[\Phi] - \tfrac{1}{2}\iint \Phi\,M^2\,\Phi + \ldots, \tag{35}$$

where $S_0 = \int -\frac{1}{2}a^{D-2}\partial\Phi\cdot\partial\Phi$ has $\delta S_0/\delta\Phi = \partial_\mu(\sqrt{-\hat g}\,\hat g^{\mu\nu}\partial_\nu\Phi)$. Then $\delta\Gamma/\delta\Phi = 0$ is (I.168), and

$$-iM^2(x;x') = \frac{\delta^2\big(i\Gamma_1[\Phi]\big)}{\delta\Phi(x)\,\delta\Phi(x')}, \tag{36}$$

where Γ_1 is the quantum part of Γ. This matches Sec. IV of Paper I, which refers to the self-mass-squared as a one-particle-irreducible (1PI) function. A local counterterm action $S_{\rm ct}$ contributes $-iM^2_{\rm ct} = i\,\delta^2 S_{\rm ct}/\delta\Phi\,\delta\Phi$. Section II G shows that this reproduces the paper's own counterterm vertices (I.130)–(I.131).

TABLE 3. Sign conventions of the printed equations. The definition sign is fixed by (I.168) and (36); the paper convention is the sign as printed, and definition sign = −1× paper convention.

Convention	Printed equations
Paper convention	4-point: (I.32), (I.33), (I.41), (I.44). 3-point: (I.47)–(I.49), (I.54), (I.57)–(I.59), (I.63)–(I.67), (I.89), Tables III–X. Renormalization: (I.160)–(I.165), (I.167). Paper II: Table 1, (II.10), Table 5.
Definition sign	(I.45), (I.46).
Consistent with the definition	counterterm vertices (I.130), (I.131), (I.145), built with the convention-free rule vertex $= i\,\delta^2 S_{\rm ct}/\delta\phi\,\delta\phi$ (Sec. II G).
Convention independent	(I.50)–(I.53), (I.55)–(I.56), (I.60)–(I.62), (I.68)–(I.88), (I.90)–(I.103) (identities without an overall coefficient); (I.166), $\alpha_i = -\beta_i + \Delta\alpha_i$.

Definition sign and paper convention. We call the sign fixed by (36), i.e. by (I.168), the *definition sign*. The *paper convention* (or Table-X convention) is the sign of the self-mass contributions as Paper I prints them. This paper establishes that the definition sign is −1 times the paper convention, Eq. (4). Our Table 3 lists which printed equations are in which convention. Both the 4-point and the 3-point contributions are reversed, so their relative sign in Paper I is consistent. The local cancellation stated after (I.67) holds in either convention. The sign of the local part of the renormalized self-mass has no physical meaning, since it is fixed by the arbitrary finite counterterms $\Delta\alpha_i$. The nonlocal part has no counterterm freedom. Once (I.168) is adopted, its sign is physical: the physical nonlocal self-mass is −(corrected Table X). We make every physical statement in the definition sign and state the convention wherever we quote a printed number. The overall sign reverses the one-loop correction to the mode function of Paper II; as shown in Sec. VII, the non-falling total of that correction is zero at late times, so only terms that fall off change sign and the late-time conclusion is unchanged.

G. From vertex operators to kernels: Lemma A and the derivative rules

Since $\delta^D(x-x')$ depends only on $x-x'$, we have $\partial'_\mu\delta^D(x-x') = -\partial_\mu\delta^D(x-x')$. A kernel acts as $(\mathcal{K}f)(x) = \int d^Dx'\,\mathcal{K}(x;x')f(x')$, and two kernels are equal if they agree on all pairs of test functions $f_1(x)f_2(x')$.

Lemma A. Let $F[\phi] = \int d^Dz\,K^{\mu\nu}(z)\,\partial_\mu\phi\,\partial_\nu\phi$ with K smooth and not necessarily symmetric, and let $\mathcal{K}^{\mu\nu} \equiv K^{\mu\nu} + K^{\nu\mu}$. Then

$$\frac{\delta^2 F}{\delta\phi(x)\,\delta\phi(x')} = -\partial_\mu\Big[\mathcal{K}^{\mu\nu}(x)\,\partial_\nu\delta^D(x-x')\Big] = +\partial_\mu\Big[\mathcal{K}^{\mu\nu}(x)\,\partial'_\nu\delta^D(x-x')\Big]. \tag{37}$$

Proof. The coefficient of $\epsilon_1\epsilon_2$ in $F[\epsilon_1 f_1 + \epsilon_2 f_2]$ is $\int d^Dx\,\mathcal{K}^{\mu\nu}\partial_\mu f_1\,\partial_\nu f_2$. Apply the right-hand side of (37) to f_2, integrate against f_1, and integrate by parts once; the boundary term vanishes by compact support. The result is the same expression. We also proved the lemma with formal δ-functions, by integrating by parts in the vertex coordinate z.

In words: one derivative is moved, by partial integration, onto the whole expression and becomes $-\partial$. The other acts on $\delta^D(x-x')$ as $+\partial$ *with respect to* x, which equals $-\partial'$. A positivity test fixes the sign. For $D=1$ and $K=c>0$, $F=\int c\,\phi'^2 \geq 0$, so the Hessian must be a positive operator. Indeed $\int f\,[-2(cf')'] = \int 2cf'^2 \geq 0$, while the opposite sign gives a negative operator.

Calibration with the counterterm vertices. Paper I prints [(I.130), (I.131)]

$$\tfrac{1}{2}\alpha_1\kappa^2\Box\phi\,\Box\phi\,a^D \;\to\; i\alpha_1\kappa^2 a^D\Box^2\delta^D(x-x'), \qquad -\tfrac{1}{2}\alpha_2\kappa^2H^2\partial_\mu\phi\,\partial^\mu\phi\,a^{D-2} \;\to\; i\alpha_2\kappa^2H^2a^D\Box\,\delta^D(x-x').$$

For the second term, Lemma A with $\mathcal{K}^{\mu\nu} = -\alpha_2\kappa^2H^2a^{D-2}\eta^{\mu\nu}$ and (12) give

$$i\frac{\delta^2 S_{\rm ct}}{\delta\Phi\,\delta\Phi} = -i\,\partial_\mu\big[-\alpha_2\kappa^2H^2a^{D-2}\partial^\mu\delta^D(x-x')\big] = +i\alpha_2\kappa^2H^2a^D\Box\,\delta^D(x-x'), \tag{38}$$

which is identical to (I.131). Because $a^D\Box$ is self-adjoint, the first term gives exactly (I.130). The definition (36) therefore reproduces the paper's own counterterm vertices. Of the two, the vertex (I.131) carries one derivative on each leg, like the 4-point vertices, and so calibrates the derivative-sign rule directly. The vertex (I.130) carries two derivatives on each leg and does not depend on that rule.

TABLE 4. Derivative rules for the outer legs (definition sign). The 3-point rule of Paper I is correct; its 4-point rule differs by the treatment of the second leg.

	3-point, (39)	4-point, (43)
outer legs	at different vertices, x and x'	at the same vertex x, with $\delta^D(x-x')$
leg 1	partial integration: $-\partial$, on everything	partial integration: $-\partial$, on everything
leg 2	partial integration: $-\partial'$, on everything	$+\partial$ (w.r.t. x) on $\delta^D(x-x')$, i.e. $-\partial'$
sign product	$(-)(-)=+$	$(-)(+)=-$
Paper I	same as here	(I.32): $\partial_2 \to +\partial'$, overall sign reversed

4-point contribution. At one loop the 4-point vertices contribute $i\Gamma_{4\mathrm{pt}} = i\int d^D z\,\langle \kappa^2 \mathcal{L}^{(2)}\rangle$. Once the two graviton fields are contracted into the coincident propagator $X_{\alpha\beta\rho\sigma}$, this is a quadratic form of the type of Lemma A in $\partial\Phi$. Lemma A and (36) then give the kernel in the form of (I.31), $-iM^2_{4\mathrm{pt}} = \sum_I U_I^{\alpha\beta\rho\sigma}\, i[_{\alpha\beta}\Delta_{\rho\sigma}](x;x)\,\delta^D(x-x')$, with normalization 1 (the scalar $\frac{1}{2}$ in (18) cancels against the factor 2 of Lemma A), provided the derivatives are read as follows:

(R1) $\partial_{1\mu} \to -\partial_\mu$, acting on everything to its right: a^{D-2}, X and the factor containing δ^D;

(R2) $\partial_{2\nu} \to +\partial_\nu$, the derivative with respect to x, acting only on $\delta^D(x-x')$; this means $-\partial'_\nu$.

The paper's rule for ∂_1 is (R1). For ∂_2, the text after (I.31) says that "∂_2" acts upon x', and (I.32) uses $\partial_2 \to +\partial'$. This is the opposite of (R2). Every U_I contains exactly one ∂_1 and one ∂_2, so the difference is a common factor -1 for the whole 4-point contribution. The explicit kernel, the resulting (I.32)–(I.33), and the independent evidence for (R2) (the field-equation route, the calibration with (I.131), test functions, a lattice Hessian and an operator-level check) are given in Sec. III A.

3-point contribution. For two cubic vertices the two outer legs sit at different vertices, x and x'. Each outer-leg derivative produces $\partial^z\delta^D(z-x)$ at its own vertex, and a partial integration in z turns it into $-\partial$ acting on everything at that vertex; the inner-leg derivatives act only on $i\Delta_A$. With the overall normalization 1 (Sec. IV A) this gives [cf. (I.45)]

$$-iM^2_{3\mathrm{pt}}(x;x') = \sum_{I=1}^{2} V_I^{\alpha\beta}(x) \sum_{J=1}^{2} V_J^{\rho\sigma}(x')\; i\big[_{\alpha\beta}\Delta_{\rho\sigma}\big](x;x')\, i\Delta_A(x;x'), \tag{39}$$

read with the following rules:

(R1) $\partial_1 \to -\partial$, and likewise (R1′) $\partial'_1 \to -\partial'$, each acting on everything at its own vertex;

(R2) $\partial_2 \to +\partial$, and likewise (R2′) $\partial'_2 \to +\partial'$, acting only on $i\Delta_A$.

This agrees with (I.45) and with the rule stated in the text of Paper I ("partially integrate any derivative that acts upon an outer leg"). The effective action behind (39), its normalization, the derivation of (I.46) and the code checks are given in Sec. IV A. Our Table 4 compares the two rules.

Origin of the overall sign. In the 4-point contribution, the printed result is reversed at its source, the reading of ∂_2 in (I.32). In the 3-point contribution the rule is correct and (I.45)–(I.46) agree with the definition sign. However, the step from (I.46) to (I.47)–(I.49) carries an overall sign slip in all nine terms. The paper's subsequent 3-point expressions evaluate left-hand sides that carry the printed signs of (I.47)–(I.49) (Secs. IV B–IV E). Both parts of the printed $-iM^2$ are therefore -1 times the quantity defined by (I.168), and their relative sign is consistent. Several independent lines of evidence support the definition sign: Lemma A (two proofs), the field-equation route, the counterterm vertex (I.131), test functions, the lattice Hessian and positivity, and the consistency of the 3-point steps.

H. Summary of this section

Everything else in this part of Paper I is confirmed as printed:

- (I.4), (I.5) and the bracket of (I.6) (exact, general D);
- Table I, (I.8), and rows 1, 3 and 4 of Table II;
- the structure and normalization of (I.31) and (I.45), and the four terms of (I.46);

TABLE 5. Differences from the printed equations in the part of Paper I covered by this section. "Typo" means that the printed value did not enter the paper's computation. S1 is the 4-point origin of the overall sign, Eq. (4); it is classified as substantive because the sign of the nonlocal part of the self-mass is physical. Its consequences are discussed in the following sections.

ID	Where	Printed	Correct	Class / evidence
C1	(I.1)	$-\frac{1}{2}\partial_\mu\phi\,\partial_\mu\phi\,g^{\mu\nu}\sqrt{-g}$	$-\frac{1}{2}\partial_\mu\phi\,\partial_\nu\phi\,g^{\mu\nu}\sqrt{-g}$	typo; free index
C2	(I.6), left side	$-\frac{1}{2}\partial_\mu\phi\,\partial_\nu g^{\mu\nu}\sqrt{-g}$	$-\frac{1}{2}\partial_\mu\phi\,\partial_\nu\phi\,g^{\mu\nu}\sqrt{-g}$	typo; right side correct
C3	(I.7), left side	$-\frac{1}{2}\kappa a^{D-2}\partial_\alpha\phi\,\partial_\beta\phi\,h^{\alpha\beta}$	$+\frac{1}{2}\kappa a^{D-2}\partial_\alpha\phi\,\partial_\beta\phi\,h^{\alpha\beta}$	typo; V_1 correct and used in (I.46)
C4	(I.14), first coefficient	$\Gamma(\frac{D}{2}-1)/(\frac{D}{2}-1)$	$\Gamma(\frac{D}{2}-1)$	typo; required by (I.26), the recursion and (I.52)
C5	Table II, U_2	$-\frac{i}{4}\kappa^2 a^{D-2}\eta^{\alpha\rho}\eta^{\beta\sigma}\partial_1\cdot\partial_2$	$+\frac{i}{4}\kappa^2 a^{D-2}\eta^{\alpha\rho}\eta^{\beta\sigma}\partial_1\cdot\partial_2$	typo; (I.33) uses $+\frac{i}{4}$
S1	rule of (I.32)	$\partial_2\to+\partial'$ on $\delta^D(x-x')$	$\partial_2\to+\partial=-\partial'$ (Lemma A)	substantive: 4-point origin of the overall sign (physical for the nonlocal part)

- (I.52), obtained along three routes.

The differences are collected in our Table 5. The labels Cn refer to the errata list of Sec. VIII (our Table 40), and S1 denotes the first substantive finding, the overall sign of the printed self-mass.

III. THE 4-POINT CONTRIBUTION

This section corresponds to Sec. III A of Paper I [8], Eqs. (I.31)–(I.44). The 4-point vertices U_I of Table II of Paper I, with the graviton loop closing on itself at a single point (Fig. 1 of Paper I), give a purely local contribution to the self-mass-squared. Three steps are needed: (i) the kernel generated by the vertices, which fixes how the derivatives of Table II have to be read; (ii) the coincidence limits of the three scalar propagators and of the four contractions of the coincident graviton propagator; (iii) the assembly into the final local form. The outcome is the following. The structure and normalization of (I.31), all D-dependent coefficients, the coincidence limits (I.34)–(I.39) and the algebra (I.41)→(I.44) are confirmed. There are two differences. First, the rule by which Paper I reads the derivative "∂_2" introduces a factor -1 relative to the paper's own definition (I.168); this is the 4-point origin of the overall sign of the printed self-mass. Second, the finite part of (I.40) has the wrong sign, which produces extra finite terms in (I.41) and (I.44); these have counterterm form and are absorbed. Full details are in Ch. 2 of the derivation notes and in the scripts of `code/02_four_point/` [17].

Conventions. We write $\partial_\mu\equiv\partial/\partial x^\mu$, $\partial'_\mu\equiv\partial/\partial x'^\mu$, and indices are raised and lowered with $\eta_{\mu\nu}=\mathrm{diag}(-1,1,\dots,1)$. We use conformal coordinates with $a=-1/(H\eta)$, so $\partial_0 a=Ha^2$. The dimension D is kept general throughout, unless a step says otherwise. We abbreviate

$$k\equiv\frac{H^{D-2}}{(4\pi)^{D/2}}\frac{\Gamma(D-1)}{\Gamma(\frac{D}{2})},\qquad h\equiv\frac{H^2}{16\pi^2},\qquad \hat\pi\equiv-\pi\cot\Big(\frac{\pi D}{2}\Big),\tag{40}$$

where k is the constant of (I.13); as $D\to4$, $k\to2h=H^2/8\pi^2$. (The constant h is not to be confused with the graviton trace $h=\eta^{\mu\nu}h_{\mu\nu}$ of Sec. II, which does not occur in this section.) Signs matter in this section. The *definition sign* is the one fixed by Paper I's own definition (I.168) through Eq. (36) (Sec. II F); the *paper convention* is the sign with which Paper I prints (I.32), (I.33), (I.41) and (I.44). All kernels derived below are in the definition sign, and we show that the paper convention is -1 times the definition sign [Eq. (4)].

A. The kernel of the 4-point vertices and the derivative rule

Vertices. The 4-point vertex operators U_I, derived from the expansion (I.6) in the normalization (18), are listed in our Table 2. A factor $\frac{1}{2}$ is extracted for the two identical scalars but not for the two identical gravitons, so that no symmetry factor is needed (Sec. II C). Table II of Paper I prints U_2 with $-\frac{i}{4}$. The printed (I.33) carries $+\frac{i}{4}$ in the corresponding term, so the derived value $+\frac{i}{4}$ was the one used in the calculation and the printed $-\frac{i}{4}$ is a typographical error.

Effective action. At order κ^2 and quadratic order in the background Φ, the 1PI contributions are (i) a single 4-point vertex with a closed graviton loop, (ii) two 3-point vertices (Sec. IV), and (iii) counterterms. A 3-point vertex with a graviton tadpole is not 1PI. For (i), $i\Gamma_{4\mathrm{pt}}[\Phi] = i\langle S_4[\Phi, h]\rangle$, with $\langle T\, h_{\alpha\beta}(x) h_{\rho\sigma}(x')\rangle = i[_{\alpha\beta}\Delta_{\rho\sigma}](x;x')$. In dimensional regularization the coincidence limit

$$X_{\alpha\beta\rho\sigma}(x) \equiv i\big[_{\alpha\beta}\Delta_{\rho\sigma}\big](x;x) \tag{41}$$

is a smooth function of x. It is symmetric within each index pair, and it is symmetric under exchange of the pairs, $X_{\alpha\beta\rho\sigma} = X_{\rho\sigma\alpha\beta}$; the latter can be read off term by term from the tensor factors (I.22)–(I.24). Writing $U_I^{\alpha\beta\rho\sigma} X_{\alpha\beta\rho\sigma} \equiv \tilde{K}_I^{\mu\nu}(z)\, \partial_{1\mu}\partial_{2\nu}$, we obtain

$$\langle S_4\rangle = \int d^D z\, \frac{1}{2i} \sum_I \tilde{K}_I^{\mu\nu}(z)\, \partial_\mu\Phi\, \partial_\nu\Phi. \tag{42}$$

Second functional derivatives. We use Lemma A of Sec. II G, Eq. (37): for a quadratic form $F = \int d^D z\, K^{\mu\nu}\partial_\mu\phi\, \partial_\nu\phi$, one of the two derivatives is integrated by parts onto the whole expression, where it becomes $-\partial$, while the other acts on $\delta^D(x-x')$ as $+\partial$ *with respect to* x, i.e. as $-\partial'$, because $\partial'_\mu \delta^D(x-x') = -\partial_\mu \delta^D(x-x')$. The sign is fixed by a positivity test (Sec. II G).

The kernel and the reading rule. Applying Eq. (36) and Lemma A to Eq. (42),

$$-iM^2_{4\mathrm{pt}}(x;x') = i\,\frac{\delta^2\langle S_4\rangle}{\delta\Phi(x)\,\delta\Phi(x')} = -\sum_I \partial_\mu\Big[\tilde{K}^{\mu\nu}_{I,s}(x)\,\partial_\nu\delta^D(x-x')\Big], \qquad \tilde{K}_s \equiv \tfrac{1}{2}\big(\tilde{K}+\tilde{K}^T\big). \tag{43}$$

The scalar factor $\frac{1}{2}$ of Eq. (18) cancels against the factor 2 from the two ways of attaching the external legs, so the normalization is 1, and Eq. (43) takes the form of (I.31):

$$-iM^2_{4\mathrm{pt}}(x;x') = \sum_{I=1}^{4} U_I^{\alpha\beta\rho\sigma}\; i\big[_{\alpha\beta}\Delta_{\rho\sigma}\big](x;x)\;\delta^D(x-x'). \tag{44}$$

This notation is given its meaning by the reading rule that follows from Eq. (43):

(R1) $\partial_{1\mu} \to -\partial_\mu$, acting on everything to its right (a^{D-2}, the coincident propagator and the factor containing δ^D);

(R2) $\partial_{2\nu} \to +\partial_\nu$, the derivative with respect to x, acting only on $\delta^D(x-x')$; equivalently $\partial_{2\nu} \to -\partial'_\nu$;

(R3) if $\tilde{K}_I$ is not symmetric, it is symmetrized. This is not needed here: $\tilde{K}_{1,2}$ are proportional to $\eta^{\mu\nu}$, $\tilde{K}_3$ is symmetric by the within-pair symmetry, and $\tilde{K}_4$ is symmetric by the pair-exchange symmetry of $X_{\alpha\beta\rho\sigma}$.

Rule (R1) is the paper's rule that "∂_1" is partially integrated back onto the entire contribution. For "∂_2", however, Paper I states that it acts upon x' and uses $\partial_2 \to +\partial'$ in (I.32). This is exactly the opposite of (R2). Since each U_I contains exactly one ∂_1 and one ∂_2, the difference is a common factor -1 for the whole 4-point contribution.

For the first vertex we obtain, in the definition sign,

$$-\tfrac{i}{8}\kappa^2 a^{D-2}\eta^{\alpha\beta}\eta^{\rho\sigma}\,\partial_1\cdot\partial_2 \times i\big[_{\alpha\beta}\Delta_{\rho\sigma}\big](x;x)\,\delta^D(x-x') \;\Rightarrow\; -\tfrac{i}{8}\kappa^2\,\partial^\mu\Big\{a^{D-2}\,i\big[^\alpha_\alpha\Delta^\rho_\rho\big](x;x)\,\partial'_\mu\delta^D(x-x')\Big\}, \tag{45}$$

whereas the right-hand side of the printed (I.32) carries $+\frac{i}{8}$. The contractions for U_2, U_3, U_4 are $\eta^{\alpha\rho}\eta^{\beta\sigma}X_{\alpha\beta\rho\sigma} = i\big[^{\alpha\beta}\Delta_{\alpha\beta}\big](x;x)$, $\eta^{\alpha\beta}X_{\alpha\beta}{}^{\rho\sigma} = i\big[^\alpha_\alpha\Delta^{\rho\sigma}\big](x;x)$ and $X^\alpha{}_\rho{}^{\rho\sigma} = i\big[^\alpha{}_\rho\Delta^{\rho\sigma}\big](x;x)$. Collecting all four terms (definition sign, general D; cf. (I.33)):

$$\begin{aligned} -iM^2_{4\mathrm{pt}}(x;x') = {} & +\tfrac{i}{8}\kappa^2\partial^\mu\big\{a^{D-2}\,i\big[^\alpha_\alpha\Delta^\rho_\rho\big](x;x)\,\partial_\mu\delta^D(x-x')\big\} - \tfrac{i}{4}\kappa^2\partial^\mu\big\{a^{D-2}\,i\big[^{\alpha\beta}\Delta_{\alpha\beta}\big](x;x)\,\partial_\mu\delta^D(x-x')\big\} \\ & -\tfrac{i}{2}\kappa^2\partial_\rho\big\{a^{D-2}\,i\big[^\alpha_\alpha\Delta^{\rho\sigma}\big](x;x)\,\partial_\sigma\delta^D(x-x')\big\} + i\kappa^2\partial^\alpha\big\{a^{D-2}\,i\big[_{\alpha\rho}\Delta^{\rho\sigma}\big](x;x)\,\partial_\sigma\delta^D(x-x')\big\}. \end{aligned} \tag{46}$$

The printed (I.33) has the coefficients $(-\frac{i}{8}, +\frac{i}{4}, +\frac{i}{2}, -i)$. Magnitudes, tensor contractions and relative signs agree; the overall sign is reversed. In compact form,

$$\begin{aligned} -iM^2_{4\mathrm{pt}} &= -\partial_\mu\big[\mathcal{K}^{\mu\nu}(x)\,\partial_\nu\delta^D(x-x')\big], \qquad \mathcal{K}^{\mu\nu} = \kappa^2 a^{D-2}\Big\{-\tfrac{i}{8}\eta^{\mu\nu}X_1 + \tfrac{i}{4}\eta^{\mu\nu}X_2 + \tfrac{i}{2}X_3^{\mu\nu} - i\,X_4^{\mu\nu}\Big\}, \\ X_1 &= i\big[^\alpha_\alpha\Delta^\rho_\rho\big](x;x), \quad X_2 = i\big[^{\alpha\beta}\Delta_{\alpha\beta}\big](x;x), \quad X_3^{\mu\nu} = i\big[^\alpha_\alpha\Delta^{\mu\nu}\big](x;x), \quad X_4^{\mu\nu} = \eta^{\mu\alpha}\,i\big[_{\alpha\rho}\Delta^{\rho\nu}\big](x;x). \end{aligned} \tag{47}$$

Physically, $\mathcal{K}^{\mu\nu}$ is the correction to the kinetic term generated by $\langle\mathcal{L}_4\rangle$.

Independent evidence for the sign. The sign of Eq. (46) is supported by several independent arguments.

1. *Field-equation route.* This route uses only the Euler–Lagrange equation and not the lemma. From $\Gamma \supset \int \frac{1}{2i}\mathcal{K}^{\mu\nu}\partial_\mu\Phi\partial_\nu\Phi$ ($\mathcal{K}$ symmetric), $\delta\Gamma/\delta\Phi = \partial_\mu(a^{D-2}\partial^\mu\Phi) + i\,\partial_\mu(\mathcal{K}^{\mu\nu}\partial_\nu\Phi) + \dots$. Comparison with (I.168) gives $\int M^2\Phi = -i\,\partial_\mu(\mathcal{K}^{\mu\nu}\partial_\nu\Phi)$, which is Eq. (47).

2. *The paper's own counterterm vertex (I.131).* Like the 4-point vertices, the counterterm $-\frac{1}{2}\alpha_2\kappa^2H^2\partial_\mu\phi\,\partial^\mu\phi\,a^{D-2}$ has one derivative on each leg, so it calibrates the derivative rule. Lemma A reproduces the printed vertex $i\alpha_2\kappa^2H^2a^D\Box\,\delta^D(x-x')$ [Eq. (38)]. Written as a vertex operator in the normalization of Eq. (18), the same term is $U_{\rm ct} = -i\alpha_2\kappa^2H^2a^{D-2}\partial_1\cdot\partial_2$. The rules (R1)–(R2) give $+i\alpha_2\kappa^2H^2a^D\Box\,\delta^D(x-x')$, while the paper's 4-point rule ($\partial_2 \to +\partial'$) gives $-i\alpha_2\kappa^2H^2a^D\Box\,\delta^D(x-x')$. Hence (I.131) and the rule of (I.32) cannot both hold in a single convention, and (I.131) is the one that agrees with (I.168). The vertex (I.130) is also reproduced, but it has two derivatives on each leg and is insensitive to the rule.

3. *Test functions, lattice Hessian and operator-level check* (script `check_4pt.py` [17]). (a) For $D = 2$, a deliberately non-symmetric polynomial $K^{\mu\nu}$ and two Gaussian-type test functions, the bilinear form is $\partial^2F/\partial\epsilon_1\partial\epsilon_2 = \frac{97\sqrt{6}\,\pi}{54}e^{-1/2} \approx 8.384$; the kernel (37) gives the same value and the opposite-sign kernel gives -8.384. (b) On a periodic one-dimensional lattice with $N = 7$ sites and $S = \sum_z c_z(\phi_{z+1}-\phi_z)^2$, the Hessian is exactly $-\nabla^-(2c\,\nabla^+)$, the lattice version of $-\partial(2c\,\partial\,\delta)$; for $c = 1$ its eigenvalues are 0, 1.506, 4.890, 7.604. (c) For $D = 3$, a general pair-symmetric $X_{\alpha\beta\rho\sigma}(x)$ built from 21 independent functions and a general $a(\eta)$, the kernel (43) equals Eq. (46) and is exactly -1 times the printed (I.33); the paper's rule combined with $U_2 = +\frac{i}{4}$ reproduces the printed (I.33) exactly, while with the printed $U_2 = -\frac{i}{4}$ it does not. (d) The lemma reproduces (I.131); the paper's 4-point rule gives its negative.

The flat-space reduction cannot discriminate between the two signs, because the coincident flat-space graviton propagator vanishes in dimensional regularization.

Relation to the 3-point part. The paper's derivative rule for the 3-point diagram ("partially integrate any derivative that acts upon an outer leg") is correct: each outer-leg derivative becomes $-\partial$ at its own vertex, as in the lemma. However, as shown in Sec. IV, the printed (I.47)–(I.49) are -1 times what follows from the paper's own (I.46). Both parts of the printed result are therefore reversed. Their relative sign is consistent, the local cancellation between (I.44) and (I.67) holds in either convention, and *the printed self-mass is* -1 *times the one defined by (I.168).* Because the nonlocal part of the self-mass has no counterterm freedom, this overall sign is physical for the nonlocal part, which is $-$(corrected Table X). It reverses the sign of the one-loop mode-function correction of Paper II; but, as shown in Sec. VII, the non-falling total of that correction is zero at late times, so only decaying terms change sign and the late-time conclusion of Paper II is unchanged.

B. Coincidence limits

These limits (cf. (I.34)–(I.40); the paper takes them from Ref. [25]) are independent of the vertices and of the derivative rule, so the sign issue of Sec. III A does not enter here.

Scalar propagators. As $x' \to x$ one has $y(x;x') \to 0$ and $\ln(aa') \to 2\ln a$. The propagators (I.13)–(I.14) and (I.26)–(I.28) are sums of powers $(y/4)^{p(D)}$ with exponents of the form $1-\frac{D}{2}$, $2-\frac{D}{2}$, $n \geq 1$ and $n-\frac{D}{2}+2$ ($n \geq 0$). All of them are positive for $\mathrm{Re}\,D < 2$, where the limit exists literally and may be taken term by term (the series converge uniformly on $0 < y \leq y_0 < 4$); the result is then continued analytically in D. Only the terms with $p \equiv 0$ survive.[1] This gives (cf. (I.34)–(I.36))

$$\lim_{x'\to x} i\Delta_A(x;x') = k\big\{\hat\pi + 2\ln a\big\} = k\Big\{-\frac{2}{D-4} + 2\ln a + O(D-4)\Big\}, \tag{48}$$

$$b \equiv \lim_{x'\to x} i\Delta_B(x;x') = -\frac{k}{D-2} \;\longrightarrow\; -h, \qquad c \equiv \lim_{x'\to x} i\Delta_C(x;x') = \frac{k}{(D-2)(D-3)} \;\longrightarrow\; +h, \tag{49}$$

where $\hat\pi = -2/(D-4) + \frac{\pi^2}{6}(D-4) + O\big((D-4)^3\big)$. The B- and C-type limits are finite at $D = 4$ and are related by

$$c = -\frac{b}{D-3}. \tag{50}$$

[1] The first coefficient of $A(y)$ in (I.14) is misprinted (it should be $\Gamma(\frac{D}{2}-1)$; erratum C4); since that term vanishes in the coincidence limit, (I.34) is not affected.

As an independent check, b and c follow from Gauss's theorem ${}_2F_1(a_1,a_2;c_1;1)=\Gamma(c_1)\Gamma(c_1-a_1-a_2)/[\Gamma(c_1-a_1)\Gamma(c_1-a_2)]$ applied to the hypergeometric forms (I.29)–(I.30) in the region $\mathrm{Re}(c_1-a_1-a_2)=1-\frac{D}{2}>0$; and at $D=0.6$, where no continuation is needed, the series approach Eqs. (48)–(49) like $(y/4)^{0.7}$ (script `check_coincidence.py`).

Tensor algebra. Let $u_\mu\equiv\delta^0_\mu$, so that $u^\mu=-\delta^\mu_0$, $u\cdot u=-1$, $\bar\eta_{\mu\nu}=\eta_{\mu\nu}+u_\mu u_\nu$, $\bar\eta_{\mu\nu}u^\nu=0$, $\eta^{\mu\nu}\bar\eta_{\mu\nu}=\bar\delta^\mu_\mu=D-1$, $u_\alpha u^\sigma=-\delta^0_\alpha\delta^\sigma_0$ and $u^\rho u^\sigma=\delta^\rho_0\delta^\sigma_0$. The tensor factors (I.22)–(I.24) of the graviton propagator, with the symmetrizations written out, are

$$\begin{aligned}
T^A_{\alpha\beta\rho\sigma}&=\bar\eta_{\alpha\rho}\bar\eta_{\sigma\beta}+\bar\eta_{\alpha\sigma}\bar\eta_{\rho\beta}-\tfrac{2}{D-3}\bar\eta_{\alpha\beta}\bar\eta_{\rho\sigma},\\
T^B_{\alpha\beta\rho\sigma}&=-\big[u_\alpha\bar\eta_{\beta\rho}u_\sigma+u_\beta\bar\eta_{\alpha\rho}u_\sigma+u_\alpha\bar\eta_{\beta\sigma}u_\rho+u_\beta\bar\eta_{\alpha\sigma}u_\rho\big],\\
T^C_{\alpha\beta\rho\sigma}&=\tfrac{2}{(D-2)(D-3)}f_{\alpha\beta}f_{\rho\sigma},\qquad f_{\alpha\beta}\equiv(D-3)u_\alpha u_\beta+\bar\eta_{\alpha\beta}.
\end{aligned}\tag{51}$$

Since these are constant tensors, the coincidence limit acts only on the scalar propagators:

$$X_{\alpha\beta\rho\sigma}=T^A_{\alpha\beta\rho\sigma}\,i\Delta_A(x;x)+T^B_{\alpha\beta\rho\sigma}\,b+T^C_{\alpha\beta\rho\sigma}\,c.\tag{52}$$

The four contractions needed in Eq. (46) then follow by elementary algebra. For example, in the double trace the T^B terms vanish because every term contains a u contracted with a $\bar\eta$, and $\eta^{\alpha\beta}f_{\alpha\beta}=2$. Exactly in D we find (cf. (I.37)–(I.40))

$$i\big[{}^{\alpha}_{\alpha}\Delta^{\rho}_{\rho}\big](x;x)=-\frac{4(D-1)}{D-3}\,i\Delta_A(x;x)+\frac{8}{(D-2)(D-3)}\,c,\tag{53}$$

$$i\big[{}^{\alpha\beta}\Delta_{\alpha\beta}\big](x;x)=\frac{(D-1)(D^2-3D-2)}{D-3}\,i\Delta_A(x;x)+2(D-1)\,b+\frac{2(D^2-5D+8)}{(D-2)(D-3)}\,c,\tag{54}$$

$$i\big[{}^{\alpha}_{\alpha}\Delta^{\rho\sigma}\big](x;x)=-\frac{4}{D-3}\,\bar\eta^{\rho\sigma}\,i\Delta_A(x;x)+\frac{4}{(D-2)(D-3)}\big[(D-3)\delta^\rho_0\delta^\sigma_0+\bar\eta^{\rho\sigma}\big]\,c,\tag{55}$$

$$\begin{aligned}
i\big[{}_{\alpha\rho}\Delta^{\rho\sigma}\big](x;x)=&\frac{D^2-3D-2}{D-3}\,\bar\delta^\sigma_\alpha\,i\Delta_A(x;x)+\big[(D-1)\delta^0_\alpha\delta^\sigma_0+\bar\delta^\sigma_\alpha\big]\,b\\
&+\frac{2}{(D-2)(D-3)}\big[(D-3)^2\delta^0_\alpha\delta^\sigma_0+\bar\delta^\sigma_\alpha\big]\,c.
\end{aligned}\tag{56}$$

Two identities hold by definition and are satisfied identically in D: the trace $\alpha=\sigma$ of Eq. (56) equals Eq. (54) (this uses only the within-pair symmetry), and $\eta_{\rho\sigma}$ times Eq. (55) equals Eq. (53). All four contractions were also verified component by component, with explicit D^4-component arrays for $D=4,5,6,7$ and symbolic $i\Delta_A$, b, c.

Finite parts. With Eq. (50) the B- and C-type (finite) parts become, exactly in D,

$$\begin{aligned}
(53):&\quad-\frac{8\,b}{(D-2)(D-3)^2},\qquad(54):\quad 2b\Big[(D-1)-\frac{D^2-5D+8}{(D-2)(D-3)^2}\Big],\\
(55):&\quad-\frac{4\,b}{(D-2)(D-3)^2}\big[(D-3)\delta^\rho_0\delta^\sigma_0+\bar\eta^{\rho\sigma}\big],\\
(56):&\quad\frac{D(D-3)}{D-2}\,b\,\delta^0_\alpha\delta^\sigma_0+b\Big[1-\frac{2}{(D-2)(D-3)^2}\Big]\bar\delta^\sigma_\alpha,
\end{aligned}\tag{57}$$

and at $D=4$ ($b\to-h$)

$$+4h,\qquad-2h,\qquad\big[2\delta^\rho_0\delta^\sigma_0+2\bar\eta^{\rho\sigma}\big]h,\qquad-2\,\delta^0_\alpha\delta^\sigma_0\,h,\tag{58}$$

respectively; the coefficient of $\bar\delta^\sigma_\alpha$ in the last one vanishes. Setting $D=4$ in the finite parts is legitimate as long as they are not multiplied by a pole: b and c enter Eq. (46) linearly and without a divergent prefactor, so the change is $O(D-4)$. The D-dependent coefficients of the divergent $i\Delta_A(x;x)$, on the other hand, must be kept exactly, as the paper does.

The comparison with Paper I is given in our Table 6. Everything agrees except the finite part of (I.40), where Paper I prints $+2\,\delta^0_\alpha\delta^\sigma_0\,h$. The printed value is inconsistent with the paper's own (I.38): the trace identity, which holds by definition, would require the trace of the finite part of (I.40), $+2h$ as printed, to equal the finite part of (I.38), which is $-2h$ as printed. The discrepancy cannot be traced to the sign of T^B or T^C; for example, reversing T^B would make the finite part of (I.38) equal to $+10h$.

TABLE 6. Coincidence limits, cf. (I.34)–(I.40). Finite parts are quoted at $D=4$ in units of $h=H^2/16\pi^2$; the coefficients of $i\Delta_A(x;x)$ are exact in D.

Equation	This work	Paper I prints	Status
(I.34)	$k\{\hat\pi+2\ln a\}$	same	agrees
(I.35), (I.36)	$b=-\frac{k}{D-2}\to -h,\ \ c=\frac{k}{(D-2)(D-3)}\to +h$	same	agree
(I.37)	$-\frac{4(D-1)}{D-3}\,i\Delta_A$; finite $+4$	same	agrees
(I.38)	$\frac{(D-1)(D^2-3D-2)}{D-3}\,i\Delta_A$; finite -2	same	agrees
(I.39)	$-\frac{4}{D-3}\bar\eta^{\rho\sigma}i\Delta_A$; finite $2\delta^\rho_0\delta^\sigma_0+2\bar\eta^{\rho\sigma}$	finite $2\delta^\rho_0\delta^\sigma_0+2\bar\eta_{\rho\sigma}$	index placement only
(I.40), $i\Delta_A$	$\frac{D^2-3D-2}{D-3}\bar\delta^\sigma_\alpha\,i\Delta_A$	same	agrees
(I.40), finite	$-2\,\delta^0_\alpha\delta^\sigma_0$	$+2\,\delta^0_\alpha\delta^\sigma_0$	sign error

C. The 4-point result

Assembly in general D. Write $\Lambda\equiv i\Delta_A(x;x)=k(\hat\pi+2\ln a)$ and split each contraction into its Λ part and its finite part. Inserting Eqs. (53)–(56) into Eq. (47) (with $\eta^{\mu\alpha}\bar\delta^\nu_\alpha=\bar\eta^{\mu\nu}$), the coefficients of $\eta^{\mu\nu}$ and $\bar\eta^{\mu\nu}$ in the Λ part are

$$\tfrac{i}{8}\cdot\tfrac{4(D-1)}{D-3}+\tfrac{i}{4}\cdot\tfrac{(D-1)(D^2-3D-2)}{D-3}=\tfrac{i}{4}D(D-1),\qquad \tfrac{i}{2}\cdot\Big(-\tfrac{4}{D-3}\Big)-i\,\tfrac{D^2-3D-2}{D-3}=-iD. \tag{59}$$

The factors $1/(D-3)$ of the graviton propagator cancel exactly. The finite parts (57) combine, with $\eta^{\mu\nu}=\bar\eta^{\mu\nu}-u^\mu u^\nu$, into a multiple of $D-4$. Altogether $\mathcal{K}=\mathcal{K}_A+\mathcal{K}_F$ with

$$\mathcal{K}^{\mu\nu}_A=i\kappa^2a^{D-2}\,\Lambda\Big[\tfrac14D(D-1)\,\eta^{\mu\nu}-D\,\bar\eta^{\mu\nu}\Big],\qquad \mathcal{K}^{\mu\nu}_F=i\kappa^2a^{D-2}\,b\,\frac{D-4}{2(D-3)}\Big[(D-2)\,\bar\eta^{\mu\nu}+D\,u^\mu u^\nu\Big]. \tag{60}$$

Thus, with the correct (I.40), the finite B- and C-type contributions are exactly $O(D-4)$. For a function $g(\eta)$ of time only one has $\partial^\mu(g\,\partial_\mu\delta)=-\partial_0(g\,\partial_0\delta)+g\nabla^2\delta$, $\partial_\mu(g\,\bar\eta^{\mu\nu}\partial_\nu\delta)=g\nabla^2\delta$ and $\partial_\mu(g\,u^\mu u^\nu\partial_\nu\delta)=\partial_0(g\,\partial_0\delta)$, so the kernel is, exactly in D and in the definition sign,

$$\begin{aligned}-iM^2_{\rm 4pt}=&-i\kappa^2\Big\{\tfrac14D(D-1)\,\partial^\mu\big(a^{D-2}\Lambda\,\partial_\mu\delta^D(x-x')\big)-D\,a^{D-2}\Lambda\,\nabla^2\delta^D(x-x')\Big\}\\&-i\kappa^2\,b\,\frac{D-4}{2(D-3)}\Big\{D\,\partial_0\big(a^{D-2}\partial_0\delta^D(x-x')\big)+(D-2)\,a^{D-2}\nabla^2\delta^D(x-x')\Big\}.\end{aligned} \tag{61}$$

The analogue of (I.41). In Eq. (61) the $k\hat\pi$ terms carry a pole and are kept exactly in D. The $2k\ln a$ terms are finite, so setting $D=4$ in them ($2k\to H^2/4\pi^2$, $a^{D-2}\to a^2$, $\delta^D\to\delta^4$) changes them only by $O(D-4)$; the paper does the same. The b term is explicitly $O(D-4)$, is not multiplied by a pole, and drops out. In the definition sign,

$$\begin{aligned}-iM^2_{\rm 4pt}(x;x')=&-i\kappa^2k\,\hat\pi\,\Big\{\tfrac14D(D-1)\,\partial^\mu\big(a^{D-2}\partial_\mu\delta^D(x-x')\big)-D\,a^{D-2}\nabla^2\delta^D(x-x')\Big\}\\&-\frac{i\kappa^2H^2}{4\pi^2}\Big\{3\,\partial^\mu\big(a^2\ln(a)\,\partial_\mu\delta^4(x-x')\big)-4\ln(a)\,a^2\nabla^2\delta^4(x-x')\Big\}+O(D-4).\end{aligned} \tag{62}$$

Paper I prints (I.41) as

$$\begin{aligned}-iM^2_{\rm 4pt}\big|_{\rm printed}=&+i\kappa^2k\,\hat\pi\,\Big\{\tfrac14D(D-1)\,\partial^\mu\big(a^{D-2}\partial_\mu\delta^D(x-x')\big)-D\,a^{D-2}\nabla^2\delta^D(x-x')\Big\}\\&+\frac{i\kappa^2H^2}{4\pi^2}\Big\{3\,\partial^\mu\big(a^2\ln(a)\,\partial_\mu\delta^4(x-x')\big)-4\ln(a)\,a^2\nabla^2\delta^4(x-x')\\&\qquad-\partial^\mu\big(a^2\partial_\mu\delta^4(x-x')\big)+a^2\nabla^2\delta^4(x-x')\Big\}+O(D-4),\end{aligned} \tag{63}$$

where we have read the unparenthesized factor $-\pi\cot(\frac{\pi}{2}D)$ of the printed equation as multiplying the prefactor, as (I.44) confirms. Apart from the overall sign, the last bracket is the only difference. Using the paper's (I.33) together with the *printed* (I.40), the finite part at $D=4$ is $4i\kappa^2h\,\partial_0(a^2\partial_0\delta^4(x-x'))=\frac{i\kappa^2H^2}{4\pi^2}\{-\partial^\mu(a^2\partial_\mu\delta^4(x-x'))+a^2\nabla^2\delta^4(x-x')\}$, which is exactly this bracket; with the correct (I.40) the same place gives zero.

Extracting the logarithm. With (I.42), $\Box = a^{-D}\partial^\mu(a^{D-2}\partial_\mu)$, one has $\partial^\mu(a^{D-2}\partial_\mu\delta^D(x-x')) = a^D\Box\,\delta^D(x-x')$. For the logarithmic terms we use the distributional identity, valid for any smooth $g(x,x')$,

$$g(x,x')\,\partial_\mu\delta^D(x-x') = g(x,x)\,\partial_\mu\delta^D(x-x') + \big[\partial'_\mu g(x,x')\big]_{x'=x}\delta^D(x-x'), \tag{64}$$

which follows by testing with $f(x')$ and integrating by parts once. For $g = \ln a(\eta')$ it gives $\ln(a')\,\partial_\mu\delta^D(x-x') = \ln(a)\,\partial_\mu\delta^D(x-x') + (\partial_\mu \ln a)\,\delta^D(x-x')$. Combining this with the product rule one finds $\partial^\mu(a^{D-2}\ln(a)\,\partial_\mu\delta^D(x-x')) = \frac{1}{2}\ln(aa')\,a^D\Box\,\delta^D(x-x') - \frac{1}{2}\,a^D(\Box\ln a)\,\delta^D(x-x')$, and $\Box\ln a = -(D-1)H^2$. This is the general-D form of (I.43),

$$\partial^\mu\big(a^{D-2}\ln(a)\,\partial_\mu\delta^D(x-x')\big) = \tfrac{1}{2}\ln(aa')\,a^D\Box\,\delta^D(x-x') + \tfrac{1}{2}(D-1)H^2a^D\,\delta^D(x-x'), \tag{65}$$

which reduces to the printed (I.43) at $D=4$. The spatial counterpart has no correction term, $\ln(a)\,\nabla^2\delta^D(x-x') = \frac{1}{2}\ln(aa')\,\nabla^2\delta^D(x-x')$, because $\ln a'$ does not depend on $\vec{x}\,'$. The symmetric $\ln(aa')$ is essential: with $\ln a^2$ in its place, Eq. (65) is false.

Final 4-point result. Inserting Eq. (65) and $\frac{H^2}{4\pi^2} = 2k + O(D-4)$ into Eq. (62) gives the corrected form of (I.44), in the definition sign:

$$\begin{aligned} -iM^2_{4\rm pt}(x;x') = \; & -i\kappa^2k\Big\{\Big[-\tfrac{1}{4}D(D-1)\pi\cot\big(\tfrac{\pi}{2}D\big) + 3\ln(aa')\Big]a^D\Box \\ & + \Big[D\pi\cot\big(\tfrac{\pi}{2}D\big) - 4\ln(aa')\Big]a^{D-2}\nabla^2 + 9H^2a^D + O(D-4)\Big\}\delta^D(x-x'), \end{aligned} \tag{66}$$

with $k = \frac{H^{D-2}}{(4\pi)^{D/2}}\frac{\Gamma(D-1)}{\Gamma(D/2)}$, the same prefactor as in the printed equation. If Eq. (65) is used in general D, no finite term needs to be evaluated at $D=4$:

$$\begin{aligned} -iM^2_{4\rm pt} = \; & -i\kappa^2k\Big\{\tfrac{1}{4}D(D-1)\big[\hat\pi + \ln(aa')\big]a^D\Box - D\big[\hat\pi + \ln(aa')\big]a^{D-2}\nabla^2 \\ & + \tfrac{1}{4}D(D-1)^2H^2a^D\Big\}\delta^D(x-x') + O(D-4), \end{aligned} \tag{67}$$

where the $O(D-4)$ is the b term of Eq. (61); at $D=4$, $\frac{1}{4}\cdot 4\cdot 3 = 3$ and $\frac{1}{4}\cdot 4\cdot 9 = 9$. In the paper convention the 4-point result is -1 times Eq. (66),

$$\begin{aligned} -iM^2_{4\rm pt}\big|_{\rm paper\ conv.} = \; & +i\kappa^2k\Big\{\Big[-\tfrac{1}{4}D(D-1)\pi\cot\big(\tfrac{\pi}{2}D\big) + 3\ln(aa')\Big]a^D\Box \\ & + \Big[D\pi\cot\big(\tfrac{\pi}{2}D\big) - 4\ln(aa')\Big]a^{D-2}\nabla^2 + 9H^2a^D + O(D-4)\Big\}\delta^D(x-x'), \end{aligned} \tag{68}$$

which is the printed (I.44) with the constant -2 removed from the $a^D\Box$ bracket and the constant $+2$ removed from the $a^{D-2}\nabla^2$ bracket. The printed (I.44) reads

$$\begin{aligned} -iM^2_{4\rm pt}\big|_{\rm printed} = \; & +i\kappa^2k\Big\{\Big[-\tfrac{1}{4}D(D-1)\pi\cot\big(\tfrac{\pi}{2}D\big) - 2 + 3\ln(aa')\Big]a^D\Box \\ & + \Big[D\pi\cot\big(\tfrac{\pi}{2}D\big) + 2 - 4\ln(aa')\Big]a^{D-2}\nabla^2 + 9H^2a^D + O(D-4)\Big\}\delta^D(x-x'). \end{aligned} \tag{69}$$

Diagnosis. The paper's inputs (the printed (I.33) and the printed (I.40)) reproduce the printed (I.41) and (I.44) exactly, and the printed (I.41) gives the printed (I.44) exactly through (I.42)–(I.43) (script `check_41_44.py`). The steps (I.33)→(I.41)→(I.44) therefore contain no new error. Both differences are inherited: the overall sign from the reading of ∂_2, and the ∓ 2 constants (together with the extra bracket of (I.41)) from the finite part of (I.40). Note that $-2a^D\Box + 2a^{D-2}\nabla^2 = 2\,\partial_0(a^{D-2}\partial_0\,\cdot\,)$, which is the form found for the extra finite part above.

Physical significance. The terms produced by the (I.40) slip are harmless. At $D=4$, $k\to H^2/8\pi^2$, the extra part of the printed (I.44) is

$$i\kappa^2k\big(-2a^D\Box + 2a^{D-2}\nabla^2\big)\delta^D(x-x') = i\Big(-\frac{1}{4\pi^2}\Big)\kappa^2H^2a^D\Box\,\delta^D(x-x') + i\Big(+\frac{1}{4\pi^2}\Big)\kappa^2H^2a^{D-2}\nabla^2\delta^D(x-x'). \tag{70}$$

This has exactly the form of the counterterm vertices (I.131) and (I.145) [Eqs. (173) and (188)]. It is therefore equivalent to the finite counterterm shifts $\delta\alpha_2 = -4$ and $\delta\alpha_3 = +4$ (paper convention), in the units $H^{D-4}/(4\pi)^{D/2}$ of Sec. I E, whose value at $D=4$ is $1/16\pi^2$; the coefficients in Eq. (70) are $\mp\frac{1}{4\pi^2} = \mp\frac{4}{16\pi^2}$. In the printed result these are contributions of -4 to β_2 and $+4$ to β_3. Removing them shifts β_2 by $+4$ and β_3 by -4. Together with a finite arithmetic slip in the 3-point local term (I.59) (Sec. IV), the corrected values are $\beta_2 = -\frac{49}{3}$ instead of $-\frac{61}{3}$ and

TABLE 7. Summary of the comparison for (I.31)–(I.44). "Definition sign" refers to (I.168); "paper convention" is the sign of the printed equations. S1 denotes the overall sign of the printed self-mass (substantive, because the sign of the nonlocal part is physical); Cn and Bn refer to our Tables 40 and 39.

Item	This work	Paper I prints	Classification
(I.31): structure, normalization	normalization 1	same	agrees
Table II, U_2	$+\frac{i}{4}\kappa^2 a^{D-2}\eta^{\alpha\rho}\eta^{\beta\sigma}\partial_1\cdot\partial_2$	$-\frac{i}{4}$	typo C5 ($+\frac{i}{4}$ used in (I.33))
Reading of ∂_2	$+\partial$ on δ^D, i.e. $-\partial'$	$+\partial'$	substantive (S1): 4-point origin of the overall sign
(I.32), (I.33) (definition sign)	$(+\frac{i}{8}, -\frac{i}{4}, -\frac{i}{2}, +i)$	$(-\frac{i}{8}, +\frac{i}{4}, +\frac{i}{2}, -i)$	S1: inherited overall sign, -1
(I.34)–(I.38)	exact in D	same	agree
(I.39)	$\bar\eta^{\rho\sigma}$ in finite part	$\bar\eta_{\rho\sigma}$	typo C6 (index placement)
(I.40), finite part	$-2\,\delta^0_\alpha\delta^\sigma_0\,h$	$+2\,\delta^0_\alpha\delta^\sigma_0\,h$	absorbable (B1)
(I.41), overall sign	$-i\kappa^2 k\,\hat\pi\{\dots\}$, $-\frac{i\kappa^2H^2}{4\pi^2}\{\dots\}$	$+$	S1: inherited overall sign
(I.41), bracket $\{-\partial^\mu(a^2\partial_\mu\delta^4)+a^2\nabla^2\delta^4\}$	absent ($O(D-4)$)	present	absorbable (B1, from (I.40))
(I.42), (I.43)	(I.43) also in general D	same	agree
(I.44): $\pi\cot$, $3\ln(aa')$, $-4\ln(aa')$, $9H^2a^D$	same structures	same	agree (overall sign: S1)
(I.44): constants -2 ($a^D\Box$), $+2$ ($a^{D-2}\nabla^2$)	absent	present	absorbable (B1, from (I.40))

$\beta_3 = -\frac{4}{D-4} + \frac{52}{3} + 2\gamma_{\rm E}$ instead of $-\frac{4}{D-4} + \frac{58}{3} + 2\gamma_{\rm E}$ (paper convention); the pole is unchanged. The present section supplies only the first of the two corrections; Sec. VI collects both. Since the finite parts of the counterterms are arbitrary, these shifts are absorbed into $\Delta\alpha_2$ and $\Delta\alpha_3$ and have no physical effect.

The overall sign is a different matter. The terms $3\ln(aa')a^D\Box$, $-4\ln(aa')a^{D-2}\nabla^2$ and $9H^2a^D$ in Eq. (66) are not of counterterm form: the first two carry a time-dependent $\ln a$, and the last is of mass type, while no mass counterterm is used. They cancel against the corresponding terms of the local 3-point contribution (I.67) (Sec. IV). Because the printed 3-point part is reversed as well, this cancellation holds in either convention, and the overall sign of the 4-point part has no consequence by itself. What remains is the overall sign of the complete self-mass, discussed at the end of Sec. III A.

Our Table 7 summarizes the comparison of this section with Paper I. All checks described in this section were run with the scripts `check_4pt.py`, `check_coincidence.py` and `check_41_44.py` in `code/02_four_point/` of Ref. [17]; their outputs agree with the logged outputs of the original working sessions.

IV. THE 3-POINT CONTRIBUTION: SETUP, LOCAL AND LOGARITHM TERMS

This section re-derives the contribution of two 3-point vertices to the one-loop scalar self-mass-squared, the diagram of Fig. 2 of Paper I, following Sec. III B of Ref. [8], Eqs. (I.45)–(I.67). We derive (i) the effective-action form (I.45)–(I.46); (ii) the nine terms produced by the graviton propagator, (I.47)–(I.49); (iii) the split into local, logarithm and normal contributions, (I.50)–(I.53), including a derivation of the distributional identity (I.52); (iv) the local contributions (I.54)–(I.59); and (v) the logarithm contributions (I.60)–(I.67), together with their cancellation against the finite local 4-point terms of (I.44). The normal contributions, (I.68) onward, are treated in Sec. V. The complete derivations and the scripts quoted below are in the public notes and code [17] (Chs. 3 and 4 of the notes; folders `code/03_three_point_setup` and `code/04_three_point_local_log`).

Sign conventions. Every equation derived in this section is in the *definition sign*, fixed by Paper I's own definition (I.168) (Sec. II F). The printed (I.45)–(I.46) agree with it. From (I.47) on, the printed 3-point expressions are in the *paper convention*, which is -1 times the definition sign: the printed (I.47)–(I.49) are -1 times what follows from the printed (I.46) (Sec. IV B), and the paper evaluates all later 3-point expressions from left-hand sides that carry these printed signs (Secs. IV D and IV E). The printed 4-point part is reversed as well (Sec. III A). Hence the printed self-mass is -1 times the one defined by (I.168); the 4-point and 3-point parts are both reversed, so their relative sign, and the local cancellation stated after (I.67), are consistent. Equations (I.50)–(I.53), (I.55)–(I.56) and (I.60)–(I.62) carry no self-mass coefficient and do not depend on the convention.

Notation. The dimension D is kept general throughout; $D=4$ is set only in finite terms that are not multiplied by a pole. We use $a=-1/(H\eta)$, so that $\partial_0 a = Ha^2$, and the abbreviations

$$k \equiv \frac{H^{D-2}}{(4\pi)^{D/2}}\,\frac{\Gamma(D-1)}{\Gamma(\frac{D}{2})} \;\xrightarrow{D\to 4}\; \frac{H^2}{8\pi^2}, \qquad \hat{\pi} \equiv -\pi\cot\Big(\frac{\pi D}{2}\Big) = -\frac{2}{D-4} + O(D-4), \tag{71}$$

together with $p \equiv D-2$, $l \equiv \ln a(\eta)$, $l' \equiv \ln a(\eta')$ and $L \equiv \ln(aa') = l + l'$. The scalar propagator is $i\Delta_A(x;x') = A(y) + k\ln(aa')$ [cf. (I.13)], with $y = aa'H^2\{\|\vec{x}-\vec{x}'\|^2 - (|\eta-\eta'|-i\delta)^2\}$ [cf. (I.11)], and the covariant d'Alembertian is $a^D\Box \equiv \partial^\mu(a^{D-2}\partial_\mu)$ [cf. (I.42)]. The coincidence limits derived for the 4-point contribution [cf. (I.34)–(I.36)] are

$$A(x;x) = k\,\hat{\pi}, \qquad i\Delta_B(x;x) = -\frac{k}{D-2}, \qquad i\Delta_C(x;x) = \frac{k}{(D-2)(D-3)}, \tag{72}$$

where $A(y)\,\delta^D(x-x') = k\hat{\pi}\,\delta^D(x-x')$ is the dimensional-regularization assumption shared with Paper I (for $\mathrm{Re}\,D<2$ all powers of y vanish at coincidence; the result is continued in D).

A. The effective action and Eqs. (I.45)–(I.46)

Vertices and effective action. The cubic interaction is [cf. Table I and (I.7)]

$$\kappa\mathcal{L}^{(1)} = \frac{1}{2}\cdot\frac{1}{i}\sum_{I=1}^{2}\Big[V_I^{\alpha\beta}\phi_1\phi_2\Big]_{\phi_1=\phi_2=\phi}h_{\alpha\beta}, \qquad V_1^{\alpha\beta} = i\kappa a^{D-2}\partial_1^\alpha\partial_2^\beta, \qquad V_2^{\alpha\beta} = -\frac{i}{2}\kappa a^{D-2}\eta^{\alpha\beta}\partial_1\cdot\partial_2. \tag{73}$$

Table I of Paper I prints these vertex operators correctly.[2] We write $\phi = \Phi + \varphi$ with background Φ and quantum field φ. Because V_I is symmetric under $1\leftrightarrow 2$ once contracted into $h_{\alpha\beta}$, the cross term is

$$S_3^{\times} = \int d^Dz\,\frac{1}{i}\sum_I\big[V_I^{\alpha\beta}(\partial_1,\partial_2)\,\Phi_1\varphi_2\big]h_{\alpha\beta}, \tag{74}$$

with leg 1 the outer leg (Φ) and leg 2 the inner leg (φ). At second order, $i\Gamma \supset \frac{1}{2}\langle(iS_{\rm int})^2\rangle_{\rm c}$. The only 1PI term that is quadratic in Φ, of order κ^2 and built from two cubic vertices is $\langle(iS_3^\times)^2\rangle$ with $\varphi\varphi$ and hh contracted; the diagrams in which the graviton line ends on a scalar loop, or on the $\langle h\rangle$ tadpole, are one-particle reducible. With $\langle T\varphi(z)\varphi(z')\rangle = i\Delta_A(z;z')$ and $\langle Th_{\alpha\beta}(z)h_{\rho\sigma}(z')\rangle = i[_{\alpha\beta}\Delta_{\rho\sigma}](z;z')$,

$$i\Gamma_{\rm 3pt}[\Phi] = \frac{1}{2}\sum_{I,J}\iint d^Dz\,d^Dz'\;V_I^{\alpha\beta}(z)\,V_J^{\rho\sigma}(z')\;\Phi(z)\,\Phi(z')\;i[_{\alpha\beta}\Delta_{\rho\sigma}](z;z')\;i\Delta_A(z;z'), \tag{75}$$

where ∂_1 (∂_1') acts on $\Phi(z)$ ($\Phi(z')$) and ∂_2 (∂_2') on the first (second) argument of $i\Delta_A$.

Normalization and derivative rules. The second functional derivative has two assignments, $(z,z')\to(x,x')$ and $(z,z')\to(x',x)$, which contribute equally because both propagators are symmetric under exchange of their arguments. The overall factor is

$$\underbrace{\tfrac{1}{2}}_{\text{cumulant}} \times \underbrace{2\times 2}_{\text{cross terms}} \times \underbrace{(2i)^{-2}}_{\text{normalization}} \times \underbrace{i^2}_{e^{iS}} \times \underbrace{2}_{\text{assignments}} = 1. \tag{76}$$

For an outer-leg derivative, $\delta/\delta\Phi(x)$ produces $\partial_\mu^z\delta^D(z-x)$, and a partial integration in z turns it into $-\partial_\mu$ acting on *everything* at the vertex x: $a^{D-2}(x)$, the graviton propagator and the inner derivative of $i\Delta_A$. The same holds at x'. Inner-leg derivatives act only on $i\Delta_A$, with a + sign. The two outer signs give $(-1)(-1)=+1$, and therefore

$$-iM^2_{\rm 3pt}(x;x') = \sum_{I=1}^{2}V_I^{\alpha\beta}(x)\sum_{J=1}^{2}V_J^{\rho\sigma}(x')\;i\big[_{\alpha\beta}\Delta_{\rho\sigma}\big](x;x')\;i\Delta_A(x;x'), \tag{77}$$

with the outer derivatives partially integrated. This is (I.45), together with the rule stated in Paper I ("partially integrate any derivative that acts upon an outer leg").

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[2] The left-hand side of (I.7) is printed with $-\frac{1}{2}\kappa a^{D-2}\partial_\alpha\phi\,\partial_\beta\phi\,h^{\alpha\beta}$; the correct sign is $+\frac{1}{2}\kappa$. The printed V_1 and the printed (I.46) are built with the correct sign. Had the printed sign been used, the second and third terms of (I.46) would be reversed.

Relation to the 4-point rule. In the 4-point contribution both outer legs sit at the same vertex. After the first partial integration no integration variable is left, and the second outer derivative acts on $\delta^D(x-x')$ with respect to x, i.e. as $-\partial'$. The rule (I.32) reads it as $+\partial'$; this is the origin of the reversed sign of the printed 4-point part (Sec. III A). The 3-point rule, with $-\partial$ on each outer leg at its own vertex, is correct. The printed 3-point part is nonetheless reversed as well, by a separate overall sign slip in the step from (I.46) to (I.47)–(I.49) (Sec. IV B).

Eq. (I.46). Expanding the four (I,J) terms of (77) with (73): the vertex coefficients are $(i\kappa)^2=-\kappa^2$, $(-\frac{i}{2}\kappa)(i\kappa)=+\frac{\kappa^2}{2}$ (twice) and $(-\frac{i}{2}\kappa)^2=-\frac{\kappa^2}{4}$; the outer derivatives give $(-\partial)(-\partial')$ in each case; and $\eta^{\alpha\beta}$ in V_2 produces a trace of the graviton propagator. Thus

$$\begin{aligned}-iM^2_{3\mathrm{pt}}(x;x') = &-\kappa^2\partial_\alpha\partial'_\rho\Big\{(aa')^{D-2}\,i[^{\alpha\beta}\Delta^{\rho\sigma]}\,\partial_\beta\partial'_\sigma i\Delta_A\Big\}+\frac{\kappa^2}{2}\partial^\mu\partial'_\rho\Big\{(aa')^{D-2}\,i[^{\alpha}{}_{\alpha}\Delta^{\rho\sigma]}\,\partial_\mu\partial'_\sigma i\Delta_A\Big\}\\ &+\frac{\kappa^2}{2}\partial_\alpha\partial'^\nu\Big\{(aa')^{D-2}\,i[^{\alpha\beta}\Delta^{\rho}{}_{\rho]}\,\partial_\beta\partial'_\nu i\Delta_A\Big\}-\frac{\kappa^2}{4}\partial^\mu\partial'^\nu\Big\{(aa')^{D-2}\,i[^{\alpha}{}_{\alpha}\Delta^{\rho}{}_{\rho]}\,\partial_\mu\partial'_\nu i\Delta_A\Big\},\end{aligned}\tag{78}$$

where all propagators have arguments $(x;x')$. The coefficients do not depend on D. Equation (78) agrees with the printed (I.46) in every coefficient, sign and index contraction, and has the same structure and signs as the flat-space 3-point expression (I.175) of Paper I.

Checks. The script `code/03_three_point_setup/check_45_46.py` [17] verifies: (a) for $D=3$ and a generic tensor $X_{\alpha\beta\rho\sigma}(x;x')$ symmetric within each pair (36 independent components), $\sum_{IJ}V_IV_JX$ reproduces (78), while $V_1\to-V_1$ (the printed left-hand side of (I.7)) reverses exactly the second and third terms; (b) the factor (76); (c) in a one-dimensional model with smooth symmetric kernels and test functions, the Gâteaux second derivative (0.779492709611) is reproduced by the rule $(-\partial)(-\partial')$ on everything, while a rule with one outer leg reversed gives the opposite value. A textbook comparison (the cubic theory $\frac{g}{2}\phi^2\chi$, whose one-loop self-energy has vertex ig and symmetry factor 1) confirms the same normalization.

B. Inserting the graviton propagator: Eqs. (I.47)–(I.49)

Inputs. The graviton propagator [cf. (I.20)–(I.24)] is $i[_{\alpha\beta}\Delta_{\rho\sigma]}=\sum_{I=A,B,C}T^I_{\alpha\beta\rho\sigma}\,i\Delta_I(x;x')$, with the constant tensors $T^{A,B,C}$ of Eqs. (24)–(25) and their single and double traces (27). Because $i\Delta_A$ depends on $\vec{x},\vec{x}'$ only through $\|\vec{x}-\vec{x}'\|$,

$$\partial'_i\,i\Delta_A=-\partial_i\,i\Delta_A,\qquad \partial_i\partial'_j\,i\Delta_A=\partial_j\partial'_i\,i\Delta_A.\tag{79}$$

Contractions. We represent the outer derivatives by formal vectors P (at x) and P' (at x'), the inner ones by Q and Q', and write $m\equiv P_0Q_0$, $\bar m\equiv\bar P\cdot\bar Q\equiv\sum_i P_iQ_i$, $n\equiv P'_0Q'_0$, $\bar n\equiv\bar P'\cdot\bar Q'$, so that $P\cdot Q=-m+\bar m$ and $P'\cdot Q'=-n+\bar n$. For the result we use the shorthand

$$\big[\!\big[\,\partial_a\partial'_c\,\big|\,\partial_b\partial'_d\,\big]\!\big]_I\equiv\partial_a\partial'_c\Big\{(aa')^{D-2}\,i\Delta_I(x;x')\,\partial_b\partial'_d\,i\Delta_A(x;x')\Big\},\tag{80}$$

with outer derivatives to the left of the bar and inner ones to the right. The first term of (78) contracts the full tensors; with $(u\cdot P)(u\cdot Q')=P_0Q'_0$ for $u_\mu=\delta^0_\mu$,

$$\begin{aligned}T^A_{\alpha\beta\rho\sigma}P^\alpha Q^\beta P'^\rho Q'^\sigma&=(\bar P\cdot\bar P')(\bar Q\cdot\bar Q')+(\bar P\cdot\bar Q')(\bar P'\cdot\bar Q)-\tfrac{2}{D-3}\,\bar m\,\bar n,\\ T^B_{\alpha\beta\rho\sigma}P^\alpha Q^\beta P'^\rho Q'^\sigma&=-\big[P_0Q'_0(\bar P'\cdot\bar Q)+Q_0Q'_0(\bar P\cdot\bar P')+P_0P'_0(\bar Q\cdot\bar Q')+Q_0P'_0(\bar P\cdot\bar Q')\big],\\ T^C_{\alpha\beta\rho\sigma}P^\alpha Q^\beta P'^\rho Q'^\sigma&=\tfrac{2}{(D-2)(D-3)}\,\big[(D-3)m+\bar m\big]\big[(D-3)n+\bar n\big].\end{aligned}\tag{81}$$

The second and third terms of (78) involve one trace and the fourth term the double trace; the T^B sector receives contributions from the first term only.

AA terms. Our Table 8 collects the T^A parts of the four terms. In the $\bar m\bar n$ row the factor $D-3$ cancels exactly, $(2-2-2+D-1)/(D-3)=1$. The total is

$$\kappa^2\Big\{-(\bar P\cdot\bar P')(\bar Q\cdot\bar Q')+\tfrac{D-1}{D-3}\,mn-\bar m\,n-m\,\bar n\Big\}+\kappa^2\Big\{\bar m\,\bar n-(\bar P\cdot\bar Q')(\bar P'\cdot\bar Q)\Big\}.\tag{82}$$

The second brace is $\big[\!\big[\,\partial_i\partial'_j\,\big|\,\partial_i\partial'_j-\partial_j\partial'_i\,\big]\!\big]_A$, which vanishes identically by (79); Paper I does not write it either.

TABLE 8. Coefficients of the AA monomials (definition sign), from the T^A parts of the four terms of Eq. (78).

Monomial	1st term	2nd term	3rd term	4th term	Total/κ^2
$\bar{m}\,\bar{n}$	$+\frac{2}{D-3}$	$-\frac{2}{D-3}$	$-\frac{2}{D-3}$	$+\frac{D-1}{D-3}$	$+1$
$m\,\bar{n}$	—	$+\frac{2}{D-3}$	—	$-\frac{D-1}{D-3}$	-1
$\bar{m}\,n$	—	—	$+\frac{2}{D-3}$	$-\frac{D-1}{D-3}$	-1
$m\,n$	—	—	—	$+\frac{D-1}{D-3}$	$+\frac{D-1}{D-3}$
$(\bar{P}\cdot\bar{P}')(\bar{Q}\cdot\bar{Q}')$	-1	—	—	—	-1
$(\bar{P}\cdot\bar{Q}')(\bar{P}'\cdot\bar{Q})$	-1	—	—	—	-1

AB and AC terms. Since the traces of T^B vanish, the AB sector is $-\kappa^2$ times the T^B line of (81). In the AC sector, with $M_1 = (D-3)m + \bar{m}$, $M_2 = P\cdot Q = -m + \bar{m}$, the analogous $N_{1,2}$, and $c_C \equiv 2\kappa^2/[(D-2)(D-3)]$, the four terms add up to

$$c_C\big[-M_1N_1 + M_2N_1 + M_1N_2 - M_2N_2\big] = -c_C\,(M_1 - M_2)(N_1 - N_2) = -c_C\,(D-2)^2\,mn. \tag{83}$$

Result. Mapping the monomials back to derivatives, the nine terms are, in the definition sign,

$$AA: \quad -\kappa^2\,[\![\,\nabla\cdot\nabla' \,|\, \nabla\cdot\nabla'\,]\!]_A + \kappa^2\Big(\frac{D-1}{D-3}\Big)[\![\,\partial_0\partial_0' \,|\, \partial_0\partial_0'\,]\!]_A - \kappa^2\,[\![\,\partial_i\partial_0' \,|\, \partial_i\partial_0'\,]\!]_A - \kappa^2\,[\![\,\partial_0\partial_i' \,|\, \partial_0\partial_i'\,]\!]_A, \tag{84a}$$

$$AB: \quad +\kappa^2\,[\![\,\partial_0\partial_0' \,|\, \nabla\cdot\nabla'\,]\!]_B + \kappa^2\,[\![\,\partial_i\partial_0' \,|\, \partial_0\partial_i'\,]\!]_B + \kappa^2\,[\![\,\partial_0\partial_i' \,|\, \partial_i\partial_0'\,]\!]_B + \kappa^2\,[\![\,\nabla\cdot\nabla' \,|\, \partial_0\partial_0'\,]\!]_B, \tag{84b}$$

$$AC: \quad -2\kappa^2\Big(\frac{D-2}{D-3}\Big)[\![\,\partial_0\partial_0' \,|\, \partial_0\partial_0'\,]\!]_C. \tag{84c}$$

These are (I.47), (I.48) and (I.49), respectively. Structures, magnitudes and the D-dependent coefficients $\frac{D-1}{D-3}$ and $\frac{2(D-2)}{D-3}$ agree with Paper I, but each of the nine printed terms has the opposite sign (our Table 9). A one-line hand check: the structure $[\![\,\nabla\cdot\nabla' \,|\, \nabla\cdot\nabla'\,]\!]_A$ can only come from the first term of (78) (coefficient $-\kappa^2$) and the $\bar{\eta}\bar{\eta}$ part of T^A (coefficient $+1$), so its coefficient is unambiguously $-\kappa^2$; Paper I prints $+\kappa^2$. The difference cannot come from the tensors (24)–(25), which reproduce the printed (I.37)–(I.38) exactly, nor from the vertices, since (78) agrees with the printed (I.46). It is an overall sign slip in the printed step from (I.46) to (I.47)–(I.49).

TABLE 9. The nine terms of (I.47)–(I.49): derived coefficients (definition sign) and printed coefficients (paper convention).

Term	Derived, Eq. (84)	Printed in Paper I
AA: $[\![\,\nabla\cdot\nabla' \,\vert\, \nabla\cdot\nabla'\,]\!]_A$	$-\kappa^2$	$+\kappa^2$
AA: $[\![\,\partial_0\partial_0' \,\vert\, \partial_0\partial_0'\,]\!]_A$	$+\kappa^2\frac{D-1}{D-3}$	$-\kappa^2\frac{D-1}{D-3}$
AA: $[\![\,\partial_i\partial_0' \,\vert\, \partial_i\partial_0'\,]\!]_A$, $[\![\,\partial_0\partial_i' \,\vert\, \partial_0\partial_i'\,]\!]_A$	$-\kappa^2$, $-\kappa^2$	$+\kappa^2$, $+\kappa^2$
AB: four terms	all $+\kappa^2$	all $-\kappa^2$
AC: $[\![\,\partial_0\partial_0' \,\vert\, \partial_0\partial_0'\,]\!]_C$	$-2\kappa^2\frac{D-2}{D-3}$	$+2\kappa^2\frac{D-2}{D-3}$
antisymmetric AA pair	0 by Eq. (79)	not written

Checks. The script `code/03_three_point_setup/check_47_49.py` [17] builds $T^{A,B,C}$ as explicit component arrays for $D = 4, 5, 6$ and tracks the outer propagator type symbolically. For each of these D it finds: the route through (78) equals the route directly from (77) and Table I; both equal (84) plus the antisymmetric pair, which vanishes with $\partial_i' = -\partial_i$ on $i\Delta_A$; and they equal -1 times the printed (I.47)–(I.49). The homogeneity relations (79) are verified for $A(y) + k\ln(aa')$.

C. The three-way split: Eqs. (I.50)–(I.53)

Each of the nine terms has one outer derivative at x, one at x', and an inner pair of the form $\partial_\rho\partial_\sigma'$, i.e. the form (I.50), $\kappa^2\partial_\mu\partial_\nu'[(aa')^{D-2} i\Delta_I\,\partial_\rho\partial_\sigma' i\Delta_A]$ with $I = A, B, C$. By (I.13), (I.27) and (I.28), $i\Delta_A = A(y) + k\ln(aa')$,

$i\Delta_B = B(y)$ and $i\Delta_C = C(y)$, which is (I.51). Since each inner pair is mixed,

$$\partial_\rho \partial'_\sigma \, k \ln(aa') = k \, \partial_\rho \partial'_\sigma \big[\ln a(\eta) + \ln a(\eta') \big] = 0, \tag{85}$$

so the inner derivatives eliminate the de Sitter breaking term, as stated in Paper I. The only nontrivial equation of this group is (I.52), derived next.

Derivation of (I.52): direct computation. For $\delta > 0$, $y = aa' H^2 \{ r^2 - \Delta\eta^2 + 2i\delta|\Delta\eta| + \delta^2 \}$ with $r \equiv \|\vec{x} - \vec{x}'\|$ and $\Delta\eta \equiv \eta - \eta'$, so $\mathrm{Im}\, y = 2aa' H^2 \delta |\Delta\eta| \geq 0$, and $y > 0$ when $\Delta\eta = 0$. Thus y never reaches zero or the negative real axis, the powers $y^{1-D/2}, y^{2-D/2}, \ldots$ in $A(y)$ are analytic, and the only non-smooth structure is the kink of $|\Delta\eta|$. With $\partial_0 |\Delta\eta| = \mathrm{sgn}(\Delta\eta)$ and $\partial'_0 \, \mathrm{sgn}(\Delta\eta) = -2\delta(\Delta\eta)$,

$$\partial_0 \partial'_0 \big(|\Delta\eta| - i\delta \big)^2 = -2 + 4i\delta \, \delta(\Delta\eta), \tag{86}$$

while $\partial'_j y = -2aa' H^2 (x - x')_j$ contains no $|\Delta\eta|$. The chain rule then gives

$$\partial_\rho \partial'_\sigma A(y) = A''(y) \, \partial_\rho y \, \partial'_\sigma y + A'(y) \big[\partial_\rho \partial'_\sigma y \big]_{\mathrm{reg}} - 4i\delta \, \delta^0_\rho \delta^0_\sigma \, a^2 H^2 A' \big(a^2 H^2 (r^2 + \delta^2) \big) \, \delta(\Delta\eta), \tag{87}$$

where $\delta(\Delta\eta)$ has been used to set $a' = a$ in the last term. In $D - 1$ dimensions

$$\lim_{\delta \to 0^+} \delta \, (r^2 + \delta^2)^{-D/2} = \frac{\pi^{D/2}}{\Gamma(\frac{D}{2})} \, \delta^{D-1}(\vec{x} - \vec{x}'), \tag{88}$$

since the total weight is $\Omega_{D-2} \int_0^\infty ds \, s^{D-2} (1 + s^2)^{-D/2} = \pi^{D/2} / \Gamma(\frac{D}{2})$. Writing the leading term of $A(y)$ as $\mathcal{N} c_0 (4/y)^{D/2-1}$ with $\mathcal{N} \equiv H^{D-2}/(4\pi)^{D/2}$, and using $\mathcal{N} \, 4^{D/2} H^{2-D} \pi^{D/2} = 1$, the last term of (87) tends to

$$i \, \frac{c_0 \, (\frac{D}{2} - 1)}{\Gamma(\frac{D}{2})} \, a^{2-D} \, \delta^D(x - x'). \tag{89}$$

The subleading terms of A' are at most as singular as $y^{1-D/2}$; their weight is $O(\delta)$ and they drop out. With $c_0 = \Gamma(\frac{D}{2} - 1)$ (see below), $c_0(\frac{D}{2} - 1) = \Gamma(\frac{D}{2})$, and

$$\partial_\rho \partial'_\sigma \, i\Delta_A(x; x') = \delta^0_\rho \delta^0_\sigma \, \frac{i}{a^{D-2}} \, \delta^D(x - x') + A''(y) \, \frac{\partial y}{\partial x^\rho} \, \frac{\partial y}{\partial x'^\sigma} + A'(y) \, \frac{\partial^2 y}{\partial x^\rho \partial x'^\sigma}, \tag{90}$$

where the last derivative is the pointwise (regular) one. This is the printed (I.52).

Independent derivations of the local coefficient. (B) *Propagator equation.* For a function of $|\Delta\eta|$ the singular parts obey $[\partial_0 \partial'_0 G]_{\mathrm{sing}} = -[\partial_0^2 G]_{\mathrm{sing}}$. In (I.12), $\partial_\mu (a^{D-2} \eta^{\mu\nu} \partial_\nu) i\Delta_A = i\delta^D$, the singular term comes only from $-a^{D-2}\partial_0^2$, so $[\partial_0 \partial'_0 i\Delta_A]_{\mathrm{sing}} = +i \, a^{2-D} \delta^D$. This route does not use the normalization of $A(y)$. (C) *Canonical commutator.* From $T\phi\phi' = \theta(\Delta\eta)\phi\phi' + \theta(-\Delta\eta)\phi'\phi$ and $\pi = a^{D-2}\dot{\phi}$, $[\phi, \pi'] = i\delta^{D-1}$,

$$\partial_0 \partial'_0 \langle T\phi(x)\phi(x') \rangle = \langle T\dot{\phi}(x)\dot{\phi}(x') \rangle + \frac{i}{a^{D-2}} \, \delta^D(x - x'). \tag{91}$$

All three routes give the coefficient $\delta^0_\rho \delta^0_\sigma \, i \, a^{2-D} \delta^D(x - x')$.

First coefficient of (I.14). Route A gives $i \, c_0(\frac{D}{2} - 1) a^{2-D} / \Gamma(\frac{D}{2})$; routes B and C then force $c_0 = \Gamma(\frac{D}{2}) / (\frac{D}{2} - 1) = \Gamma(\frac{D}{2} - 1)$, in agreement with the most singular part (I.26) and with the recursion obeyed by the series (Sec. II E). The printed $c_0 = \Gamma(\frac{D}{2} - 1)/(\frac{D}{2} - 1)$ in (I.14) is therefore a typographical error (C4): the printed (I.52) carries the correct local coefficient, whereas the printed c_0 would multiply it by $\frac{2}{D-2}$, which is 1 at $D = 4$ but would produce a finite error when multiplied by the pole of $i\Delta_A(x; x)$.

The split. Inserting (90) into (I.50) gives three parts:

$$\kappa^2 \partial_\mu \partial'_\nu \Big[(aa')^{D-2} \, i\Delta_I \Big(\underbrace{\delta^0_\rho \delta^0_\sigma \, \tfrac{i}{a^{D-2}} \, \delta^D}_{\text{(i) local}} + \mathcal{R}_{\rho\sigma} \Big) \Big], \qquad i\Delta_I \, \mathcal{R}_{\rho\sigma} = \begin{cases} \underbrace{A(y) \, \mathcal{R}_{\rho\sigma}}_{\text{(iii) normal}} + \underbrace{k \ln(aa') \, \mathcal{R}_{\rho\sigma}}_{\text{(ii) logarithm}}, & I = A, \\ \underbrace{I(y) \, \mathcal{R}_{\rho\sigma}}_{\text{(iii) normal}}, & I = B, C, \end{cases} \tag{92}$$

with $\mathcal{R}_{\rho\sigma} \equiv A'' \partial_\rho y \, \partial'_\sigma y + A' \partial_\rho \partial'_\sigma y$; for $I = B, C$, $I(y)$ stands for $B(y)$ or $C(y)$ (the propagator label I is unrelated to the integration operator $I[f]$ of Sec. V). Part (iii) is (I.53). Following Paper I, the local part for $I = A$ uses only $A(y)$, and the piece $k \ln(aa') \times \delta^D$ is counted with the logarithm contributions (which then use the full $\partial\partial' i\Delta_A$). With this assignment the split is complete and free of overlap. Equations (I.50)–(I.53) are statements of form without overall coefficients; they do not decide the sign question of Sec. IV B.

Checks. The script `code/03_three_point_setup/check_52.py` [17] verifies $\partial_0\partial_0' \ln(aa') = 0$; the kink identity (86); the vanishing of the $\eta^{\rho\sigma}$-contracted regular part in flat space for general D; the weight in (88) in closed form for $D = 3, \dots, 6$ and for general D, and numerically at $D = 4$ (9.650, 9.847, 9.867 for $\delta = 10^{-2}, 10^{-3}, 10^{-4}$, against $\pi^2 = 9.8696$); the local coefficient $i\,a^{2-D}$ with $c_0 = \Gamma(\frac{D}{2} - 1)$ (and $\frac{2i}{D-2} a^{2-D}$ with the printed c_0); the recursion for c_0 of Sec. II E; and $\Box f(y) = H^2[(4y - y^2)f'' + D(2 - y)f']$ for $D = 4, 5$.

D. Local contributions: Eqs. (I.54)–(I.59)

Only the three terms of (84) with the inner pair $\partial_0\partial_0'$ give local contributions: the second term of (84a), the fourth term of (84b), and (84c). In the definition sign,

$$\begin{aligned} \mathcal{J}_{AA} &= +\kappa^2\,\frac{D-1}{D-3}\,\partial_0\partial_0'\Big[(aa')^{D-2}\,i\Delta_A\,\partial_0\partial_0'\,i\Delta_A\Big], \\ \mathcal{J}_{AB} &= +\kappa^2\,\nabla\cdot\nabla'\Big[(aa')^{D-2}\,i\Delta_B\,\partial_0\partial_0'\,i\Delta_A\Big], \\ \mathcal{J}_{AC} &= -2\kappa^2\,\frac{D-2}{D-3}\,\partial_0\partial_0'\Big[(aa')^{D-2}\,i\Delta_C\,\partial_0\partial_0'\,i\Delta_A\Big]. \end{aligned} \tag{93}$$

The text of Paper I before (I.54) attributes the AA local term to "the 4th term of (47)"; as the left-hand side of (I.54) shows, the second term is meant (typographical, C7).

Rules. (R1) With the local part of (90) and the coincidence limits (72),

$$\begin{gathered} (aa')^{D-2}\,i\Delta_I(x;x')\,\frac{i}{a^{D-2}}\,\delta^D(x-x') = i\,\Lambda_I\,a^{D-2}\,\delta^D(x-x'), \\ \Lambda_A = k\hat{\pi}, \qquad \Lambda_B = -\frac{k}{D-2}, \qquad \Lambda_C = \frac{k}{(D-2)(D-3)}, \end{gathered} \tag{94}$$

where for $I = A$ only $A(y)$ is kept, and derivatives are taken after the coincidence limit. (R2) Acting on a test function, the primed derivative acts only on the delta function, and with $a^D\Box f = -\partial_0(a^{D-2}\partial_0 f) + a^{D-2}\nabla^2 f$,

$$\begin{aligned} \partial_0\partial_0'\big[a^{D-2}\delta^D\big] &= -\partial_0\big(a^{D-2}\partial_0\delta^D\big) = \big(a^D\Box - a^{D-2}\nabla^2\big)\delta^D, \\ \nabla\cdot\nabla'\big[a^{D-2}\delta^D\big] &= -a^{D-2}\nabla^2\delta^D. \end{aligned} \tag{95}$$

The printed (I.55)–(I.56) is the first relation multiplied by -1 and is correct.

The three local terms. Using $(D-1)\Gamma(D-1) = \Gamma(D)$,

$$\mathcal{J}^{\rm loc}_{AA} = \frac{i\kappa^2 H^{D-2}}{(4\pi)^{D/2}}\,\frac{\Gamma(D)}{(D-3)\Gamma(\frac{D}{2})}\Big(-\pi\cot\big(\tfrac{\pi D}{2}\big)\Big)\big\{+a^D\Box - a^{D-2}\nabla^2\big\}\delta^D(x-x'), \tag{96}$$

$$\mathcal{J}^{\rm loc}_{AB} = +\frac{i\kappa^2 k}{D-2}\,a^{D-2}\nabla^2\delta^D(x-x') = +\frac{i\kappa^2 H^2}{16\pi^2}\,a^2\nabla^2\delta^4(x-x') + O(D-4), \tag{97}$$

$$\mathcal{J}^{\rm loc}_{AC} = -\frac{2i\kappa^2 k}{(D-3)^2}\big(a^D\Box - a^{D-2}\nabla^2\big)\delta^D(x-x') = -\frac{i\kappa^2 H^2}{4\pi^2}\big\{a^4\Box - a^2\nabla^2\big\}\delta^4(x-x') + O(D-4), \tag{98}$$

which correspond to (I.54), (I.57) and (I.58). In (97) and (98) the coefficients are finite and are not multiplied by a pole, so setting $D = 4$ there changes only terms of order $D - 4$; the pole coefficient in (96) is kept at exact D.

The sum, Eq. (I.59). At exact D, in units of $i\kappa^2 k$,

$$-iM^2_{\rm 3pt,loc} = i\kappa^2 k\,\Big\{\Big[\frac{D-1}{D-3}\,\hat{\pi} - \frac{2}{(D-3)^2}\Big]a^D\Box + \Big[-\frac{D-1}{D-3}\,\hat{\pi} + \frac{1}{D-2} + \frac{2}{(D-3)^2}\Big]a^{D-2}\nabla^2\Big\}\delta^D(x-x'). \tag{99}$$

Setting $D = 4$ in the finite parts, $\frac{2}{(D-3)^2} \to 2$ and $\frac{1}{D-2} \to \frac{1}{2}$:

$$\begin{aligned} -iM^2_{\rm 3pt,loc}(x;x') = \frac{i\kappa^2 H^{D-2}}{(4\pi)^{D/2}}\,\frac{\Gamma(D-1)}{\Gamma(\frac{D}{2})}\,\Big\{\Big[-\frac{D-1}{D-3}\,\pi\cot\big(\tfrac{\pi}{2}D\big) - 2\Big]a^D\Box \\ + \Big[\frac{D-1}{D-3}\,\pi\cot\big(\tfrac{\pi}{2}D\big) + \frac{5}{2}\Big]a^{D-2}\nabla^2 + O(D-4)\Big\}\delta^D(x-x') \end{aligned} \tag{100}$$

(definition sign). In the paper convention the two brackets read $[\frac{D-1}{D-3}\pi\cot(\frac{\pi}{2}D) + 2]$ and $[-\frac{D-1}{D-3}\pi\cot(\frac{\pi}{2}D) - \frac{5}{2}]$. Paper I prints (I.59) with $-\frac{7}{2}$ in place of $-\frac{5}{2}$; all other entries agree up to the overall sign (our Table 10).

TABLE 10. Local 3-point contributions. Derived values are in the definition sign, printed values in the paper convention. Common prefactors are those of Eqs. (96)–(100).

Item	Derived	Printed in Paper I
(I.54)	$\{+a^D\Box - a^{D-2}\nabla^2\}\delta^D$	$\{-a^D\Box + a^{D-2}\nabla^2\}\delta^D$
(I.55)–(I.56)	Eq. (95)	same
(I.57)	$+\frac{i\kappa^2H^2}{16\pi^2}\,a^2\nabla^2\delta^4$	$-\frac{i\kappa^2H^2}{16\pi^2}\,a^2\nabla^2\delta^4$
(I.58)	$-\frac{i\kappa^2H^2}{4\pi^2}\{a^4\Box - a^2\nabla^2\}\delta^4$	$+\frac{i\kappa^2H^2}{4\pi^2}\{a^4\Box - a^2\nabla^2\}\delta^4$
(I.59), $a^D\Box$	$-\frac{D-1}{D-3}\pi\cot(\frac{\pi}{2}D) - 2$	$+\frac{D-1}{D-3}\pi\cot(\frac{\pi}{2}D) + 2$
(I.59), $a^{D-2}\nabla^2$	$+\frac{D-1}{D-3}\pi\cot(\frac{\pi}{2}D) + \frac{5}{2}$	$-\frac{D-1}{D-3}\pi\cot(\frac{\pi}{2}D) - \frac{7}{2}$

Diagnosis. (i) *Overall sign.* The left-hand sides of the printed (I.54), (I.57) and (I.58) carry the signs of the printed (I.47)–(I.49), and their right-hand sides are correct evaluations of those left-hand sides. The sign slip of the step (I.46)→(I.47)–(I.49) was therefore carried into the rest of the 3-point calculation, which establishes the convention statement made at the beginning of this section. (ii) *Arithmetic of (I.59).* The paper's own (I.57) and (I.58) give $-\frac{1}{2} - 2 = -\frac{5}{2}$ for the finite part of the $a^{D-2}\nabla^2$ coefficient (paper convention). The printed (I.59) minus the sum of the paper's own terms is

$$-\frac{i\kappa^2H^2}{8\pi^2}\,a^2\nabla^2\delta^4(x-x'), \tag{101}$$

which has the form of the noninvariant counterterm vertex (I.145), $i\alpha_3\kappa^2H^2a^{D-2}\nabla^2\delta^D(x-x')$. Since $-\frac{1}{8\pi^2} = -\frac{2}{16\pi^2}$, it is equivalent to a finite counterterm shift $\delta\alpha_3 = -2$ in the units $H^{D-4}/(4\pi)^{D/2}$ of Sec. I E (paper convention). This finite error is absorbed by the arbitrary finite part of the counterterm. Its only trace in the renormalized result is a shift of the finite part of β_3 by $+2$ (paper convention); together with the correction of (I.40) in the 4-point part, this changes β_3 from $-\frac{4}{D-4} + \frac{58}{3} + 2\gamma_{\rm E}$ to $-\frac{4}{D-4} + \frac{52}{3} + 2\gamma_{\rm E}$ (paper convention, units of $H^{D-4}/(4\pi)^{D/2}$), with the pole unchanged; see Sec. VI E.

Checks. The script `code/04_three_point_local_log/check_54_59.py` [17] checks: (A) the local polynomial extracted directly from (77) and Table I, without the intermediate step (84), is $\frac{D-1}{D-3}P_0P_0'\,d_A + (\bar P\cdot\bar P')\,d_B - \frac{2(D-2)}{D-3}P_0P_0'\,d_C$ for $D = 4, 5, 6$, i.e. exactly the local part of (93); (B) the rules (95) and (I.55)–(I.56) on a test function, with symbolic D in the powers of a; (C) the sum (99) with finite parts -2 and $+\frac{5}{2}$, the paper's own terms adding up to $-\frac{5}{2}$ rather than the printed $-\frac{7}{2}$, and the printed (I.54), (I.57), (I.58) being -1 times the derived ones term by term ((I.54) at symbolic D; (I.57)–(I.58), printed only at $D = 4$, through their $D = 4$ coefficients).

E. Logarithm contributions: Eqs. (I.60)–(I.67)

The logarithm contributions come from the AA terms (84a) with the outer $i\Delta_A$ replaced by $k\ln(aa')$, and the *full* $\partial\partial' i\Delta_A$ inside (including the local part of (90)). With $\mathcal{P} \equiv (aa')^{D-2}\,k\,L$, they are $-\kappa^2\mathcal{V}_1 + \kappa^2\frac{D-1}{D-3}\mathcal{W} - \kappa^2\mathcal{V}_3 - \kappa^2\mathcal{V}_4$, where

$$\begin{aligned}\mathcal{V}_1 &\equiv \nabla\cdot\nabla'\big[\mathcal{P}\,\nabla\cdot\nabla' i\Delta_A\big], \quad \mathcal{W} \equiv \partial_0\partial_0'\big[\mathcal{P}\,\partial_0\partial_0' i\Delta_A\big],\\ \mathcal{V}_3 &\equiv \partial_i\partial_0'\big[\mathcal{P}\,\partial_i\partial_0' i\Delta_A\big], \qquad \mathcal{V}_4 \equiv \partial_0\partial_i'\big[\mathcal{P}\,\partial_0\partial_i' i\Delta_A\big].\end{aligned} \tag{102}$$

The printed (I.63)–(I.66) have the opposite signs, $+,-,+,+$, inherited from the printed (I.47).

Eqs. (I.60)–(I.62). Since $\sqrt{-g}\,g^{\mu\nu} = a^{D-2}\eta^{\mu\nu}$ and $\nabla^2[k\ln(aa')] = 0$, the propagator equation (I.12) gives

$$\partial_0\big(a^{D-2}\partial_0\, i\Delta_A\big) = -i\delta^D(x-x') + a^{D-2}\nabla^2A(y), \qquad \partial_0'\big(a'^{D-2}\partial_0'\, i\Delta_A\big) = -i\delta^D(x-x') + a'^{D-2}\nabla^2A(y), \tag{103}$$

which are (I.60) and (I.61). With $\partial_0(a^p\partial_0 l) = (D-1)H^2a^D$ and $A = i\Delta_A - kL$, applying ∇^2 gives the corollary

$$\partial_0\big(a^p\partial_0\nabla^2A\big) = -i\nabla^2\delta^D + a^p\nabla^4A, \qquad \partial_0'\big(a'^p\partial_0'\nabla^2A\big) = -i\nabla^2\delta^D + a'^p\nabla^4A. \tag{104}$$

For $D = 4+\epsilon$, the first three terms of (I.14) (with $c_0 = \Gamma(\frac{D}{2}-1)$) expand as $4/y$, $\frac{4}{\epsilon} + 3 - 2\gamma_{\rm E} + 2\ln\frac{4}{y}$ and $-\frac{4}{\epsilon} - 4 + 2\gamma_{\rm E}$, all times $\mathcal{N} \to H^2/16\pi^2$; the poles cancel and the n-th series term vanishes identically at $D = 4$. Hence

$$A(y) = \frac{H^2}{16\pi^2}\Big\{\frac{4}{y} - 2\ln\frac{y}{4} - 1 + O(D-4)\Big\}, \tag{105}$$

which is (I.62). (The printed c_0 would differ only at order $D-4$ here.) As a check, $(4y-y^2)A''+4(2-y)A'=3k$, i.e. $\Box A=(D-1)kH^2$ at $D=4$.

Distributional tools. For a function f of time only, $f(\eta')\partial_0\delta^D=f(\eta)\partial_0\delta^D+f'(\eta)\delta^D$ and $f(\eta')\partial_0^2\delta^D=f\partial_0^2\delta^D+2f'\partial_0\delta^D+f''\delta^D$. As kernels, $\partial_0'[F(\eta')\partial_0'\delta^D]\equiv\partial_0[F(\eta)\partial_0\delta^D]$ (both act on a test function $g(x')$ as $\partial_0(F\partial_0 g)$), and $f(\eta,\eta')\nabla^2\delta^D=f(\eta,\eta)\nabla^2\delta^D$; in particular $L\nabla^2\delta^D=2l\nabla^2\delta^D$. From these and (95) follows the lemma

$$-2i\,\partial_0\big[a^p l\,\partial_0\delta^D\big]=i\,L\,\big(a^D\Box-a^{D-2}\nabla^2\big)\delta^D+i(D-1)H^2a^D\delta^D, \tag{106}$$

obtained from $L\,\partial_0(a^p\partial_0\delta^D)=2\,\partial_0[a^p l\,\partial_0\delta^D]+(D-1)H^2a^D\delta^D$, which in turn follows from the first identity with $f=l(\eta')$, $f'=Ha$, $f''=H^2a^2$.

The four terms at general D. (1) By (79), $\nabla\cdot\nabla' i\Delta_A=-\nabla^2 A$, and $\mathcal{P}$ is constant in space:

$$\mathcal{V}_1=k\,(aa')^{D-2}L\,\nabla^4 A. \tag{107}$$

(2) For $\mathcal{W}$ split $L=l+l'$. In the l part, $a^p l$ depends only on η and $a'^p\partial_0\partial_0' i\Delta_A=\partial_0(a'^p\partial_0' i\Delta_A)$, so (103) gives $\partial_0[a^p l\,\partial_0(-i\delta^D+a'^p\nabla^2 A)]$; the l' part is its mirror image under $x\leftrightarrow x'$. The delta-function pieces combine through the kernel symmetry and (106); the A pieces follow from (104) with $\partial_0 l=Ha$:

$$\frac{\mathcal{W}}{k}=iL\big(a^D\Box-2a^{D-2}\nabla^2\big)\delta^D+i(D-1)H^2a^D\delta^D+(aa')^{D-2}L\,\nabla^4A+H(aa')^{D-2}\big(a\partial_0+a'\partial_0'\big)\nabla^2A. \tag{108}$$

(3)–(4) In $\mathcal{V}_3$ the outer ∂_i passes through $\mathcal{P}$ and meets the inner ∂_i, so $\mathcal{V}_3=k\,\partial_0'[(aa')^{D-2}L\,\partial_0'\nabla^2A]$; the primed corollary and $\partial_0'l'=Ha'$ give

$$\frac{\mathcal{V}_3}{k}=-ia^{D-2}L\,\nabla^2\delta^D+(aa')^{D-2}L\,\nabla^4A+H(aa')^{D-2}a'\partial_0'\nabla^2A, \tag{109}$$

and $\mathcal{V}_4$ is its mirror image, with $a'\partial_0'\to a\partial_0$ in the last term. At $D=4$ the four contributions are

$$-\kappa^2\mathcal{V}_1=-\frac{\kappa^2H^2}{8\pi^2}\,L\,(aa')^2\nabla^4A, \tag{110}$$

$$3\kappa^2\mathcal{W}=\frac{3i\kappa^2H^2}{8\pi^2}\,L\,\{a^4\Box-2a^2\nabla^2\}\delta^4+\frac{3\kappa^2H^2}{8\pi^2}\,L\,(aa')^2\nabla^4A \\ +\frac{9i\kappa^2H^4}{8\pi^2}\,a^4\delta^4+\frac{3\kappa^2H^3}{8\pi^2}\,(aa')^2\big(a\partial_0+a'\partial_0'\big)\nabla^2A, \tag{111}$$

$$-\kappa^2\mathcal{V}_3=+\frac{i\kappa^2H^2}{8\pi^2}\,L\,a^2\nabla^2\delta^4-\frac{\kappa^2H^2}{8\pi^2}\,L\,(aa')^2\nabla^4A-\frac{\kappa^2H^3}{8\pi^2}\,(aa')^2a'\partial_0'\nabla^2A, \tag{112}$$

and $-\kappa^2\mathcal{V}_4$ equals (112) with $a'\partial_0'\to a\partial_0$; these correspond to (I.63)–(I.66), each of which is printed with the opposite overall sign.

The sum. Our Table 11 collects the coefficients at general D. Altogether,

$$-iM^2_{3\mathrm{pt,log}}=\kappa^2k\Big\{\ln(aa')\Big[\frac{D-1}{D-3}\,a^D\Box-\frac{4}{D-3}\,a^{D-2}\nabla^2\Big]i\delta^D+\frac{(D-1)^2}{D-3}\,H^2a^D\,i\delta^D \\ -\frac{2(D-4)}{D-3}\,(aa')^{D-2}\ln(aa')\,\nabla^4A+\frac{2}{D-3}\,H(aa')^{D-2}\big(a\partial_0+a'\partial_0'\big)\nabla^2A\Big\}. \tag{113}$$

Each term is either δ^D-local with a finite coefficient, or a smooth function times derivatives of a *single* propagator. Since $A\sim y^{1-D/2}$ is locally integrable, its derivatives are well-defined distributions without poles near $D=4$ (unlike the products of two propagators met in the normal contributions). Thus the ∇^4A term, whose coefficient is $O(D-4)$, vanishes, and setting $D=4$ elsewhere changes only terms of order $D-4$. This confirms the statement of Paper I that all logarithm contributions are finite. The result is

$$-iM^2_{3\mathrm{pt,log}}(x;x')=\frac{\kappa^2H^2}{8\pi^2}\Big\{\ln(aa')\big[3a^4\Box-4a^2\nabla^2\big]i\delta^4(x-x')+9H^2a^4\,i\delta^4(x-x') \\ +2H(aa')^2\big(a\partial_0+a'\partial_0'\big)\nabla^2A(y)+O(D-4)\Big\} \tag{114}$$

(definition sign). Paper I prints (I.67) with the bracket $\{\ln(aa')[-3a^4\Box+4a^2\nabla^2]i\delta^4-9H^2a^4i\delta^4-2H(aa')^2(a\partial_0+a'\partial_0')\nabla^2A\}$, i.e. exactly -1 times (114); the cancellation of the ∇^4A terms and the coefficients $3,4,9,2$ agree. No new error occurs in (I.60)–(I.67): the printed (I.67) is the correct sum of the printed (I.63)–(I.66), whose left-hand sides carry the signs of the printed (I.47). The nonlocal term of (114) is processed together with the normal contributions in Sec. V B, where it yields (I.89) and the corresponding rows of Table III.

TABLE 11. Coefficients of the logarithm contributions in units of $\kappa^2 k$ (definition sign), from Eqs. (107)–(109). In the last row $-\mathcal{V}_4$ contributes to the $a\partial_0$ term and $-\mathcal{V}_3$ to its mirror $a'\partial_0'$, so each receives $\frac{D-1}{D-3}-1$.

Structure	$-\mathcal{V}_1$	$\frac{D-1}{D-3}\mathcal{W}$	$-\mathcal{V}_3$	$-\mathcal{V}_4$	Total	$D=4$
$L\,a^D\Box\, i\delta^D$	—	$\frac{D-1}{D-3}$	—	—	$\frac{D-1}{D-3}$	3
$L\,a^{D-2}\nabla^2\, i\delta^D$	—	$-\frac{2(D-1)}{D-3}$	$+1$	$+1$	$-\frac{4}{D-3}$	-4
$H^2a^D\, i\delta^D$	—	$\frac{(D-1)^2}{D-3}$	—	—	$\frac{(D-1)^2}{D-3}$	9
$(aa')^{D-2}L\,\nabla^4 A$	-1	$\frac{D-1}{D-3}$	-1	-1	$-\frac{2(D-4)}{D-3}$	0
$H(aa')^{D-2}a\partial_0\nabla^2 A$ (and mirror)	—	$\frac{D-1}{D-3}$	-1	-1	$\frac{2}{D-3}$	2

Cancellation against the local 4-point terms. The corrected 4-point result at general D [Eq. (67); definition sign] is

$$-iM^2_{4\mathrm{pt}} = -i\kappa^2 k\Big\{\tfrac{1}{4}D(D-1)\big[\hat\pi+\ln(aa')\big]a^D\Box - D\big[\hat\pi+\ln(aa')\big]a^{D-2}\nabla^2 + \tfrac{1}{4}D(D-1)^2H^2a^D\Big\}\delta^D + O(D-4). \quad (115)$$

Adding the local terms of (113) to its $\ln(aa')$ and H^2 terms, and using $\frac{1}{D-3}-\frac{D}{4} = -\frac{(D-4)(D+1)}{4(D-3)}$ and $D-\frac{4}{D-3} = \frac{(D-4)(D+1)}{D-3}$, gives

$$-i\kappa^2 k\,\frac{(D-4)(D+1)}{4(D-3)}\Big\{(D-1)\ln(aa')\,a^D\Box - 4\ln(aa')\,a^{D-2}\nabla^2 + (D-1)^2H^2a^D\Big\}\delta^D \xrightarrow{D\to4} 0. \quad (116)$$

The remainder is not multiplied by a pole, so it vanishes at $D=4$: the terms $3\ln(aa')a^D\Box$, $-4\ln(aa')a^{D-2}\nabla^2$ and $9H^2a^D$, which are not of counterterm form, cancel exactly between (I.44) and (I.67). With the opposite relative sign they would add instead. Since both printed parts are reversed, the statement of Paper I after (I.67) holds in the paper convention as well. (The divergent $\hat\pi$ terms of (I.44) do not take part in this cancellation. The constants ∓2 in the brackets of the printed (I.44), which originate from the finite-part error in (I.40), are absent from the corrected 4-point result; they are of counterterm form.)

Checks. The script `code/04_three_point_local_log/check_60_67.py` [17] checks: (a) the expansion (105) from (I.14), with the printed c_0 differing only at $O(D-4)$; (b) the differential equation satisfied by (105); (c) the lemma (106) and the time-only rule for $L\nabla^2\delta^D$ on a test function at general D; (d) a pointwise check of the nonlocal parts of $\mathcal{V}_1,\mathcal{W},\mathcal{V}_3,\mathcal{V}_4$ at $D=4$, by brute-force differentiation of $A(y)+k\ln(aa')$ with the actual $A(y)$ of (105), at three random non-coincident points with 30-digit precision (exact agreement; this check is numerical, at sample points, and at $D=4$ only); (e) the general-D coefficients of our Table 11 and the remainder (116); (f) the printed (I.63)–(I.66) are -1 times the derived terms, and their sum is the printed (I.67).

F. Status of Eqs. (I.45)–(I.67)

Our Table 12 summarizes the comparison. Apart from the overall sign, the only item in this part that changes a number entering the renormalized result is the $-\frac{7}{2}\to-\frac{5}{2}$ slip in (I.59), and it is absorbable (B2). The overall sign rests on the paper's own definition (I.168); once that definition is adopted it is physical for the nonlocal part, which has no counterterm freedom: the physical nonlocal self-mass is -1 times the (corrected) Table X. In Paper II [12] this flips every entry of Table 5 and the sign of the one-loop particular solution; since the non-falling total is zero, only decaying terms change sign, and the late-time conclusion of Paper II is unchanged (Sec. VII).

V. THE 3-POINT CONTRIBUTION: NORMAL TERMS

This section corresponds to Sec. III B.3 of Paper I ("Normal contributions"), Eqs. (I.68)–(I.125) and Tables III–V, together with its Appendix "Extracting derivatives", Eqs. (I.A1)–(I.A38) and Tables XI–XX. The setup, the local and the logarithm terms of the 3-point contribution were treated in Sec. IV.

TABLE 12. Status of the equations of this part. "S1 (inherited sign)": the printed expression is -1 times the definition sign, inherited from the step (I.46)→(I.47)–(I.49); this is the 3-point part of the overall sign S1, classified as substantive because the sign of the nonlocal part of the self-mass is physical. Cn and Bn refer to our Tables 40 and 39.

Equations	Finding	Class
(I.7), left side	$-\frac{1}{2}\kappa \to +\frac{1}{2}\kappa$; (I.46) uses the correct vertex	typo (C3)
(I.45)–(I.46)	reproduced exactly (definition sign)	confirmed
(I.47)–(I.49)	structures, magnitudes, D-dependence agree; overall sign reversed in all nine terms	S1 (origin of the 3-point sign)
(I.14), first coefficient	$\Gamma(\frac{D}{2}-1)/(\frac{D}{2}-1) \to \Gamma(\frac{D}{2}-1)$; (I.52) not affected	typo (C4)
(I.50)–(I.53)	confirmed; (I.52) derived along three independent routes	confirmed
text before (I.54)	"4th term of (47)" → "2nd term of (47)"	typo (C7)
(I.54), (I.57), (I.58)	reproduced up to the overall sign	S1 (inherited sign)
(I.55)–(I.56)	reproduced	confirmed
(I.59)	overall sign; finite constant $-\frac{7}{2} \to -\frac{5}{2}$ (paper convention), equivalent to $\delta\alpha_3 = -2$ in units of $H^{D-4}/(4\pi)^{D/2}$	S1 (inherited sign); absorbable (B2)
(I.60)–(I.62)	reproduced	confirmed
(I.63)–(I.67)	reproduced up to the overall sign; no new error	S1 (inherited sign)
cancellation after (I.67)	confirmed at general D, Eq. (116)	confirmed

A. Normal contributions: extracting derivatives, Eqs. (I.68)–(I.80)

The *normal* contributions are the 3-point terms in which a pair of inner derivatives acts on one propagator function $A(y)$ and the result multiplies a second propagator function $f(y) \in \{A, B, C\}$, after the local and logarithm pieces of (I.52) have been split off, cf. (I.53). Paper I [8] reduces them by extracting the derivatives *generically*, so that only functions of y remain, acted upon by ten external operators. In this and the following subsections we follow this reduction through Tables III, IV and V of Paper I. The detailed derivations and the audit are in Chapters 5 and 6 of the derivation notes, and the scripts quoted below are in `code/05_normal_terms_tableIII/` and `code/06_tables_IV_V/` of the repository [17].

Notation. We write $\Delta x^\mu \equiv x^\mu - x'^\mu$, $\Delta\eta \equiv \eta - \eta'$, and $\Delta x_\sigma \equiv \eta_{\sigma\nu}\Delta x^\nu$, so that $\Delta x_0 = -\Delta\eta$. The length function is $y = aa'H^2\{\|\Delta\vec{x}\|^2 - (|\Delta\eta| - i\delta)^2\}$, cf. (I.11), with $a = -1/(H\eta)$ and $\partial_0 a = Ha^2$. Primes on functions of y denote d/dy; ∂ and ∂' act on x and x'. The constants are

$$\mathcal{N} \equiv \frac{H^{D-2}}{(4\pi)^{D/2}}, \qquad k \equiv \mathcal{N}\frac{\Gamma(D-1)}{\Gamma(\frac{D}{2})} \xrightarrow{D\to 4} \frac{H^2}{8\pi^2}, \qquad \hat{\pi} \equiv -\pi\cot\Big(\frac{\pi D}{2}\Big) = -\frac{2}{D-4} + O(D-4), \qquad h \equiv \frac{H^2}{16\pi^2}. \tag{117}$$

The leading term of $A(y)$ is $A_{\rm lead} \equiv \mathcal{N}\,\Gamma(\frac{D}{2}-1)\,(4/y)^{\frac{D}{2}-1}$. Here the first coefficient of (I.14) is read as $\Gamma(\frac{D}{2}-1)$ (the printed value is a misprint, C4; Sec. II E). At $D=4$, $A = h\{4/y - 2\ln(y/4) - 1\}$, cf. (I.62), and $B = C = 4h/y$, because every series term of (I.27)–(I.28) vanishes there. Table numbers without qualification (Table III, IV, …) refer to Paper I; tables of the present paper are referred to as "our Table".

Conventions. The identities (I.68)–(I.88) and (I.90)–(I.103) are exact in general D and do not depend on the overall sign of the self-mass. Only (I.89), Table III, and the tables assembled from the parts $P_1, \ldots, P_7$ defined below (Tables IV and V) carry that sign. As shown in Secs. IV B–IV E, the printed 3-point terms (I.47)–(I.49) are -1 times those that follow from the definition (I.168); the 4-point part is reversed in the same way, so the relative sign of the two parts is consistent. We give (I.89) and Table III in both conventions. Tables IV and V are quoted in the paper convention, which allows an entry-by-entry comparison; in the definition sign every entry is multiplied by -1.

Indefinite integrals, cf. Eq. (I.68).

$$I[f](y) \equiv \int^y dy'\, f(y'), \qquad \frac{d}{dy}I[f] = f, \qquad I^n \equiv I\circ\cdots\circ I. \tag{118}$$

The integral $I[f]$ is defined up to a constant and $I^2[f]$ up to $c_1 y + c_0$. In each identity below this ambiguity cancels when the constants are chosen consistently; e.g. in (121) the shift $I^2[fA''] \to I^2[fA''] + c_1 y$ is compensated by $I[f'A'] \to I[f'A'] - c_1$.

Chain rule and delta-function content, cf. Eq. (I.69). For smooth F the chain rule holds pointwise,

$$\partial_\rho \partial'_\sigma F(y) = F''(y)\,\frac{\partial y}{\partial x^\rho}\,\frac{\partial y}{\partial x'^\sigma} + F'(y)\,\frac{\partial^2 y}{\partial x^\rho \partial x'^\sigma}. \tag{119}$$

Paper I states that two derivatives acting on a product of two propagators can never produce a delta function. We find that this holds for the *naive* (pointwise) derivative. At fixed $\delta > 0$ the kink of $|\Delta\eta|$ produces in addition the singular term $\delta^0_\rho \delta^0_\sigma\,(-4i\delta\, a^2 H^2)\,F'(y)|_{\Delta\eta=0}\,\delta(\Delta\eta)$, where $y|_{\Delta\eta=0} = a^2H^2(r^2+\delta^2)$ and $r = \|\Delta\vec{x}\|$. For a component $F \supset y^s$ its spatial integral scales as $\delta \int d^{D-1}x\,(r^2+\delta^2)^{s-1} \propto \delta^{D+2s-2}$. This vanishes as $\delta \to 0$ (by analytic continuation in D) unless $s = 1 - \frac{D}{2}$, in which case a finite δ^D term survives. Hence

$$\big[\partial_\rho \partial'_\sigma F\big]_{\rm distr} = \big[\partial_\rho \partial'_\sigma F\big]_{\rm naive} + \delta^0_\rho \delta^0_\sigma\,\gamma_F\,\frac{i}{a^{D-2}}\,\delta^D(x-x'), \qquad F \supset \gamma_F\, A_{\rm lead}, \tag{120}$$

normalized so that $F = A_{\rm lead}$ reproduces the local coefficient $i\,a^{2-D}$ of (I.52).

The basic identity, cf. Eq. (I.70). With $G = I^2[fA'']$ (so $G'' = fA''$) in (119) and the integration by parts $I[fA''] = fA' - I[f'A']$,

$$f(y)\Big\{A''(y)\frac{\partial y}{\partial x^\rho}\frac{\partial y}{\partial x'^\sigma} + A'(y)\frac{\partial^2 y}{\partial x^\rho \partial x'^\sigma}\Big\} = \partial_\rho \partial'_\sigma I^2[fA''](y) + \frac{\partial^2 y}{\partial x^\rho \partial x'^\sigma}\,I[f'A'](y), \tag{121}$$

identical to the printed (I.70).

Delta content and the naive-derivative rule. Write $f = c_f + (\text{powers of } y)$, where c_f is the constant part of f: $c_f = k\hat{\pi}$ for $f = A$, and $c_f = b$, c, the coincidence limits of $i\Delta_B$ and $i\Delta_C$, for $f = B$, C. In general D the D-dependent powers do not mix with the integer powers, so the $y^{1-D/2}$ component of $I^2[fA'']$ comes only from $c_f\,A''_{\rm lead}$:

$$I^2[fA''] \supset c_f\,A_{\rm lead}\ (+\text{linear in } y) \qquad \Longrightarrow \qquad \gamma_{I^2[fA'']} = c_f. \tag{122}$$

The function $I[f'A']$ has no $y^{-D/2}$ component, so its product with the $O(\delta)$ singular part of $\partial_0\partial'_0 y$ gives no delta function. Consequently (121) holds both with naive derivatives on both sides and with distributional derivatives on both sides: the distributional left-hand side gains $f\,i\,a^{2-D}\delta^D = c_f\,i\,a^{2-D}\delta^D$ (in dimensional regularization $f(y)\delta^D = f(0)\delta^D = c_f\delta^D$), and by (120)–(122) the right-hand side gains the same term. This delta content is exactly the local contribution (I.54)–(I.58) computed separately in Sec. IV D; for $f = A$ it is (I.54), with its $k\hat{\pi}$ pole. The inner derivatives of the normal contributions must therefore be taken *naively*, as the footnote to (I.75) states; otherwise the local term would be counted twice. The corresponding question for the *outer* operators is taken up in Sec. V H.

Mixed derivatives of y, cf. Eqs. (I.71)–(I.73). For the regular part ($\delta \to 0$),

$$\partial_0 y = Ha\,y + 2aa'H^2\Delta x_0, \qquad \partial'_0 y = Ha'\,y - 2aa'H^2\Delta x_0, \qquad \partial_i y = 2aa'H^2\Delta x_i = -\partial'_i y, \tag{123}$$

and one more derivative gives the single expression

$$\frac{\partial^2 y}{\partial x^\rho \partial x'^\sigma} = H^2 aa'\big\{y\,\delta^0_\rho\delta^0_\sigma - 2a\,\delta^0_\rho H\Delta x_\sigma + 2a'H\Delta x_\rho\,\delta^0_\sigma - 2\eta_{\rho\sigma}\big\}, \tag{124}$$

which is (I.71), provided Δx_σ is read with a lowered index (Paper I does not state this explicitly). With $H(a-a') = H^2aa'\Delta\eta$ [Eq. (133) below] the components are

$$\begin{aligned}
&\frac{\partial^2 y}{\partial x^0 \partial x'^0} = H^2aa'\big[2 - y + 2aa'H^2\|\Delta\vec{x}\|^2\big], \qquad \frac{\partial^2 y}{\partial x^0 \partial x'^j} = -2H^3a^2a'\Delta x_j,\\
&\frac{\partial^2 y}{\partial x^i \partial x'^0} = 2H^3aa'^2\Delta x_i, \qquad \frac{\partial^2 y}{\partial x^i \partial x'^j} = -2H^2aa'\eta_{ij},
\end{aligned} \tag{125}$$

in agreement with (I.72)–(I.73).

Spatial identity, cf. Eq. (I.74). From (119) and (123), $\nabla\cdot\nabla'\,I^2[f] = -4(aa'H^2)^2\|\Delta\vec{x}\|^2 f - 2(D-1)aa'H^2\,I[f]$. Since aa' is constant in space,

$$aa'H^2\|\Delta\vec{x}\|^2 f(y) = -\tfrac{1}{2}(D-1)\,I[f](y) - \frac{\nabla\cdot\nabla'}{4aa'H^2}\,I^2[f](y), \tag{126}$$

which is (I.74).

Component identities, cf. Eqs. (I.75)–(I.78). These are the components of (121) with (125). For $\rho=\sigma=0$ the piece $2H^2aa'\cdot aa'H^2\|\Delta\vec{x}\|^2\,I[f'A']$ is converted with (126) applied to $I[f'A']$. The result is

$$f\,\partial_0\partial_0' A=\partial_0\partial_0' I^2[fA'']-\tfrac{1}{2}\nabla\cdot\nabla'\,I^3[f'A']+H^2aa'\big\{(2-y)\,I[f'A']-(D-1)\,I^2[f'A']\big\}, \tag{127}$$

$$f\,\partial_0\partial_j' A=\partial_0\partial_j' I^2[fA'']+Ha\,\partial_j' I^2[f'A'], \tag{128}$$

$$f\,\partial_i\partial_0' A=\partial_i\partial_0' I^2[fA'']+Ha'\,\partial_i I^2[f'A'], \tag{129}$$

$$f\,\partial_i\partial_j' A=\partial_i\partial_j' I^2[fA'']-2H^2aa'\eta_{ij}\,I[f'A'], \tag{130}$$

identical to (I.75)–(I.78). On the left the second derivatives are naive (footnote to (I.75)).

The first A term, cf. Eqs. (I.79)–(I.80). Take $f=A$ and the trace of (130). Passing the outer $\nabla\cdot\nabla'$ through the spatially constant $(aa')^{D-2}$ and using $\nabla\cdot\nabla' F(y)=-\nabla^2F(y)$,

$$\nabla\cdot\nabla'\big[(aa')^{D-2}A\,\nabla\cdot\nabla' A\big]=(aa')^{D-2}\nabla^4 I^2[AA''](y)+2(D-1)H^2(aa')^{D-1}\nabla^2 I[A'^2](y), \tag{131}$$

in agreement with (I.79)–(I.80). In (I.47) this bracket carries the coefficient $+\kappa^2$ as printed, and $-\kappa^2$ in the definition sign.

Since y^s is locally integrable in D dimensions if and only if $s>-\frac{D}{2}$, the powers present are

$$I^2[AA'']:\ y^{2-D},\,y^{3-D},\,y^{4-D},\,y^{1-\frac{D}{2}},\,y^{2-\frac{D}{2}},\ldots;\qquad I[A'^2]:\ y^{1-D},\,y^{2-D},\,y^{3-D},\,y^{1-\frac{D}{2}},\,y^{2-\frac{D}{2}},\ldots. \tag{132}$$

At $D=4$ only y^{2-D} (marginal) in $I^2[AA'']$, and y^{1-D}, y^{2-D} in $I[A'^2]$, fail to be integrable. This confirms the divergence count after (I.80). One qualification: in $I^2[AA'']$ the coefficients of $y^{1-D/2}$ and y^{3-D} each carry a pole, but their sum is finite at $D=4$ and gives a logarithm, as in (I.62); "setting $D=4$ in the higher terms" means taking these two terms together.

Code check. The script `check_68_80.py` uses $f=y^\alpha$ and $A=y^\beta$ with generic exponents (so no integration constants arise), explicit conformal coordinates and the exact $y(x;x')$. It confirms (121)–(131) at $D=4$ and $D=5$ at random rational points to 25 digits, verifies (122) symbolically for a model f, A, and generates the power lists above with symbolic D. No discrepancy with Paper I was found in (I.68)–(I.80).

B. Time derivatives, Eq. (I.89) and Table III, Eqs. (I.81)–(I.93)

First derivatives, cf. Eqs. (I.81)–(I.83). Because y depends on the spatial coordinates only through $\Delta\vec{x}$, $\partial_i' f(y)=-\partial_i f(y)$ (I.81). From $a=-1/(H\eta)$,

$$H\Delta\eta=\frac{a-a'}{aa'},\qquad a'H\Delta\eta=1-\frac{a'}{a},\qquad aH\Delta\eta=\frac{a}{a'}-1, \tag{133}$$

and therefore

$$\frac{\partial y}{\partial x^0}=Ha\Big(y-2+2\frac{a'}{a}\Big),\qquad \frac{\partial y}{\partial x'^0}=Ha'\Big(y-2+2\frac{a}{a'}\Big), \tag{134}$$

which are (I.82)–(I.83).

Three identities, cf. Eqs. (I.84)–(I.88). Adding the two time derivatives,

$$(\partial_0+\partial_0')f(y)=H(a+a')\,y\,f'(y). \tag{135}$$

The weighted combination gives $(a'\partial_0+a\partial_0')f=2Haa'\cdot aa'H^2\|\Delta\vec{x}\|^2 f'$ (I.85), where (133) was used in the form $2H\Delta\eta\,(a-a')=2H^2aa'\Delta\eta^2$. Applying (126) to f' (the integration constants drop out because $\nabla\cdot\nabla' y=-2(D-1)aa'H^2$) and $\nabla\cdot\nabla'\to-\nabla^2$,

$$(a'\partial_0+a\partial_0')f(y)=-(D-1)Haa'f(y)+\frac{\nabla^2}{2H}\,I[f](y). \tag{136}$$

Finally, $a\partial_0+a'\partial_0'=(a+a')(\partial_0+\partial_0')-(a'\partial_0+a\partial_0')$ (I.87) gives

$$(a\partial_0+a'\partial_0')f(y)=H(a+a')^2y\,f'(y)+(D-1)Haa'f(y)-\frac{\nabla^2}{2H}\,I[f](y). \tag{137}$$

Equations (135)–(137) agree with (I.84)–(I.88). They hold in general D and do not depend on the integration constant.

TABLE 13. Table III of Paper I (nonlocal logarithm contributions), coefficients of $\kappa^2H^4/(4\pi)^4$, $x\equiv y/4$. The derived table in the paper convention coincides with the printed one; in the definition sign every entry is reversed (the inherited overall sign, not a new error).

External operator	Derived, definition sign	Derived, paper convention	Printed (Table III)
$(aa')^3H^2\nabla^2$	$+\frac{12}{x}-24\ln x$	$-\frac{12}{x}+24\ln x$	$-\frac{12}{x}+24\ln x$
$(aa')^2(a+a')^2H^2\nabla^2$	$-\frac{4}{x}$	$+\frac{4}{x}$	$+\frac{4}{x}$
$(aa')^2\nabla^4$	$-8\ln x+16x\ln x$	$+8\ln x-16x\ln x$	$+8\ln x-16x\ln x$

The nonlocal logarithm term, cf. Eq. (I.89). In the definition sign, the re-derived (I.67) contains the nonlocal term

$$-iM^2_{\rm 3pt}\supset\kappa^2k\,\frac{2}{D-3}\,H(aa')^{D-2}\big(a\partial_0+a'\partial_0'\big)\nabla^2A(y)\ \xrightarrow{D\to4}\ \frac{\kappa^2H^2}{8\pi^2}\,2H(aa')^2\big(a\partial_0+a'\partial_0'\big)\nabla^2A(y),\tag{138}$$

which Paper I prints with $-2H$ in place of $2H$ (paper convention). Since $a\partial_0+a'\partial_0'$ commutes with ∇^2, Eq. (137) gives, in general D,

$$\kappa^2k\,\frac{2}{D-3}\,H(aa')^{D-2}\,\nabla^2\Big[H(a+a')^2yA'+(D-1)Haa'A-\frac{\nabla^2}{2H}\,I[A]\Big],\tag{139}$$

and at $D=4$ (definition sign)

$$\frac{\kappa^2H^2}{8\pi^2}\,2H(aa')^2\big(a\partial_0+a'\partial_0'\big)\nabla^2A=\frac{\kappa^2H^2}{16\pi^2}\Big\{12(aa')^3H^2\nabla^2A+4H^2(aa')^2(a+a')^2\nabla^2(yA')-2(aa')^2\nabla^4I[A]\Big\}.\tag{140}$$

Paper I prints, in the paper convention,

$$\frac{\kappa^2H^2}{8\pi^2}\,(-2H)(aa')^2\big(a\partial_0+a'\partial_0'\big)\nabla^2A=\frac{\kappa^2H^2}{16\pi^2}\Big\{-12(aa')^3H^2\nabla^2A-4(aa')^2(a+a')^2\nabla^2(yA')+2(aa')^2\nabla^4I[A]\Big\}.$$

Apart from the inherited overall sign, the second term lacks a factor H^2: as printed it has dimension L^{-2} times that of A, whereas the other two terms have L^{-4}. Table III carries the H^2, so this is a misprint (C8). A numerical check at test points with $H\neq1$ confirms that (I.89) holds with H^2 restored and fails without it.

Table III. With $A=h\,\hat A$, $\hat A\equiv4/y-2\ln(y/4)-1$, the prefactor becomes $\frac{\kappa^2H^2}{16\pi^2}\,h=\frac{\kappa^2H^4}{(4\pi)^4}$, and the coefficients are read off from the brace of (140) with $A\to\hat A$ and $x\equiv y/4$. Constants are dropped under ∇^2 and ∇^4, and since $\nabla^2y=2(D-1)aa'H^2$ is constant in space, $\nabla^4y=0$ (the "terms which eventually vanish such as ∇^4y" of Paper I). Thus $12\hat A\to\frac{12}{x}-24\ln x$; $y\hat A'=-\frac{4}{y}-2$, so $4y\hat A'\to-\frac{4}{x}$; and $I[\hat A]=4\ln y-2y\ln\frac{y}{4}+y$, so $-2I[\hat A]\to-8\ln x+16x\ln x$. The result is our Table 13. The same computation applied to the printed (I.89), with H^2 restored, reproduces the printed Table III exactly.

The functions $1/x$, $\ln x$ and $x\ln x$ are locally integrable at $D=4$, and the external operators contain only spatial derivatives, so they produce no δ^D term: Table III is purely nonlocal, as its caption states. Paper I states that Table X includes Table III. The first two rows are indeed added to the corresponding rows of Table X, but the third row, $8\ln x-16x\ln x$ (paper convention), is missing from the $(aa')^2\nabla^4$ row of Table X. This omission is non-analytic and changes the nonlocal self-mass; it is treated together with Table X in Sec. VI F.

Covariant d'Alembertian, cf. Eqs. (I.90)–(I.93). From $a^D\Box=\partial_\mu(a^{D-2}\eta^{\mu\nu}\partial_\nu)$ (I.42),

$$\Box=-\frac{1}{a^2}\partial_0^2-\frac{(D-2)H}{a}\partial_0+\frac{1}{a^2}\nabla^2.\tag{141}$$

For the mixed A term with inner $\partial_0\partial_i'$ (the fourth term of (I.47)), Eq. (128) with $f=A$, summed over i, and moving ∂_i' outside gives, with $G\equiv I^2[AA'']$ and $K\equiv I^2[A'^2]$,

$$\partial_0\partial_i'\big[(aa')^{D-2}A\,\partial_0\partial_i'A\big]=\nabla'^2\partial_0\big[(aa')^{D-2}\partial_0G\big]+H\nabla'^2\partial_0\big[a^{D-1}a'^{D-2}K\big]\tag{142}$$

and, with (I.42) and $\nabla'^2=\nabla^2$,

$$\partial_0\partial_i'\big[(aa')^{D-2}A\,\partial_0\partial_i'A\big]=-a^Da'^{D-2}\nabla^2\Box G+(aa')^{D-2}\nabla^4G+Ha^{D-1}a'^{D-2}\nabla^2\partial_0K+(D-1)H^2a^Da'^{D-2}\nabla^2K.\tag{143}$$

On functions of y alone, $\eta^{\mu\nu}\partial_\mu y\,\partial_\nu y/a^2=H^2(4y-y^2)$ and $\Box y=DH^2(2-y)$, so

$$\Box f(y)=H^2\big[(4y-y^2)f''(y)+D(2-y)f'(y)\big]=\Box'f(y),\tag{144}$$

the equality with $\Box'$ following from $y(x;x')=y(x';x)$. Equations (141)–(144) agree with (I.90)–(I.93).

Distributional meaning of Eqs. (I.92)–(I.93). The inner pair $\partial_0\partial_i'$ in (142) is mixed and contains no delta function even for the full $i\Delta_A$, so the step (142)→(143) is exact in the distributional sense and the $\Box$ in (143) is a *distributional* operator. Since $G \supset c_f A_{\rm lead}$ with $c_f = k\hat\pi$ [Eq. (122)] and $a^D\Box A_{\rm lead} \supset i\delta^D$, it contains the local term

$$-a^D a'^{D-2}\nabla^2\Box I^2[AA''] \supset -ik\hat\pi\, a^{D-2}\nabla^2\delta^D(x-x'), \tag{145}$$

which is not part of (I.54)–(I.59) (those contain only the inner $(0,0)$ terms). By contrast, (144) is a pointwise identity and never produces δ^D.[3] If $\Box I^2[AA'']$ in (143) is evaluated with (144), the term (145) is lost. Conversely, if a naive inner $(0,0)$ derivative, as in (127), is recombined with outer derivatives into a distributional $\Box$, its local part is counted a second time. The consistent rule is that outer operators are distributional while the local piece $c_f\, i\, a^{2-D}\delta^D$, already counted in (I.54)–(I.58), is removed from inner $(0,0)$ derivatives. Whether the reduction of Paper I respects this rule is discussed in Sec. V H.

Code check. The script `check_81_93.py` verifies (I.81)–(I.88) symbolically or at random points (25 digits), (I.90) in general D, and (I.91)–(I.93) at $D = 4, 5$ by brute-force differentiation. With the true $A(y)$ of (I.62), at three test points with $H \neq 1$, the derived nonlocal term of (I.67) plus the printed Table III vanishes exactly (so derived $=$ −printed), (I.89) with H^2 restored equals -1 times the derived term, and the printed (I.89) without H^2 fails at every point.

C. Reduction to ten external operators, Eqs. (I.94)–(I.115)

The second mixed A term, cf. Eqs. (I.94)–(I.96). Equation (129) with $f = A$, summed over i, gives

$$\partial_i\partial_0'\big[(aa')^{D-2}A\,\partial_i\partial_0'A\big] = \nabla^2\partial_0'\big[(aa')^{D-2}\partial_0'G\big] + H\nabla^2\partial_0'\big[a^{D-2}a'^{D-1}K\big] \tag{146}$$

$$= -a^{D-2}a'^D\nabla^2\Box' G + (aa')^{D-2}\nabla^4 G + Ha^{D-2}a'^{D-1}\nabla^2\partial_0' K + (D-1)H^2a^{D-2}a'^D\nabla^2 K. \tag{147}$$

With $\Box' G = \Box G$ and Eq. (136) for $f = K$, $a\partial_0' K = -a'\partial_0 K - (D-1)Haa'K + \frac{\nabla^2}{2H}I[K]$, the last term of (147) cancels and

$$\partial_i\partial_0'\big[(aa')^{D-2}A\,\partial_i\partial_0'A\big] = -a^{D-2}a'^D\nabla^2\Box G + (aa')^{D-2}\nabla^4 G - Ha^{D-3}a'^D\nabla^2\partial_0 K + \tfrac{1}{2}a^{D-3}a'^{D-1}\nabla^4 I^3[A'^2]. \tag{148}$$

Equations (146)–(148) agree with (I.94)–(I.96). Adding (143), the sum P_2 of the two mixed A terms takes the symmetric form [operators defined in Eq. (155) below]

$$P_2 = (-\delta + 2\zeta)\,G + (D-1)\,\epsilon_2 K + H(aa')^{D-2}\big(a\partial_0 + a'\partial_0'\big)\nabla^2 K$$

$$= (-\delta+2\zeta)\,I^2[AA''] + (D-1)(\epsilon_1+\epsilon_2)\,I^2[A'^2] + \epsilon_3\, yI[A'^2] - \tfrac{1}{2}\zeta\, I^3[A'^2], \tag{149}$$

where (137) was used for $f = K$; this agrees with (I.A8)–(I.A9).

The ubiquitous reduction, cf. Eqs. (I.97)–(I.101). Equation (141) and its primed counterpart give

$$\partial_0\big[(aa')^{D-2}\partial_0 f(y)\big] = -a^D a'^{D-2}\Box f + (aa')^{D-2}\nabla^2 f, \qquad \partial_0'\big[(aa')^{D-2}\partial_0' f(y)\big] = -a^{D-2}a'^D\Box f + (aa')^{D-2}\nabla^2 f. \tag{150}$$

Writing $\partial_0\partial_0' = \frac{1}{2}(\partial_0+\partial_0')^2 - \frac{1}{2}\partial_0^2 - \frac{1}{2}\partial_0'^2$ (I.99) inside

$$\partial_0\partial_0'\big[(aa')^p f\big] = (aa')^p\partial_0\partial_0' f + pH(aa')^p(a'\partial_0 + a\partial_0')f + p^2H^2(aa')^{p+1}f, \qquad p \equiv D-2, \tag{151}$$

and using $(\partial_0+\partial_0')^2 f = H^2(a^2+a'^2)yf' + H^2(a+a')^2(yf'+y^2f'')$, $-\frac{1}{2}(\partial_0^2+\partial_0'^2)f = \frac{1}{2}(a^2+a'^2)\Box f + \frac{1}{2}pH(a\partial_0 + a'\partial_0')f - \nabla^2 f$, and (136), (137), we obtain

$$\partial_0\partial_0'\big[(aa')^{D-2}f(y)\big] = \tfrac{1}{2}(aa')^{D-2}(a^2+a'^2)\big[\Box f + H^2yf'\big] - (aa')^{D-2}\nabla^2 f + \tfrac{1}{4}(D-2)(aa')^{D-2}\nabla^2 I[f]$$
$$+ \tfrac{1}{2}(D-2)(D-3)H^2(aa')^{D-1}f + \tfrac{1}{2}H^2(a+a')^2(aa')^{D-2}\big[(D-1)yf' + y^2f''\big], \tag{152}$$

as printed in (I.101).

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[3] Pointwise only the leading power cancels for $f = y^{1-D/2}$; one finds $\Box\, y^{1-D/2} = \frac{D(D-2)}{4}H^2y^{1-D/2}$, a locally integrable function.

The two B terms, cf. Eqs. (I.102)–(I.103). Expanding $\partial_i(\partial_0+\partial'_0)[X(\partial_0+\partial'_0)\partial'_iA]$, $X\equiv(aa')^{D-2}B$, and using $\partial'_i\to-\partial_i$ on functions of y,

$$\begin{aligned}&\partial_i\partial'_0\big[(aa')^{D-2}B\,\partial_0\partial'_iA\big]+\partial_0\partial'_i\big[(aa')^{D-2}B\,\partial_i\partial'_0A\big]\\&\quad=\partial_i(\partial_0+\partial'_0)\big[(aa')^{D-2}B\,(\partial_0+\partial'_0)\partial'_iA\big]-\partial_i\partial_0\big[(aa')^{D-2}B\,\partial_0\partial'_iA\big]-\partial_i\partial'_0\big[(aa')^{D-2}B\,\partial'_0\partial'_iA\big],\end{aligned}\tag{153}$$

as in (I.102). The first piece is reduced with (135) and $B\,\partial'_i(yA')=\partial'_iI[A'B+yA''B]$; the two subtracted pieces with (128)–(129) for $f=B$, (150) and (137). Collecting terms,

$$\begin{aligned}=\;&-(a^2+a'^2)(aa')^{D-2}\nabla^2\big\{\Box I^2[A''B]+H^2I[A'B+yA''B]\big\}+(aa')^{D-2}\nabla^4\big\{2I^2[A''B]-\tfrac12I^3[A'B']\big\}\\&-H^2(a+a')^2(aa')^{D-2}\nabla^2\big\{(D-2)I[A'B+yA''B]+yA'B+y^2A''B-yI[A'B']\big\}\\&+(D-1)H^2(a^2+aa'+a'^2)(aa')^{D-2}\nabla^2I^2[A'B'],\end{aligned}\tag{154}$$

which agrees term by term with (I.103).

External operators, cf. Eqs. (I.104)–(I.113). The reduced normal terms are functions of y acted upon by

$$\begin{aligned}&\alpha\equiv(aa')^D\Box^2, && \beta\equiv(aa')^{D-1}(a^2+a'^2)H^2\Box,\\&\gamma_1\equiv(aa')^DH^4, && \gamma_2\equiv(aa')^{D-1}(a^2+a'^2)H^4,\\&\gamma_3\equiv(aa')^{D-1}(a+a')^2H^4=2\gamma_1+\gamma_2, && \delta\equiv(aa')^{D-2}(a^2+a'^2)\nabla^2\Box,\\&\epsilon_1\equiv(aa')^{D-1}H^2\nabla^2, && \epsilon_2\equiv(aa')^{D-2}(a^2+a'^2)H^2\nabla^2,\\&\epsilon_3\equiv(aa')^{D-2}(a+a')^2H^2\nabla^2=2\epsilon_1+\epsilon_2, && \zeta\equiv(aa')^{D-2}\nabla^4.\end{aligned}\tag{155}$$

Under $\eta\to\lambda\eta$, $\vec{x}\to\lambda\vec{x}$ one has $a\to a/\lambda$ and $\nabla^2\to\nabla^2/\lambda^2$, while H, y and $\Box$ are invariant; all ten operators have weight λ^{-2D}. The header of the appendix Table XII of Paper I prints $\beta=(aa')^{D-2}(a^2+a'^2)H^2\Box$, of weight λ^{-2D+2}; the correct factor is $(aa')^{D-1}$, as in (I.105), which the global check below also confirms.

Differences and products, cf. Eqs. (I.114)–(I.115). Paper I introduces $\Delta B\equiv B-A$ and $\Delta C\equiv2\frac{D-2}{D-3}(C-A)$. The normalization of ΔC is chosen because the $\partial_0\partial'_0$ terms of (I.47) combine as $-\frac{D-1}{D-3}A+\frac{2(D-2)}{D-3}C=A+\Delta C$. The appendix totals then contain only the eight products

$$A'^2,\quad AA'',\quad A'B',\quad A''B,\quad A'\Delta B',\quad A''\Delta B,\quad A'\Delta C',\quad A''\Delta C.\tag{156}$$

D. Expansions and the α example, Eqs. (I.116)–(I.125)

With $z\equiv4/y$ and $p\equiv D-2$, only the two leading terms of each propagator function reach the powers z^D and z^{D-1} of the products:

$$A\simeq\mathcal{N}\big[c_0z^{p/2}+a_1z^{p/2-1}\big],\quad B\simeq\mathcal{N}\big[c_0z^{p/2}+b_1z^{p/2-1}\big],\quad C\simeq\mathcal{N}\big[c_0z^{p/2}+c_1z^{p/2-1}\big],\tag{157}$$

with $c_0=\Gamma(\frac{D}{2}-1)$, $a_1=\Gamma(\frac{D}{2}+1)/(\frac{D}{2}-2)$, $b_1=\Gamma(\frac{D}{2})$ and $c_1=-(3-\frac{D}{2})\Gamma(\frac{D}{2}-1)$. Using $\frac{d}{dy}z^q=-\frac{q}{4}z^{q+1}$ one obtains the coefficients of our Table 14, in the unit $\frac{\Gamma^2(D/2)}{16}\frac{H^{2D-4}}{(4\pi)^D}$ of (I.116)–(I.123). For example, with $q\equiv\frac{D}{2}-1$ the coefficient of z^{D-1} in $A'B'$ is $(q-1)(a_1+b_1)/\Gamma(q+1)=(q+1)+(q-1)=D-2$.

The product A'^2 appears in the appendix totals only as $yI[\cdot]$, $y^2\cdot$ and $I^2[\cdot]$, where $z^{D-1}\to z^{D-3}$ is not divergent; the $4z^3$ of (I.116) is the $D=4$ value of Dz^{D-1}. The products $A'B'$, $A'\Delta B'$ and $A'\Delta C'$ appear with a single integration, so their exact-D coefficients of z^{D-1} are needed, and Paper I gives them exactly. For $A''B$ the exact-D coefficient of z^{D-1} is D while the $D=4$ form has $2z^3$; there is no conflict, since the constant term of B times the leading term of $A''\propto z^{D/2+1}$ is a separate power in general D that coincides with z^3 at $D=4$, where it contributes -2. Only the $D=4$ value is needed, because $A''B$ is always shielded by at least two powers of y.

The α example, cf. Eqs. (I.124)–(I.125). The total of the appendix Table XI (operator α) is $I^2[AA'']+I^2[\Delta C\,A'']$. With our Table 14,

$$\kappa^2\big\{I^2[AA'']+I^2[A''\Delta C]\big\}=\frac{\Gamma^2(\frac{D}{2})}{16}\frac{\kappa^2H^{2D-4}}{(4\pi)^D}\,I^2\Big[\frac{D}{D-2}z^D+12z^3L+8z^3+12z^2L+6z^2+O\Big(\frac{D-4}{y^3}\Big)\Big],\tag{158}$$

as printed. With $I^2[z^D]=\frac{16}{(D-1)(D-2)}z^{D-2}$ for the exact-D term, and two integrations at $D=4$ for the rest,

$$=\frac{\kappa^2H^{2D-4}}{(4\pi)^D}\Gamma^2\Big(\frac{D}{2}\Big)\Big\{\frac{D}{(D-1)(D-2)^2}\Big(\frac{4}{y}\Big)^{D-2}+6\Big(\frac{4}{y}\Big)\ln\frac{y}{4}+13\Big(\frac{4}{y}\Big)-6\ln^2\frac{y}{4}-18\ln\frac{y}{4}+O\Big(\frac{D-4}{y}\Big)\Big\},\tag{159}$$

in agreement with (I.125) up to the constant $(24\ln2-36)\ln2$, which $\alpha=(aa')^D\Box^2$ annihilates.

TABLE 14. Leading coefficients of the eight products (156) in units of $\frac{\Gamma^2(D/2)}{16}\frac{H^{2D-4}}{(4\pi)^D}$, and their $D=4$ closed forms (in units of $h^2/16$), with $L\equiv\ln(y/4)$ and $z=4/y$. All entries agree with (I.116)–(I.123). *The z^{D-1} coefficient carries a pole; it combines with the constant ($\hat\pi$) term into z^3L at $D=4$. These products enter Table IV only through z^D, because $I^2[z^{D-1}]\sim z^{D-3}$ is not divergent.

Product	z^D (exact D)	z^{D-1} (exact D)	$D=4$ closed form ($\times h^2/16$)
A'^2	1	D	$z^4+4z^3+4z^2$
AA''	$\frac{D}{D-2}$	$\frac{D(D-2)}{D-4}$*	$2z^4-4z^3L-4z^2L-2z^2$
$A'B'$	1	$D-2$	z^4+2z^3
$A''B$	$\frac{D}{D-2}$	D	$2z^4+2z^3$
$A'\Delta B'$	0	-2	$-2z^3-4z^2$
$A''\Delta B$	0	$-\frac{2D}{D-4}$*	$4z^3L+2z^3+4z^2L+2z^2$
$A'\Delta C'$	0	-8	$-8z^3-16z^2$
$A''\Delta C$	0	$-\frac{8D}{D-4}$*	$16z^3L+8z^3+16z^2L+8z^2$

Code check. The script `check_94_125.py` confirms (I.94)–(I.103) in explicit coordinates at $D=4$ and $D=5$ with $A=y^\beta$, $B=y^\gamma$, $f=y^\alpha$ (generic rational exponents, 30-digit arithmetic), (I.99), (I.108), (I.112), the exact-D leading coefficients and $D=4$ closed forms of our Table 14, and (I.124)–(I.125) (exact-D term at $D=5,6,7,\frac{9}{2}$; finite part up to the constant above).

E. The appendix decomposition and the scope of its audit

Paper I splits the normal contribution into seven parts, (I.A1)–(I.A7) of its appendix "Extracting derivatives":

$$\begin{aligned}
&P_1\equiv\nabla\cdot\nabla'\big[(aa')^{D-2}A\,\nabla\cdot\nabla'A\big],\qquad P_2\equiv\partial_i\partial_0'\big[(aa')^{D-2}A\,\partial_i\partial_0'A\big]+\partial_0\partial_i'\big[(aa')^{D-2}A\,\partial_0\partial_i'A\big],\\
&P_3\equiv\partial_0\partial_0'\big[(aa')^{D-2}A\,\partial_0\partial_0'A\big],\qquad P_4\equiv-\partial_0\partial_0'\big[(aa')^{D-2}B\,\nabla\cdot\nabla'A\big],\\
&P_5\equiv-\partial_i\partial_0'\big[(aa')^{D-2}B\,\partial_0\partial_i'A\big]-\partial_0\partial_i'\big[(aa')^{D-2}B\,\partial_i\partial_0'A\big],\\
&P_6\equiv-\nabla\cdot\nabla'\big[(aa')^{D-2}B\,\partial_0\partial_0'A\big],\qquad P_7\equiv\partial_0\partial_0'\big[(aa')^{D-2}\Delta C\,\partial_0\partial_0'A\big],
\end{aligned}\tag{160}$$

with naive inner derivatives and the factors κ^2 suppressed. Their sum is the normal part of the *printed* (I.47)–(I.49) (paper convention), because $-\frac{D-1}{D-3}P_3+\frac{2(D-2)}{D-3}\partial_0\partial_0'[(aa')^{D-2}C\,\partial_0\partial_0'A]=P_3+P_7$; in the definition sign the normal part is $-\sum_k P_k$. The appendix uses (I.A17)–(I.A18), i.e. (127) and (130); the generic reductions (I.A35)–(I.A38); and (I.A9), (I.A15) for P_2 and P_5. Each pair (part, operator) gives one row of one of the Tables XI–XX. As an example of a generic reduction, (I.A36) reads

$$\partial_0\partial_0'\big[(aa')^{D-1}H^2f(y)\big]=\Big\{\frac{\beta}{2}+\frac{(D-1)(D-2)}{2}\gamma_1+\frac{\gamma_2}{2}\,y\frac{\partial}{\partial y}+\frac{\gamma_3}{2}\Big[(D-1)y\frac{\partial}{\partial y}+y^2\frac{\partial^2}{\partial y^2}\Big]+\epsilon_1\Big[-1+\frac{D}{4}I\Big]\Big\}f(y).\tag{161}$$

For instance, the part P_{3c} contains $f=-(D-1)K-yI[A'^2]$ under $\partial_0\partial_0'[(aa')^{D-1}H^2\,\cdot\,]$, so by (161) its γ_3 row is $\frac12[(D-1)yf'+y^2f'']=-\frac12D(D-1)\,yI[A'^2]-D\,y^2A'^2-\frac12y^3(A'^2)'$, which is the P_{3c} row of Table XV. Similarly, the rows P_{3b} and P_{4b} of the β table give $I[A'^2]+(D-1)I[A'B']=DI[A'B']-I[A'\Delta B']$.

The ten totals. Denoting by $\mathcal{F}_{\rm op}$ the total of the appendix table for operator op (Tables XI–XX, as printed after the five corrections of our Table 15), the content of the appendix is

$$\sum_{k=1}^{7}P_k=\alpha\,\mathcal{F}_\alpha+\beta\,\mathcal{F}_\beta+\sum_{i=1}^{3}\gamma_i\,\mathcal{F}_{\gamma_i}+\delta\,\mathcal{F}_\delta+\sum_{i=1}^{3}\epsilon_i\,\mathcal{F}_{\epsilon_i}+\zeta\,\mathcal{F}_\zeta,\tag{162}$$

TABLE 15. Typographical errors in the appendix of Paper I (numbering of our Table 40). None of them affects any downstream result (Tables IV–V are consistent with the corrected totals).

	Location	Printed	Correct	Reason
C9	Table XII header	$\beta = (aa')^{D-2}(a^2+a'^2)H^2\Box$	$(aa')^{D-1}(a^2+a'^2)H^2\Box$, as in (I.105)	weight counting; global check
C10	(I.A18)	$-2(D-1)aa'H^2\, I[f'A]$	$-2(D-1)aa'H^2\, I[f'A']$	trace of (I.78)
C11	Table XIII, row P_{7c}	repeats part of the total	$-\frac{1}{2}(D-1)(D-2)\, yI[A'\Delta C'] - \frac{1}{2}(D-1)^2(D-2)\, I^2[A'\Delta C']$	printed total correct
C12	Table XV, row P_{4b}	$(D-1)\, y'(A'B')'$	$(D-1)\, y^2(A'B')'$	from (I.A36)
C13	Table XVIII, total	$\frac{1}{4}\, yI^2[A'^2\Delta C']$	$\frac{1}{4}\, yI^2[A'\Delta C']$	row P_{7d}
C23	text before (I.A31)	"… to P_1 and P_6"	"… to P_1 and P_4"	subparts $P_{1a},\dots,P_{4a},\dots$

with

$$
\begin{aligned}
\mathcal{F}_\alpha &= I^2[AA''] + I^2[\Delta C\, A''],\\
\mathcal{F}_\beta &= DI[A'B'] - I[A'\Delta B'] - \tfrac{1}{2}yI[A'^2] - \tfrac{D-1}{2}I^2[A'^2] + I[A'\Delta C'] - \tfrac{1}{2}yI[A'\Delta C'] - \tfrac{D-1}{2}I^2[A'\Delta C'],\\
\mathcal{F}_{\gamma_1} &= (D-1)(D-2)\,\mathcal{F}_\beta,\\
\mathcal{F}_{\gamma_2} &= DyA'B' - yA'\Delta B' - \tfrac{1}{2}y^2A'^2 - \tfrac{D}{2}yI[A'^2] + yA'\Delta C' - \tfrac{D}{2}yI[A'\Delta C'] - \tfrac{1}{2}y^2A'\Delta C',\\
\mathcal{F}_{\gamma_3} &= (D-1)D\,yA'B' + Dy^2(A'B')' - (D-1)yA'\Delta B' - y^2(A'\Delta B')' - Dy^2A'^2 - \tfrac{1}{2}y^3(A'^2)'\\
&\quad - \tfrac{1}{2}(D-1)D\,yI[A'^2] + (D-1)yA'\Delta C' + y^2(A'\Delta C')' - Dy^2A'\Delta C' - \tfrac{1}{2}y^3(A'\Delta C')'\\
&\quad - \tfrac{1}{2}(D-1)D\,yI[A'\Delta C'],\\
\mathcal{F}_\delta &= 2I^2[\Delta B\, A''] + \tfrac{1}{4}I^3[A'^2] - I^2[\Delta C\, A''] + \tfrac{1}{4}I^3[A'\Delta C'],\\
\mathcal{F}_{\epsilon_1} &= -2(D-2)I[A'\Delta B'] - yI[A'\Delta B'] - \tfrac{5D-4}{2}I^2[A'\Delta B'] + \tfrac{D^2}{2}I^2[A'B'] - \tfrac{D}{4}yI^2[A'^2] + (D-2)^2I^2[BA'']\\
&\quad - \tfrac{3}{4}(D-2)I^3[A'^2] - 2I[A'\Delta C'] + yI[A'\Delta C'] + \tfrac{3D-2}{2}I^2[A'\Delta C']\\
&\quad - \tfrac{D}{4}yI^2[A'\Delta C'] - \tfrac{3}{4}(D-2)I^3[A'\Delta C'],\\
\mathcal{F}_{\epsilon_2} &= 2yI[BA''] + I^2[A'B'] - (D-1)I^2[A'\Delta B'] + \tfrac{1}{4}yI^2[A'^2] + \tfrac{1}{4}yI^2[A'\Delta C'],\\
\mathcal{F}_{\epsilon_3} &= 2y^2BA'' + 2(D-1)yI[BA''] + (D-2)I^2[A'B'] + yI[A'^2] + \tfrac{D-1}{4}yI^2[A'^2] + \tfrac{1}{4}y^2I[A'^2]\\
&\quad + \tfrac{D-1}{4}yI^2[A'\Delta C'] + \tfrac{1}{4}y^2I[A'\Delta C'],\\
\mathcal{F}_\zeta &= -4I^2[\Delta B\, A''] + I^3[A'\Delta B'] + \tfrac{D-2}{8}I^4[A'^2] + I^2[\Delta C\, A''] - \tfrac{1}{2}I^3[A'\Delta C'] + \tfrac{D-2}{8}I^4[A'\Delta C'].
\end{aligned}
\tag{163}
$$

Global pointwise check. The ten operators were applied to the totals (163) and compared pointwise with the direct sum of $P_1,\dots,P_7$ (naive inner derivatives), (i) with generic powers $A = y^{-7/3}$, $B = y^{-5/4}$, $\Delta C = y^{2/7}$ at $D = 5$ and $D = 4$ (two random spacelike points each), and (ii) with the actual $D = 4$ propagators (three random points), where the constants of the first integrals must be linked, $I[fA''] = fA' - I[f'A']$ (scripts `check_global.py`, `check_tables45.py`). In every case the difference vanishes to the 30-digit working precision, once the five misprints C9–C13 of our Table 15 are corrected (C23 in that table is a wording slip that does not enter the totals).

Scope of the audit. The appendix "Extracting derivatives" was *audited*, not redone from scratch row by row. What is established is: (i) the worked examples above (the P_{3c} row of the γ_3 table and the β and γ_1 totals) were checked by hand; (ii) the definitions (I.A1)–(I.A7), the reduction (I.A8)–(I.A9) and Tables XII (β) and XIV (γ_2) were transcribed directly from the printed appendix; (iii) the remaining tables (XI, XIII, XV–XX) were used through their totals, which were coded line by line and verified by the global pointwise test. This test is numerical, at a few random sample points with 30-digit arithmetic, and it checks only the sum over operators; it is not a symbolic identity. An earlier working report stated that every row had been reproduced by hand; that was an overstatement. Finally, Tables IV and V below were computed from the appendix totals after these totals had been checked, so for this step the printed appendix (not the target tables) is an input.

TABLE 16. Table IV of Paper I (divergent normal contributions) derived in general D, in units of $\Gamma^2(\frac{D}{2})\,\kappa^2 H^{2D-4}/(4\pi)^D$, paper convention (signs of $P_1,\dots,P_7$). The last column compares with the printed table, column by column.

Op.	Coef. of $(\frac{4}{y})^{D-1}$	Coef. of $(\frac{4}{y})^{D-2}$	Printed
α	0	$\frac{D}{(D-1)(D-2)^2}$	agrees, agrees
β	$-\frac{D}{4(D-1)}$	$-\frac{D^3-3D^2-4D+8}{4(D-1)(D-2)}$	agrees, agrees
γ_1	$-\frac{D(D-2)}{4}$	$-\frac{D^3-3D^2-4D+8}{4}$	agrees, agrees
γ_2	$\frac{D}{4}$	$+\frac{D^3-3D^2-4D+8}{4(D-1)}$	agrees; prints $-\frac{D^3-3D^2-4D+8}{4(D-1)}$
γ_3	$-\frac{D}{4}$	0	agrees, agrees
δ	0	0	agrees, agrees
ϵ_1	0	$\frac{D^2-6D+4}{2(D-1)(D-2)}$	agrees, agrees
ϵ_2	0	$\frac{1-2D}{(D-1)(D-2)}$	agrees, agrees
ϵ_3	0	0	agrees, agrees
ζ	0	0	agrees, agrees

F. Table IV in general D

The coefficients of z^{D-1} and z^{D-2} in each total follow from our Table 14 by the rules

$$I[z^q]=-\frac{4\,z^{q-1}}{q-1},\qquad y\,z^q=4z^{q-1},\qquad \frac{d}{dy}z^q=-\frac{q}{4}z^{q+1}. \tag{164}$$

The unit of Table IV is $\Gamma^2(\frac{D}{2})\kappa^2H^{2D-4}/(4\pi)^D$, so the factor $\frac{1}{16}$ of the products is divided out at the end.

Example: β. In $\mathcal{F}_\beta$ the terms reaching z^{D-1} or z^{D-2} are $DI[A'B'] \to -\frac{4D}{D-1}z^{D-1}-4Dz^{D-2}$, $-I[A'\Delta B'] \to -\frac{8}{D-2}z^{D-2}$, $-\frac{1}{2}yI[A'^2]\to+\frac{8}{D-1}z^{D-2}$, $-\frac{D-1}{2}I^2[A'^2]\to-\frac{8}{D-2}z^{D-2}$ and $I[A'\Delta C']\to+\frac{32}{D-2}z^{D-2}$. Dividing by 16,

$$z^{D-1}:\ -\frac{D}{4(D-1)},\qquad z^{D-2}:\ \frac{1}{16}\Big[-4D+\frac{16}{D-2}+\frac{8}{D-1}\Big]=-\frac{D^3-3D^2-4D+8}{4(D-1)(D-2)}, \tag{165}$$

as printed. Since $\mathcal{F}_{\gamma_1}=(D-1)(D-2)\mathcal{F}_\beta$, the γ_1 row follows.

Example: γ_2. In $\mathcal{F}_{\gamma_2}$ the contributions are $DyA'B'\to 4Dz^{D-1}+4D(D-2)z^{D-2}$, $-yA'\Delta B'\to+8z^{D-2}$, $-\frac{1}{2}y^2A'^2\to-8z^{D-2}$, $-\frac{D}{2}yI[A'^2]\to+\frac{8D}{D-1}z^{D-2}$ and $yA'\Delta C'\to-32z^{D-2}$; the last two terms of $\mathcal{F}_{\gamma_2}$ start at z^{D-3}. The first column is $4D/16=D/4$, as printed, and the second is

$$\frac{1}{16}\Big[4D(D-2)+\frac{8D}{D-1}-32\Big]=\frac{D(D-1)(D-2)+2D-8(D-1)}{4(D-1)}=+\frac{D^3-3D^2-4D+8}{4(D-1)}. \tag{166}$$

Paper I prints $-\frac{D^3-3D^2-4D+8}{4(D-1)}$. The other rows were obtained mechanically with the same rules (`check_table4.py`); the α row also follows from (159). The result is our Table 16: 19 of the 20 entries agree with Paper I in general D. The printed sign of the γ_2 entry is a misprint (C15): Paper I's own Tables VI–IX use the + sign, and shift symmetry independently requires it (with the printed sign a forbidden term $-\frac{16}{3}$ would remain in Table IX at $D=4$); this is discussed in Sec. VI D.

G. Table V at $D=4$ and the missing $72\gamma_2$

The totals (163) were evaluated exactly at $D=4$ with $A=h\{4/y-2\ln(y/4)-1\}$ and $B=C=4h/y$ (linked first integrals, higher integrals as `sympy` antiderivatives, $\ln y=\ln x+\ln 4$). The resulting functions are listed in our Table 17. Their $1/x^3$ and $1/x^2$ terms are Table IV at $D=4$ (with the derived γ_2 entry $+\frac{2}{3}$); the rest is to be compared with Table V.

TABLE 17. Derived operator functions at $D = 4$ (coefficients of $\kappa^2 H^4/(4\pi)^4$, paper convention, $x = y/4$), with the integration constants as produced by `check_tables45b.py`. Polynomial terms (constants, x, x^2 and the $\ln 2$ terms) depend on the integration-constant convention.

Op.	Derived $D = 4$ function
α	$\frac{1}{3x^2} + \frac{13}{x} + \frac{6\ln x}{x} - 6\ln^2 x - 18\ln x + 24\ln^2 2 - 36\ln 2$
β	$-\frac{1}{3x^3} - \frac{1}{3x^2} + \frac{5}{x} - 18\ln x - 6 - 36\ln 2$
γ_1	$-\frac{2}{x^3} - \frac{2}{x^2} + \frac{30}{x} - 108\ln x - 36 - 216\ln 2$
γ_2	$\frac{1}{x^3} + \frac{2}{3x^2} - \frac{5}{x} - 18$
γ_3	$-\frac{1}{x^3} - \frac{5}{x} - 36$
δ	$-\frac{4\ln x}{x} - \frac{49}{6x} + 4\ln^2 x + 10\ln x + 12x\ln x - 12x + 24x\ln 2 - 16\ln^2 2 + 20\ln 2$
ϵ_1	$-\frac{1}{3x^2} - \frac{26}{3x} + 60\ln x - 120x\ln x + 72x + 20 + 120\ln 2 - 240x\ln 2$
ϵ_2	$-\frac{7}{6x^2} + \frac{13}{6x} - 12\ln x + 12x\ln x - 2 - 24\ln 2 + 24x\ln 2$
ϵ_3	$-\frac{11}{6x} + 36x\ln x + 12x - 8 + 72x\ln 2$
ζ	$\frac{10}{3}\ln x - 24x\ln x + 24x^2\ln x - 36x^2 + 24x + 48x^2\ln 2 - 48x\ln 2 + \frac{20}{3}\ln 2$

TABLE 18. Per-operator differences at $D = 4$, derived minus [printed Table IV($D = 4$) + printed Table V], in units of $\kappa^2 H^4/(4\pi)^4$, paper convention (`check_tables45.py`).

Op.	Difference	Effect
α	$24\ln^2 2 - 36\ln 2$	constant, annihilated by $\Box^2$
β	$-36\ln 2 - 6$	constant, annihilated by $\Box$
γ_1	$-216\ln 2$	convention; cancels against ϵ_1, ζ, ϵ_3
γ_2	$\frac{4}{3x^2}$	sign of the Table IV entry, Eq. (166)
γ_3	0	—
δ	$-12x + 24x\ln 2 - 16\ln^2 2 + 20\ln 2$	$\ln 2$ part: convention; $-12x$ gives $+72\gamma_2[1]$
ϵ_1	$-240x\ln 2 + 20 + 120\ln 2$	convention
ϵ_2	$24x\ln 2 - 24\ln 2 - 2$	convention
ϵ_3	$72x\ln 2 - 8$	convention
ζ	$48x^2\ln 2 - 48x\ln 2 + 24x + \frac{20}{3}\ln 2$	$\nabla^4 x = 0$; $x^2\ln 2$: convention

Comparison operator by operator. Our Table 18 lists the derived function minus [printed Table IV at $D = 4$ + printed Table V]. All logarithmic and inverse-power terms ($1/x$, $\ln x/x$, $\ln x$, $\ln^2 x$, $x\ln x$, $x^2\ln x$) agree. Apart from the γ_2 entry of Table IV, the differences are polynomials. To see their effect we need, at $D = 4$,

$$\nabla^2 x = \tfrac{3}{2}aa'H^2, \qquad \nabla^4 x = 0, \qquad \nabla^4 x^2 = \tfrac{15}{2}(aa'H^2)^2, \qquad \Box x = H^2(2-4x) \;\Rightarrow\; \nabla^2\Box x = -6aa'H^4, \tag{167}$$

the last from (144). Then: (i) constants are annihilated by $\Box$, ∇^2 and ∇^4; (ii) all $\ln 2$ terms cancel globally, using $\gamma_3 = 2\gamma_1 + \gamma_2$ and $\epsilon_3 = 2\epsilon_1 + \epsilon_2$: in units of $\ln 2$, the γ_1 structure receives -216 (from γ_1), -360 (from ϵ_1), $+360$ (from ζ) and $+216$ (from ϵ_3), and the γ_2 structure receives -144 (from δ), $+36$ (from ϵ_2) and $+108$ (from ϵ_3); (iii) what remains is the $-12x$ of the δ row,

$$\delta[-12x] = (aa')^2(a^2 + a'^2)(-12)\,\nabla^2\Box x = 72\,(aa')^3(a^2 + a'^2)H^4 = 72\,\gamma_2[1]. \tag{168}$$

The choice of integration constants does not change the sum over operators (every consistent convention gives exact identities), so this remainder is convention independent.

Global test. The direct sum $\sum_k P_k$, with the actual $D = 4$ propagators, was compared at three random spacelike points with several versions of Tables IV+V acted upon by the ten operators (our Table 19). Only the version with both corrections agrees, to the 30-digit working precision.

Corrected Tables IV and V. Thus the printed Tables IV+V reproduce the direct sum only with two corrections: the sign of the second column of the γ_2 row of Table IV, and a finite term equivalent to $72\,\gamma_2$ acting on a constant. The latter is most naturally read as a term $-12x$ dropped from the δ row of Table V (Paper I may have assumed

TABLE 19. Residual $\sum_k P_k - \sum_{\rm op}$ op [Table IV + Table V] at three random spacelike points (`check_tables45b.py`); "0" means zero to 30 digits.

Version of Tables IV+V	Residual at the three points
as printed	0.0356, 17.05, 0.607
γ_2 sign of Table IV corrected only	0.0351, 16.97, 0.602
$-12x$ added to the δ row only	5×10^{-4}, 0.074, 0.0047
γ_2 sign corrected + $-12x$ added to the δ row	0, 0, 0
γ_2 sign corrected + γ_2 constant $-18\to+54$ (equivalent)	0, 0, 0

TABLE 20. Table V of Paper I (finite normal contributions), corrected, coefficients of $\kappa^2H^4/(4\pi)^4$, $x=y/4$, paper convention. The only change with respect to the printed table is the term $-12x$ in the δ row (equivalently, the γ_2 row $-\frac{5}{x}+54$ instead of the printed $-\frac{5}{x}-18$). Together with the corrected γ_2 entry of Table IV, $(\frac{D}{4},\ +\frac{D^3-3D^2-4D+8}{4(D-1)})$, this reproduces the direct sum pointwise.

Op.	Coefficient of $\kappa^2H^4/(4\pi)^4$
α	$\frac{6\ln x}{x}+\frac{13}{x}-6\ln^2x-18\ln x$
β	$\frac{5}{x}-18\ln x$
γ_1	$\frac{30}{x}-108\ln x-36$
γ_2	$-\frac{5}{x}-18$
γ_3	$-\frac{5}{x}-36$
δ	$-\frac{4\ln x}{x}-\frac{49}{6x}+4\ln^2x+10\ln x+12x\ln x$ **$-12x$**
ϵ_1	$-\frac{26}{3x}+60\ln x-120x\ln x+72x$
ϵ_2	$\frac{13}{6x}-12\ln x+12x\ln x$
ϵ_3	$-\frac{11}{6x}+36x\ln x+12x$
ζ	$\frac{10}{3}\ln x-24x\ln x+24x^2\ln x-36x^2$

that $\nabla^2\Box x$ vanishes); equivalently the γ_2 row becomes $-\frac{5}{x}+54$. Where the term belongs is an interpretation: the tests fix only the sum. The corrected Table V is given in our Table 20; in the definition sign all entries of Tables IV and V are multiplied by -1.

Physical significance of $72\gamma_2$. The term $72\gamma_2[1]$ (S3) is independent of y, i.e. analytic. In the Schwinger–Keldysh formalism of Paper II [12] it is the same in the $++$ and $+-$ kernels and therefore drops out of the retarded self-mass-squared identically; the same holds for all polynomial (integration-constant) ambiguities of Table V. It reappears in Sec. VI F as the constant of the row $(aa')^3(a^2+a'^2)H^4$ of Table X, $-54\to+18$ (paper convention). There, the global pointwise test of Table X against the direct sum of the normal contributions succeeds only when both this correction and the restoration of the missing Table III row in the $(aa')^2\nabla^4$ row are applied; neither correction alone suffices. The $72\gamma_2$ correction has no effect on any retarded quantity, and hence none on the late-time mode function.

H. Distributional content of the external operators

By Eq. (122), the function multiplying α, $I^2[(A+\Delta C)A'']$, contains the component $c_{A+\Delta C}\,A_{\rm lead}$ with

$$c_{A+\Delta C}=k\hat\pi+\frac{2(D-2)}{D-3}\big(c-k\hat\pi\big)=-\frac{D-1}{D-3}\,k\hat\pi+\frac{2(D-2)}{D-3}\,c,\qquad c=\frac{k}{(D-2)(D-3)},\tag{169}$$

which is exactly the combination in the local contribution (I.54)+(I.58). The generic reduction (I.A35) is correct pointwise. However, if the $\Box$ in $\alpha=(aa')^D\Box^2$ and in δ is applied outside the integral as a distribution, as Paper I states, then $\Box A_{\rm lead}\supset i\delta^D/a^D$ would reproduce (I.54)+(I.58) a second time; in the $D=4$ tables this content sits in the $1/x$ and $\ln x/x$ terms of α, β and δ. The local term (145) raises the complementary question.

This point is only partially resolved. In the counterterm analysis (Sec. VI D) we find that in Paper I divergent local terms arise only from Table IV, while in the finite Table V functions the $c_fA_{\rm lead}$ component has combined with

TABLE 21. Status of Eqs. (I.68)–(I.125) and Tables III–V of Paper I. "Inherited sign": the printed quantity is −1 times the definition-sign result, the overall sign established for (I.47)–(I.49).

Item	Result	Status
(I.68)–(I.80)	derived; (I.69) holds for naive derivatives; divergence count confirmed	agrees
(I.81)–(I.88), (I.90)–(I.93)	derived in general D; $\Box$ in (I.92) distributional, (I.93) pointwise	agrees
(I.89)	second term lacks H^2; overall sign	misprint (C8); inherited sign
Table III	all three rows reproduced	agrees (inherited sign)
(I.94)–(I.115), (I.A8)–(I.A9)	derived; weight counting consistent	agrees
(I.116)–(I.125)	leading coefficients and $D = 4$ forms; (I.125) up to a constant annihilated by α	agrees
Appendix, Tables XI–XX	totals verified by the global pointwise test (audit, not row-by-row re-derivation)	misprints C9–C13; wording slip C23
Table IV	19/20 entries; γ_2, second column, sign reversed	misprint (C15)
Table V	all non-analytic terms agree; finite term $72\gamma_2$ missing globally	S3: analytic; no retarded effect
Outer $\Box$ as a distribution	possible double counting of a local term of counterterm form	partially resolved

the pole-carrying z^{D-3} terms into finite logarithms. Any double counting then has exactly the form $c(D)\,(a^D\Box - a^{D-2}\nabla^2)\delta^D(x-x')$, with a constant $c(D)$, which may contain a pole, and no $\ln a$; this is the form of the counterterms (I.131) and (I.145). Both the pole and the finite part of $c(D)$ are absorbed into α_2 and α_3, and the renormalized result changes only through the arbitrary $\Delta\alpha_i$. Statements about the poles and finite parts of β_2 and β_3 (Sec. VI E) therefore hold within Paper I's treatment of the outer operators. (The term (145) is likewise of the form $a^{D-2}\nabla^2\delta^D$ of (I.145).) It has not been shown that the outer-operator reduction (I.A35)–(I.A38) produces no *other* local difference at coincidence, for example a finite $H^4a^D\delta^D$; that would require a full distributional analysis. Partial support comes from the vanishing $a^D\Box^2$ total of Table VIII and the finiteness of Table IX.

I. Summary of this section

Our Table 21 summarizes the comparison of Eqs. (I.68)–(I.125) and Tables III–V with Paper I.

Within Eqs. (I.68)–(I.125) the misprints in (I.89) (C8), in Table IV (C15) and in the appendix (C9–C13, C23) have no effect on the printed results downstream, and the overall sign is the one inherited from (I.47)–(I.49). Only the $72\gamma_2$ term (S3) changes a quantity used later (the constant of one row of Table X); it is analytic and has no retarded effect. The substantive consequence of this part, S2, the omission of the third row of Table III from the $(aa')^2\nabla^4$ row of Table X, which is non-analytic and changes the nonlocal self-mass (but not the late-time mode function), arises only at the assembly of Table X.

VI. RENORMALIZATION

This section corresponds to Sec. IV of Paper I [8], Eqs. (I.126)–(I.167) and Tables VI–X. It has three parts. First we determine the counterterms that the symmetries allow: two de Sitter-invariant ones and, because the gauge-fixing functional (I.15) breaks de Sitter invariance, exactly one noninvariant one (Secs. VI A–VI C). Second, we separate the divergences of the normal 3-point contributions of Table IV into local terms and finite nonlocal terms (Sec. VI D, Tables VI–IX). Third, we collect all local terms into the coefficients $\beta_1, \beta_2, \beta_3$ of the three counterterm structures, collect all finite nonlocal terms into Table X, and renormalize (Secs. VI E–VI G). Every step is written out in Chapters 7 and 8 of the derivation notes; the scripts are in `code/07_desitter_counterterms/` and `code/08_renormalization_tableX/` of Ref. [17].

In brief, the counterterm analysis (I.126)–(I.145), the separation of divergences (I.146)–(I.159) with Tables VI–IX, the vanishing of β_1, and the pole of β_3 are confirmed. The finite parts of β_2 and β_3 shift because of the two

finite local errors in (I.40) and (I.59) found in Secs. III and IV; the arbitrary finite counterterms absorb these shifts. Table X needs two corrections: a non-analytic contribution is missing from the $(aa')^2\nabla^4$ row, and the constant of the $(aa')^3(a^2+a'^2)H^4$ row changes. Only with both corrections does Table X agree with a direct pointwise sum of the normal contributions. In the definition sign the physical nonlocal part of the renormalized self-mass-squared is $-$(corrected Table X).

Conventions for this section. Tables VI–X and the coefficients β_i are first given in the *paper convention*, the sign of the printed (I.47)–(I.49) and Tables III–X, so that they can be compared entry by entry with Paper I. Values in the *definition sign*, fixed by (I.168), are -1 times these (Sec. I E). Counterterm vertices are computed with the rule "vertex $= i\,\delta^2 S_{\rm ct}/\delta\phi(x)\,\delta\phi(x')$". This rule reproduces the printed (I.130), (I.131) and (I.145), and it is the rule consistent with (I.168). The overall sign of the printed self-mass-squared therefore affects only the fitted values of the counterterm coefficients α_i, not the structure of the counterterms. All consistency checks on Tables VI–IX below are linear and homogeneous, so they hold in either convention. Following Paper I, the letters $\alpha, \beta, \gamma_i, \delta, \epsilon_i, \zeta$ denote the external operators of the normal contributions, while α_i and β_i denote counterterm coefficients. We write Euler's constant as $\gamma_{\rm E}$ (it is γ in Paper I) to avoid a clash with the operators γ_i. A bare table number ("Table VI") refers to Paper I; tables of the present paper are cited as "our Table". The coefficients β_i and α_i are in units of $H^{D-4}/(4\pi)^{D/2}$; the entries of Tables VII and X are coefficients of $\kappa^2H^4/(4\pi)^4$, functions of $x \equiv y/4$, with $y(x;x')$ the de Sitter length function (I.11). We also use

$$z \equiv \frac{4}{y}, \qquad k \equiv \frac{H^{D-2}}{(4\pi)^{D/2}}\,\frac{\Gamma(D-1)}{\Gamma(\frac{D}{2})} \equiv \frac{H^{D-2}}{(4\pi)^{D/2}}\,\bar{k}, \qquad \hat{\pi} \equiv -\pi\cot\Big(\frac{\pi D}{2}\Big), \qquad L \equiv \ln(aa'), \tag{170}$$

so that $\bar{k} \to 2$ at $D = 4$.

A. Invariant counterterms

This subsection corresponds to (I.126)–(I.131). A one-loop counterterm for the scalar self-mass-squared is proportional to $\kappa^2\phi\phi$. Since $[\kappa^2] = -2$ and $[\phi] = 1$ in $D = 4$, four further units of mass are needed, in the form m^4, $m^2\partial^2$ or ∂^4 (a curvature counts as two derivatives). This is the superficial degree of divergence four stated in Paper I: the vertices $\kappa^n h^n \partial\phi\partial\phi$ carry two derivatives, like the $\kappa^n h^n \partial h\partial h$ vertices of pure gravity. The candidate invariants are $\kappa^2 m^4\phi^2\sqrt{-g}$ (I.126); none with one or three derivatives; $\partial\phi\cdot\partial\phi$ and $\phi^2 R$ with two derivatives (I.127) (after $\phi\Box\phi \simeq -\partial\phi\cdot\partial\phi$, where $\simeq$ denotes equality up to a total derivative); and, with four derivatives,

$$(\Box\phi)^2, \quad R\,\partial\phi\cdot\partial\phi, \quad R^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi, \quad \phi^2R^2, \quad \phi^2R_{\mu\nu}R^{\mu\nu}, \quad \phi^2R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \quad \phi^2\Box R. \tag{171}$$

To reach this list we used $\phi\Box^2\phi \simeq (\Box\phi)^2$, $\phi\,\partial\phi\cdot\partial R \simeq -\frac{1}{2}\phi^2\Box R$, the contracted Bianchi identity, and the commutator identity $\phi_{;\mu\nu}\phi^{;\mu\nu} = (\Box\phi)^2 - R_{\mu\nu}\phi^{;\mu}\phi^{;\nu} + \nabla_\mu\big(\phi^{;\nu}\phi_{;\nu}{}^{;\mu} - \phi^{;\mu}\Box\phi\big)$.[4] Three arguments shorten the list. (i) The Lagrangian (I.1) is invariant under $\phi \to \phi + {\rm const}$, and dimensional regularization preserves this symmetry; every term with an undifferentiated ϕ is excluded. (ii) The scalar is massless and mass is renormalized multiplicatively in dimensional regularization, so no term carries m. (iii) On the background, $\hat{R}_{\mu\nu} = (D-1)H^2\hat{g}_{\mu\nu}$, so $R\,\partial\phi\cdot\partial\phi$ and $R^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ become degenerate. With $\sqrt{-\hat{g}} = a^D$ and $\sqrt{-\hat{g}}\,\hat{g}^{\mu\nu} = a^{D-2}\eta^{\mu\nu}$, two invariant counterterms remain (cf. (I.129)),

$$\tfrac{1}{2}\alpha_1\kappa^2\,\Box\phi\,\Box\phi\,a^D \qquad \text{and} \qquad -\tfrac{1}{2}\alpha_2\kappa^2H^2\,\partial_\mu\phi\,\partial_\nu\phi\,\eta^{\mu\nu}a^{D-2}. \tag{172}$$

Lemma A of Sec. II G, Eq. (37), together with $a^D\Box = \partial_\mu(a^{D-2}\eta^{\mu\nu}\partial_\nu)$, gives the vertices (cf. (I.130)–(I.131))

$$\tfrac{1}{2}\alpha_1\kappa^2\Box\phi\,\Box\phi\,a^D \;\longrightarrow\; i\alpha_1\kappa^2a^D\Box^2\delta^D(x-x'), \qquad -\tfrac{1}{2}\alpha_2\kappa^2H^2\partial\phi\cdot\partial\phi\,a^{D-2} \;\longrightarrow\; i\alpha_2\kappa^2H^2a^D\Box\,\delta^D(x-x'), \tag{173}$$

exactly as printed. Had the gauge condition been de Sitter invariant, these two counterterms would absorb every divergence. It is not, so we must determine which part of the de Sitter group the gauge respects.

B. The de Sitter group and the residual symmetries of the gauge

This subsection corresponds to (I.132)–(I.135) and the text that follows them. Paper I states that the $\frac{1}{2}D(D+1)$ de Sitter transformations split into four classes, and that the gauge (I.15) breaks only the fourth, with a reference to the literature [5]. We prove both statements directly.

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[4] The list (I.128) does not contain the last two terms of Eq. (171) (erratum C14). Both carry an undifferentiated ϕ and are removed by the shift symmetry, so (I.129) is unaffected.

TABLE 22. The four classes of de Sitter transformations in conformal coordinates, from the general Killing vector (175). The last column states whether the gauge-fixing term (I.15) is invariant (Eq. (180)).

Parameter	Generator	Number	Paper I	$\mathcal{L}_{\rm GF}$
c^i	$P_i = \partial_i$	$D-1$	(i) spatial translations, (I.132)	invariant
ω_{ij}	$M_{ij} = x_i\partial_j - x_j\partial_i$	$\frac{1}{2}(D-1)(D-2)$	(ii) rotations, (I.133)	invariant
λ	$\mathcal{D} = \eta\partial_0 + x^i\partial_i$	1	(iii) dilatation, (I.134)	invariant
b^i	$K_i = 2x_i(\eta\partial_0 + x^j\partial_j) - (x\cdot x)\partial_i$	$D-1$	(iv) spatial special conformal, (I.135)	broken

Killing vectors. With $\hat{g}_{\mu\nu} = a^2\eta_{\mu\nu}$, $a = -1/(H\eta)$, $\eta < 0$ and $\xi_\mu \equiv \eta_{\mu\nu}\xi^\nu$ (index lowered with the flat metric), the Killing equation reads

$$\pounds_\xi \hat{g}_{\mu\nu} = a^2\Big[\partial_\mu\xi_\nu + \partial_\nu\xi_\mu - \frac{2\xi^0}{\eta}\,\eta_{\mu\nu}\Big] = 0. \tag{174}$$

Its components are $\partial_0\xi^0 = \xi^0/\eta$, $\partial_0\xi^i = \partial_i\xi^0$ and $\partial_i\xi^j + \partial_j\xi^i = 2(\xi^0/\eta)\delta_{ij}$. The first gives $\xi^0 = \eta f(\vec{x})$ and the second $\xi^i = \frac{1}{2}\eta^2\partial_i f + g^i(\vec{x})$. In the third, the coefficient of η^2 must vanish separately, so $\partial_i\partial_j f = 0$ and $f = \lambda + 2\vec{b}\cdot\vec{x}$. What remains, $\partial_i g^j + \partial_j g^i = 2f\delta_{ij}$, is the conformal Killing equation of $\mathbb{R}^{D-1}$. Its general solution gives

$$\xi^0 = \eta\,(\lambda + 2\vec{b}\cdot\vec{x}), \qquad \xi^i = c^i + \omega^i{}_j x^j + \lambda x^i + 2(\vec{b}\cdot\vec{x})\,x^i - (x\cdot x)\,b^i, \qquad x\cdot x \equiv \|\vec{x}\|^2 - \eta^2, \tag{175}$$

with $\omega_{ij} = -\omega_{ji}$. This is the general solution, so there are no other Killing vectors. The four parameter sets generate the four classes of Paper I (our Table 22), and their number is

$$(D-1) + \tfrac{1}{2}(D-1)(D-2) + 1 + (D-1) = \tfrac{1}{2}D(D+1) = \dim SO(D,1). \tag{176}$$

As a cross-check, Eq. (174) says that ξ is a conformal Killing vector of Minkowski space with conformal factor ξ^0/η. Of the $\frac{1}{2}(D+1)(D+2)$ flat conformal generators this condition removes time translation, the $D-1$ boosts and the temporal special conformal transformation, leaving $\frac{1}{2}D(D+1)$. The flows of c, ω and λ give (I.132)–(I.134), with the dilatation factor $s = e^\lambda$ (called k in (I.134); it is unrelated to the propagator constant k of Eq. (170)). The factor must be positive, since $s < 0$ would map the patch $\eta < 0$ to $\eta > 0$. A special conformal transformation is an inversion, followed by a translation by $\theta^\mu = (0,\vec{\theta})$, followed by an inversion:

$$x'^\mu = \frac{x^\mu - \theta^\mu\, x\cdot x}{\Omega}, \qquad \Omega \equiv 1 - 2\,\vec{\theta}\cdot\vec{x} + \|\vec{\theta}\|^2\, x\cdot x, \qquad dx'\cdot dx' = \frac{dx\cdot dx}{\Omega^2}, \qquad \eta' = \frac{\eta}{\Omega}. \tag{177}$$

Hence $ds^2 = dx\cdot dx/(H^2\eta^2)$ is invariant. This is (I.135), in which $x\cdot x$ must be read as the Minkowski product $\|\vec{x}\|^2 - \eta^2$. The generators close into the conformal algebra of $\mathbb{R}^{D-1}$, isomorphic to $\mathfrak{so}(D,1)$, with $[\mathcal{D},P_i] = -P_i$, $[\mathcal{D},K_i] = K_i$ and $[K_i,P_j] = -2(\delta_{ij}\mathcal{D} + M_{ij})$.

The gauge condition. The gauge-fixing term (I.15) is $\mathcal{L}_{\rm GF} = -\frac{1}{2}a^{D-2}\eta^{\mu\nu}F_\mu F_\nu = -\frac{1}{2}\sqrt{-\hat{g}}\,\hat{g}^{\mu\nu}F_\mu F_\nu$, with $F_\mu = \eta^{\rho\sigma}\big(h_{\mu\rho,\sigma} - \frac{1}{2}h_{\rho\sigma,\mu} + (D-2)Ha\,h_{\mu\rho}\delta^0_\sigma\big)$. Let $\psi_{\mu\nu} \equiv a^2 h_{\mu\nu}$, which transforms as a tensor under background isometries, and let $u^\mu \equiv \delta^\mu_0/a$ be the unit normal of the surfaces $\eta = $ const (in this subsection only; it differs from the vector $u_\mu = \delta^0_\mu$, $u^\mu = -\delta^\mu_0$ of Secs. II–V). Computing the background covariant derivatives with $\hat{\Gamma}^\lambda_{\mu\nu} = Ha(\delta^\lambda_\mu\delta^0_\nu + \delta^\lambda_\nu\delta^0_\mu + \eta_{\mu\nu}\delta^\lambda_0)$, we find

$$F_\mu = G_\mu + E_\mu, \qquad G_\mu = \hat{g}^{\rho\sigma}\big(\hat{\nabla}_\sigma\psi_{\mu\rho} - \tfrac{1}{2}\hat{\nabla}_\mu\psi_{\rho\sigma}\big), \qquad E_\mu = H\big(2\psi_{\mu\nu}u^\nu - u_\mu\psi\big), \tag{178}$$

with $\psi \equiv \hat{g}^{\rho\sigma}\psi_{\rho\sigma}$. The vector G_μ is the de Donder vector of the background and is covariant under every isometry; the whole de Sitter-breaking part E_μ enters through u^μ, that is, through the equal-time slicing. For a Killing vector ξ, $\delta F_\mu = \pounds_\xi F_\mu + \Delta_\mu$, and since $\nabla\cdot\xi = 0$,

$$\delta\mathcal{L}_{\rm GF} = \partial_\lambda\big(\xi^\lambda\mathcal{L}_{\rm GF}\big) - \sqrt{-\hat{g}}\,\hat{g}^{\mu\nu}F_\mu\Delta_\nu, \qquad \Delta_\mu = -H\big(2\psi_{\mu\nu}w^\nu - w_\mu\psi\big), \qquad w \equiv \pounds_\xi u = [\xi,u]. \tag{179}$$

An isometry is therefore a symmetry of the gauge-fixing term if and only if it preserves u. With $u^\mu = -H\eta\,\delta^\mu_0$ and Eq. (175),

$$[\xi,u]^\mu = -H\big(\xi^0\delta^\mu_0 - \eta\,\partial_0\xi^\mu\big) = \big(0,\ 2H\eta^2\,b^i\big), \tag{180}$$

TABLE 23. Reduction of candidate counterterm densities (each multiplied by κ^2) to the basis (183), in general D, up to total derivatives. All fifteen candidates of the form (182), including these eleven, were also reduced by computing their Euler–Lagrange operators at $D = 4, 5, 6$; all agree.

Candidate	Reduction ($\simeq$)
$a^D(a^{-2}\partial_0^2\varphi)^2$	$B_1 - 2B_3 + B_4 + (D-2)(B_2 - B_5)$
$a^D(a^{-2}\partial_0^2\varphi)(a^{-2}\nabla^2\varphi)$	$-B_3 + B_4 - \frac{1}{2}(D-2)(D-3)B_5$
$a^{D-4}(\nabla^2\varphi)^2$, $a^{D-4}(\partial_i\partial_j\varphi)^2$	B_4
$a^{D-4}(\partial_0\partial_i\varphi)^2$	$-B_3 + B_4 - (D-3)B_5$
$a^{D-4}\partial_0\varphi\,\partial_0^3\varphi$	$-B_1 + 2B_3 - B_4 - \big[(D-2) + \frac{1}{2}(D-3)(D-4)\big](B_2 - B_5)$
$Ha^{D-3}\partial_0\varphi\,\partial_0^2\varphi$	$\frac{1}{2}(D-3)(B_2 - B_5)$
$Ha^{D-3}\partial_0\varphi\,\nabla^2\varphi$	$\frac{1}{2}(D-3)B_5$
$Ha^{D-3}\partial_i\varphi\,\partial_0\partial_i\varphi$	$-\frac{1}{2}(D-3)B_5$
$Ha^{D-1}\partial_0\varphi\,\Box\varphi$	$\frac{1}{2}(D-1)B_2 - B_5$
$H^2a^{D-2}(\partial_0\varphi)^2$	$-B_2 + B_5$

because the only η-dependence of ξ^i is in $-(x\cdot x)b^i$. Classes (i)–(iii) are symmetries of $\mathcal{L}_{\rm GF}$, and class (iv) is broken. The explicit breaking is

$$\Delta_\mu = -\big(4\,h_{\mu i}b^i - 2\,\delta^i_\mu b_i\,h^\rho{}_\rho\big), \tag{181}$$

with the graviton trace $h^\rho{}_\rho = \eta^{\rho\sigma}h_{\rho\sigma}$, independent of H. The breaking is real: on the gauge surface $F_\mu = 0$ one has $\delta F_\mu = \Delta_\mu \neq 0$, and the change of the action is not a boundary term, because its Euler–Lagrange derivatives with respect to h_{00} and h_{12} are nonzero (checked symbolically in $D = 3$). Geometrically, (i)–(iii) map the slices $\eta = \text{const}$ into themselves, whereas (iv) bends them, since $\eta' = \eta/\Omega$ depends on $\vec{x}$.[5]

C. Noninvariant counterterms

This subsection corresponds to (I.136)–(I.145). We assume, as Paper I does, that dimensional regularization preserves the symmetries of the gauge-fixed action, so that the divergences and the counterterms respect (i)–(iii). Translations forbid explicit x^i dependence and rotations require contracted spatial indices. Under a dilatation $d^Dx \to s^Dd^Dx$, $a \to a/s$ and $\partial_\mu \to s^{-1}\partial_\mu$, so the invariant building blocks are $a^{-1}\partial_\mu$ and H, with an overall measure factor a^D. With the shift symmetry and the dimension counting, every allowed counterterm density has the form

$$\mathcal{L}_{\rm ct} = \kappa^2 a^D\,Q\big(a^{-1}\partial_0,\ a^{-1}\partial_i,\ H\big)[\varphi,\varphi], \tag{182}$$

where Q is a rotation-invariant polynomial of total weight four, bilinear in the two fields, with each field differentiated at least once. (Following Paper I, the scalar is written φ in this subsection.) We reduce all such densities to the basis

$$\begin{aligned} &B_1 \equiv a^D(\Box\varphi)^2, \qquad B_2 \equiv H^2a^{D-2}\eta^{\mu\nu}\partial_\mu\varphi\,\partial_\nu\varphi, \quad B_3 \equiv a^{D-2}\,\Box\varphi\,\nabla^2\varphi,\\ &B_4 \equiv a^{D-4}(\nabla^2\varphi)^2, \quad B_5 \equiv H^2a^{D-2}\,\nabla\varphi\cdot\nabla\varphi, \end{aligned} \tag{183}$$

of which B_1 and B_2 are the invariant counterterms (172). The reduction tools are the identities (cf. (I.136)–(I.143); $\partial_0 a = Ha^2$)

$$\frac{1}{a^2}\partial_0^2 = -\Box - (D-2)\frac{H}{a}\partial_0 + \frac{\nabla^2}{a^2}, \qquad \frac{(\partial_0\varphi)^2}{a^2} = -\partial_\mu\varphi\,\partial_\nu\varphi\,g^{\mu\nu} + \frac{1}{a^2}\nabla\varphi\cdot\nabla\varphi, \tag{184}$$

$$Ha^{D-1}\partial_0\varphi\,\Box\varphi \simeq -Ha^{D-3}\partial_\mu\partial_0\varphi\,\partial_\nu\varphi\,\eta^{\mu\nu} - H^2a^{D-2}(\partial_0\varphi)^2 \simeq \tfrac{1}{2}(D-1)B_2 - B_5, \tag{185}$$

$$Ha^{D-3}\partial_0\varphi\,\nabla^2\varphi \simeq -\tfrac{1}{2}Ha^{D-3}\partial_0(\nabla\varphi\cdot\nabla\varphi) \simeq \tfrac{1}{2}(D-3)B_5. \tag{186}$$

[5] The term $k\ln(aa')$ of the scalar propagator shifts by a constant under a dilatation. This comes from the infrared state of the massless minimally coupled scalar, not from the gauge functional; a constant drops out under any derivative and does not affect the ultraviolet divergences.

In each step marked $\simeq$ the discarded total derivative was written out explicitly; for example, the first step of Eq. (185) discards $\partial_\mu\big(Ha^{D-3}\partial_0\varphi\,\partial^\mu\varphi\big)$. The second identity of Eq. (184) gives $H^2a^{D-2}(\partial_0\varphi)^2=-B_2+B_5$. Our Table 23 lists the resulting general-D reductions of eleven candidate densities.

Every allowed density lies in $\mathrm{span}\{B_1,\dots,B_5\}$, and the five are independent, because their Euler–Lagrange operators are linearly independent. Removing the invariant B_1 and B_2 leaves exactly three noninvariant counterterms (cf. (I.144)),

$$\tfrac12\kappa^2a^{D-2}\,\Box\varphi\,\nabla^2\varphi,\qquad \tfrac12\kappa^2a^{D-4}\,\nabla^2\varphi\,\nabla^2\varphi,\qquad -\tfrac12\kappa^2H^2a^{D-2}\,\nabla\varphi\cdot\nabla\varphi, \tag{187}$$

as printed. General-D formulas were derived for the eleven densities of our Table 23; the identities (184)–(186) were verified with symbolic D, while the completeness of the basis (fifteen candidates) and the independence of $B_1,\dots,B_5$ were verified by code at $D=4,5,6$ only.

Flat-space limit. For $H\to0$ at fixed comoving time, $a\to1$ and $E_\mu\to0$ in Eq. (178): F_μ becomes the flat de Donder vector and the gauge-fixed theory becomes Poincaré invariant. Assuming, as Paper I does, that the ultraviolet divergences depend polynomially on H, the coefficients of noninvariant counterterms must vanish in this limit. The first two terms of Eq. (187) have dimensionless coefficients and tend to non-Poincaré-invariant densities, so they are excluded; $B_5\propto H^2$ survives. The single noninvariant counterterm and its vertex are therefore (cf. (I.145))

$$-\tfrac12\alpha_3\kappa^2H^2a^D\,\frac{\nabla\varphi}{a}\cdot\frac{\nabla\varphi}{a}\;\longrightarrow\; i\alpha_3\kappa^2H^2a^{D-2}\,\nabla^2\delta^D(x-x'), \tag{188}$$

exactly as printed. This also confirms the two statements after (I.145). No local term $Ha^{D-3}\nabla^2\partial_0\delta^D$ can arise: the only candidate, $\int c(\eta)\,\partial_0\varphi\,\nabla^2\varphi$, has the Euler–Lagrange operator $-c'(\eta)\nabla^2\varphi$, in which the time derivative has dropped out. No divergence $H^3a^{D-1}\partial_0\delta^D$ can arise either, because it would require an undifferentiated field, which the shift symmetry forbids. The checks of this subsection and the previous one are in `check_126_135.py` and `check_136_145.py` [17].

D. Divergences of the normal contributions

This subsection corresponds to (I.146)–(I.159) and Tables VI–IX. The 4-point contribution and the local 3-point contribution are already local. The normal 3-point contributions of Table IV contain the powers

$$z^{D-1}=\Big(\frac4y\Big)^{D-1}\qquad\text{and}\qquad z^{D-2}=\Big(\frac4y\Big)^{D-2}, \tag{189}$$

which are not integrable at $D=4$ (cf. (I.146)): y^s is locally integrable only for $s>-\frac D2$, so the first diverges quadratically and the second logarithmically. We write c_1 and c_2 for the coefficients of z^{D-1} and z^{D-2} in Table IV, in units of $\kappa^2H^{2D-4}\Gamma^2(\frac D2)/(4\pi)^D$, for each external operator ((I.104)–(I.113); general-D forms in our Table 24). For the γ_2 operator we use $c_2=+\frac{q}{4(D-1)}$, $q\equiv D^3-3D^2-4D+8$, as derived in Sec. V F; Table IV prints $-\frac{q}{4(D-1)}$, a misprint (C15) that the paper did not carry into its later tables (see below). Following the partial-integration technique used in Paper I [20], we extract d'Alembertians until the powers become integrable.

The key identity. Let $\mathcal{L}_y[f]\equiv(4y-y^2)f''+D(2-y)f'$, so that the pointwise part of (I.93) is $\Box f(y)=H^2\mathcal{L}_y[f]$. In distributional derivatives a delta function arises only from the component of f proportional to $y^{1-D/2}$; all other powers give zero by analytic continuation. The coefficient follows from the propagator equation $a^D\Box\,i\Delta_A=i\delta^D(x-x')$ and the leading term $\frac{H^{D-2}}{(4\pi)^{D/2}}\Gamma(\frac D2-1)z^{D/2-1}$ of $i\Delta_A$. The result is (cf. (I.147))

$$\Box f(y)=H^2\mathcal{L}_y[f]+\mathrm{Res}\Big[y^{\frac D2-2}f\Big]\,\frac{4\pi^{D/2}H^{2-D}}{\Gamma(\frac D2-1)}\,\frac{i\,\delta^D(x-x')}{\sqrt{-\hat g}}, \tag{190}$$

where Res denotes the coefficient of $1/y$, and $\sqrt{-\hat g}=a^D$ is the background determinant (written $\sqrt{-g}$ in Paper I). For powers, $\mathcal{L}_y[z^q]=q\big[\{4(q+1)-2D\}/y+(D-q-1)\big]z^q$, with $z^q/y=\frac14z^{q+1}$. For $q=D-2$ and $q=D-3$ the residue vanishes for generic D; for $q=\frac D2-1$ the first bracket vanishes and the residue is $4^{D/2-1}$. This gives (cf. (I.148)–(I.150))

$$z^{D-1}=\frac{2}{(D-2)^2}\,\frac{\Box}{H^2}z^{D-2}-\frac{2}{D-2}\,z^{D-2},\qquad z^{D-2}=\frac{2}{(D-3)(D-4)}\,\frac{\Box}{H^2}z^{D-3}-\frac{4}{D-4}\,z^{D-3}, \tag{191}$$

$$0=\frac{\Box}{H^2}z^{\frac D2-1}-\frac D2\Big(\frac D2-1\Big)z^{\frac D2-1}-C_\delta\,\frac{i\,\delta^D(x-x')}{a^D},\qquad C_\delta\equiv\frac{(4\pi)^{D/2}H^{-D}}{\Gamma(\frac D2-1)}. \tag{192}$$

TABLE 24. Table VI of Paper I, derived from Table IV (paper convention), with $q \equiv D^3 - 3D^2 - 4D + 8$; (c_1, c_2) are the Table IV coefficients after eliminating γ_3 and ϵ_3. The operators δ and ζ have $c_1 = c_2 = 0$. Units as in Eq. (196). Every row agrees with the printed Table VI.

Operator	(c_1, c_2)	Table VI coefficient
$\alpha = (aa')^D \Box^2$	$\big(0, \frac{D}{(D-1)(D-2)^2}\big)$	$\frac{D}{(D-1)(D-2)}$
$\beta = (aa')^{D-1}(a^2 + a'^2) H^2 \Box$	$\big(-\frac{D}{4(D-1)}, -\frac{q}{4(D-1)(D-2)}\big)$	$-\frac{D}{2(D-1)(D-2)}\frac{\Box}{H^2} - \frac{(D+2)(D-4)}{4}$
$\gamma_1 = (aa')^D H^4$	$\big(-\frac{D^2}{4}, -\frac{q}{4}\big)$	$-\frac{D^2}{2(D-2)}\frac{\Box}{H^2} - \frac{D^4-5D^3+16D-16}{4}$
$\gamma_2 = (aa')^{D-1}(a^2 + a'^2) H^4$	$\big(0, +\frac{q}{4(D-1)}\big)$	$+\frac{(D-2)q}{4(D-1)}$
$\epsilon_1 = (aa')^{D-1} H^2 \nabla^2$	$\big(0, \frac{D^2-6D+4}{2(D-1)(D-2)}\big)$	$\frac{D^2-6D+4}{2(D-1)}$
$\epsilon_2 = (aa')^{D-2}(a^2 + a'^2) H^2 \nabla^2$	$\big(0, \frac{1-2D}{(D-1)(D-2)}\big)$	$\frac{1-2D}{D-1}$

Segregating the pole. Subtracting $\frac{2}{(D-3)(D-4)}$ times Eq. (192) from the second identity of Eq. (191) gives (cf. (I.151))

$$z^{D-2} = \frac{2}{(D-3)(D-4)}\Big\{C_\delta \frac{i\,\delta^D}{a^D} + \frac{\Box}{H^2}\big[z^{D-3} - z^{\frac{D}{2}-1}\big]\Big\} - \frac{4}{D-4}\Big\{z^{D-3} - \frac{D(D-2)}{8(D-3)}\, z^{\frac{D}{2}-1}\Big\}. \tag{193}$$

At $D = 4$, z^{D-3} and $z^{D/2-1}$ are the same function, so both braces are $O(D-4)$ and the pole sits entirely on the delta function. Expanding in $D-4$ gives (cf. (I.152), (I.153))

$$z^{D-2} = \frac{iH^{-D}(4\pi)^{D/2}}{(D-3)(D-4)\Gamma(\frac{D}{2})}\,(D-2)\,\frac{\delta^D(x-x')}{a^D} - \frac{\Box}{H^2}\Big\{\frac{4}{y}\ln\frac{y}{4}\Big\} + 2\,\frac{4}{y}\ln\frac{y}{4} - \frac{4}{y} + O(D-4), \tag{194}$$

$$\begin{aligned} z^{D-1} = {} & \frac{iH^{-D}(4\pi)^{D/2}}{(D-3)(D-4)\Gamma(\frac{D}{2})}\Big\{\frac{2}{D-2}\frac{\Box}{H^2} - 2\Big\}\frac{\delta^D(x-x')}{a^D} - \frac{1}{2}\frac{\Box^2}{H^4}\Big\{\frac{4}{y}\ln\frac{y}{4}\Big\} \\ & + \frac{\Box}{H^2}\Big\{2\,\frac{4}{y}\ln\frac{y}{4} - \frac{1}{2}\,\frac{4}{y}\Big\} - 2\,\frac{4}{y}\ln\frac{y}{4} + \frac{4}{y} + O(D-4), \end{aligned} \tag{195}$$

in agreement with Paper I. These are equalities of distributions: $[z^{D-3} - z^{D/2-1}]/(D-4)$ converges to $\frac{1}{2} z \ln z$ in $L^1_{\rm loc}$, and the outer d'Alembertians act distributionally. The flat Euclidean analogue of Eqs. (191)–(193), $r^{-2(D-2)} = [\nabla^2(r^{-2(D-3)} - r^{2-D}) - (D-2)\Omega_D \delta^D]/[2(D-3)(D-4)]$ with $\Omega_D = 2\pi^{D/2}/\Gamma(\frac{D}{2})$, was checked exactly in D against the test function e^{-r^2}.

Table VI: local divergent terms. Keeping the local parts of Eqs. (194)–(195), each operator contributes

$$c_1\Big\{\frac{2}{D-2}\frac{\Box}{H^2} - 2\Big\} + c_2\,(D-2) \qquad \text{times} \qquad \frac{i\kappa^2 H^{D-4}}{(4\pi)^{D/2}}\,\frac{\Gamma(\frac{D}{2})}{(D-3)(D-4)}\,\frac{\delta^D(x-x')}{a^D}, \tag{196}$$

after the redundant operators have been eliminated with $\gamma_3 = 2\gamma_1 + \gamma_2$ and $\epsilon_3 = 2\epsilon_1 + \epsilon_2$, which follow from $(a+a')^2 = a^2 + a'^2 + 2aa'$. Our Table 24 gives the result; it agrees with the printed Table VI row by row. The printed γ_2 row, $+\frac{(D-2)q}{4(D-1)}$, follows from $c_2 = +\frac{q}{4(D-1)}$; the sign printed in Table IV would give the opposite sign. So Paper I used the correct sign in its calculation, and the sign in Table IV is only a misprint.

Table VII: finite nonlocal terms. At $D = 4$, with $x = y/4$, the nonlocal parts of Eqs. (194) and (195) are

$$N_2 = -\frac{\Box}{H^2}\Big[\frac{\ln x}{x}\Big] + \frac{2\ln x}{x} - \frac{1}{x}, \qquad N_1 = -\frac{1}{2}\frac{\Box^2}{H^4}\Big[\frac{\ln x}{x}\Big] + \frac{\Box}{H^2}\Big[\frac{2\ln x}{x} - \frac{1}{2x}\Big] - \frac{2\ln x}{x} + \frac{1}{x}, \tag{197}$$

and each operator contributes $c_1 N_1 + c_2 N_2$ with its $D = 4$ values of (c_1, c_2). The results (our Table 25) agree with the printed Table VII. The header of Table VII reads "Coefficient of $i\kappa^2 H^4/(4\pi)^4$"; the factor i is a misprint (C16), since the units of Table IV carry no i, the nonlocal parts of (I.152)–(I.153) are real, and the header of Table X, which contains Table VII, has no i.

Converting a' into a. To compare the local terms with the counterterms, the factors a' of the external operators are converted into a with the delta function, and all scale factors are moved to the left. Acting on a test function, $\int d^D x'\, a^p(x)\, a'^q\, \Box^n\big[a^{-D}\delta^D(x-x')\big]\phi(x') = a^p \Box^n\big[a^{q-D}\phi\big](x)$, so a'^q becomes a^q *inside* the d'Alembertians. For α, γ_1,

TABLE 25. Table VII of Paper I, derived (paper convention); coefficients of $\kappa^2 H^4/(4\pi)^4$, $x = y/4$. All rows agree with the printed table. In the β row the terms without $\Box$ cancel.

Operator	$(c_1, c_2)_{D=4}$	Finite nonlocal contribution
α	$(0, \frac{1}{3})$	$\frac{\Box}{H^2}\big[-\frac{\ln x}{3x}\big] + \frac{2\ln x}{3x} - \frac{1}{3x}$
β	$(-\frac{1}{3}, -\frac{1}{3})$	$\frac{\Box^2}{H^4}\big[\frac{\ln x}{6x}\big] + \frac{\Box}{H^2}\big[-\frac{\ln x}{3x} + \frac{1}{6x}\big]$
γ_1	$(-4, -2)$	$\frac{\Box^2}{H^4}\big[\frac{2\ln x}{x}\big] + \frac{\Box}{H^2}\big[-\frac{6\ln x}{x} + \frac{2}{x}\big] + \frac{4\ln x}{x} - \frac{2}{x}$
γ_2	$(0, \frac{2}{3})$	$\frac{\Box}{H^2}\big[-\frac{2\ln x}{3x}\big] + \frac{4\ln x}{3x} - \frac{2}{3x}$
ϵ_1	$(0, -\frac{1}{3})$	$\frac{\Box}{H^2}\big[\frac{\ln x}{3x}\big] - \frac{2\ln x}{3x} + \frac{1}{3x}$
ϵ_2	$(0, -\frac{7}{6})$	$\frac{\Box}{H^2}\big[\frac{7\ln x}{6x}\big] - \frac{7\ln x}{3x} + \frac{7}{6x}$

TABLE 26. Table VIII of Paper I, derived (paper convention): local normal contributions to the three counterterm structures. All entries are multiplied by $\frac{i\kappa^2 H^{D-4}}{(4\pi)^{D/2}} \frac{\Gamma(D/2)}{(D-3)(D-4)}$. Rows and totals agree with the printed table.

From	$a^D\Box^2\delta^D$	$a^D H^2\Box\delta^D$	$a^{D-2}H^2\nabla^2\delta^D$
α	$\frac{D}{(D-1)(D-2)}$	0	0
$\beta\,\Box/H^2$ (Eq. (198))	$-\frac{D}{(D-1)(D-2)}$	$\frac{6D}{(D-1)(D-2)}$	$-\frac{4D}{(D-1)(D-2)}$
β (Eq. (199))	0	$-\frac{(D+2)(D-4)}{2}$	0
$\gamma_1\,\Box/H^2$	0	$-\frac{D^2}{2(D-2)}$	0
ϵ_1	0	0	$\frac{D^2-6D+4}{2(D-1)}$
ϵ_2	0	0	$\frac{2-4D}{D-1}$
Total	0	$\frac{(D-4)(-D^3+D-4)}{2(D-1)(D-2)}$	$\frac{D^3-16D^2+28D-16}{2(D-1)(D-2)}$

γ_2, ϵ_1, ϵ_2 and $\gamma_1\Box/H^2$ this is immediate (for example $\gamma_2 \to 2a^D H^4$ and $\epsilon_2 \to 2a^{D-2}H^2\nabla^2$). For the two operators that contain $(a^2 + a'^2)$ together with $\Box$ we find, using $\Box a = -DH^2 a$ and $\Box(1/a) = (D-2)H^2/a$ (cf. (I.154)–(I.155)),

$$(aa')^{D-1}(a^2+a'^2)\Box^2\Big[\frac{\delta^D(x-x')}{a^D}\Big] = \big[2a^D\Box^2 - 12H^2a^D\Box + 8a^{D-2}H^2\nabla^2 + 2(D^2-2D+2)H^4a^D\big]\delta^D(x-x'), \tag{198}$$

$$(aa')^{D-1}(a^2+a'^2)H^2\Box\Big[\frac{\delta^D(x-x')}{a^D}\Big] = 2a^D\big(H^2\Box - H^4\big)\delta^D(x-x'), \tag{199}$$

as printed; both were verified symbolically for general D on an arbitrary test function. The term $8a^{D-2}H^2\nabla^2$ has the form of the noninvariant counterterm (188).

Table VIII and $\beta_1 = 0$. Converting each row of Table VI in this way gives our Table 26, which agrees with the printed Table VIII row by row and in the totals. The total of the $a^D\Box^2$ column vanishes. Paper I explains this by the vanishing of this counterterm in flat space, and we confirmed that statement independently. For $H \to 0$ the $a^D\Box^2$ coefficient must equal the divergence of the flat-space one-loop self-energy. In flat space the 4-point contribution vanishes in dimensional regularization, and in the 3-point contribution the vertex $V^{\alpha\beta} \propto p^{(\alpha}q^{\beta)} - \frac{1}{2}\eta^{\alpha\beta}p\cdot q$ (with p the external and q the loop momentum) contracted with the de Donder tensor $P = \frac{1}{2}(\eta\eta + \eta\eta) - \frac{1}{D-2}\eta\eta$ gives $V\cdot P\cdot V = \frac{1}{2}p^2q^2$, so the self-energy is proportional to $\frac{1}{2}p^2\int d^Dq/(p-q)^2 = 0$, a scaleless integral.

Table IX and shift symmetry. The terms proportional to $a^D H^4\delta^D$ (our Table 27) agree with the printed Table IX. A counterterm $H^4a^D\phi^2$ is forbidden by the shift symmetry, so their total must be finite. It is: the $D=4$ values of the four rows are $-\frac{20}{3} + 0 + 4 + \frac{8}{3} = 0$, so the factor $(D-4)$ of the total cancels the pole of the units. With the γ_2 sign printed in Table IV the sum would be $-\frac{20}{3} + 4 - \frac{8}{3} = -\frac{16}{3}$, a forbidden divergence. This confirms the γ_2 sign independently of its direct computation.

Since the total is finite we may take $D = 4$ (cf. (I.157)–(I.158)):

$$\text{Table IX} = \frac{i\kappa^2 H^{D-4}}{(4\pi)^{D/2}}\,\frac{(-D^4+5D^3-16D+16)\,\Gamma(\frac{D}{2})}{4(D-2)(D-3)}\,a^D H^4\delta^D(x-x') \xrightarrow{D=4} \frac{\kappa^2H^4}{8\pi^2}\,ia^4\delta^4(x-x'). \tag{200}$$

TABLE 27. Table IX of Paper I, derived (paper convention): coefficients of $a^D H^4 \delta^D(x-x')$, in the units of our Table 26; $q \equiv D^3 - 3D^2 - 4D + 8$. Rows and total agree with the printed table.

From	Coefficient	Origin
$\beta\,\Box/H^2$	$-\frac{D(D^2-2D+2)}{(D-1)(D-2)}$	H^4 term of Eq. (198)
β	$\frac{(D+2)(D-4)}{2}$	H^4 term of Eq. (199)
γ_1	$-\frac{D^4-5D^3+16D-16}{4}$	$\gamma_1 \to a^D H^4$
γ_2	$\frac{(D-2)q}{2(D-1)}$	$\gamma_2 \to 2a^D H^4$
Total	$\frac{(D-4)(-D^4+5D^3-16D+16)}{4(D-2)}$	finite

TABLE 28. Destination of the parts of the regulated self-mass-squared.

Part	Local ($\to \beta_i$)	Finite nonlocal ($\to$ Table X)
4-point, (I.44)	$\hat{\pi}$ terms; constants ∓ 2 from (I.40); $3L$, $-4L$, $9H^2$ terms (cancel)	—
3-point local, (I.59)	$\hat{\pi}$ terms and finite constants	—
3-point logarithm, (I.67)	L and $9H^2$ terms (cancel)	$2H(aa')^2(a\partial_0 + a'\partial'_0)\nabla^2 A \to$ Table III
Normal, Table IV	Table VI $\to$ Table VIII; Table IX $\to$ Eq. (202)	Table VII; Eq. (202)
Normal, Table V	—	Table V

This finite local term is rewritten in nonlocal form. The propagator equation, with $i\Delta_A = A + k\ln(aa')$ and $\Box \ln a = -(D-1)H^2$, gives (cf. (I.156))

$$ia^D\delta^D(x-x') = (aa')^D\big\{\Box A(y) - (D-1)kH^2\big\}, \tag{201}$$

where a^D may be replaced by $(aa')^D$ because the bracket is supported at coincidence, and $\Box A$ is distributional. At $D=4$, $k = H^2/8\pi^2$ and $A = \frac{H^2}{16\pi^2}\{\frac{4}{y} - 2\ln\frac{y}{4} - 1\}$, so (cf. (I.159))

$$\frac{\kappa^2 H^4}{8\pi^2}\, ia^4\delta^4(x-x') = \frac{\kappa^2 H^4}{(4\pi)^4}\Big\{(aa')^4 H^2 \Box\Big[\frac{2}{x} - 4\ln x\Big] - 12\,(aa')^4 H^4\Big\}. \tag{202}$$

Equations (189)–(202) and Tables VI–IX were checked by `check_146_IX.py` [17].

A possible extra local term. The question raised in Sec. V H, whether the outer d'Alembertians of the operators α and δ count the local term (I.54)+(I.58) a second time, bears on this subsection. In Paper I the divergent local terms come only from Table IV through Eqs. (194)–(195); in the finite functions of Table V the relevant component has combined with the pole-carrying z^{D-3} terms into finite logarithms. Any double counting would then be a constant $c(D)$, which may contain a pole, times $(a^D\Box - a^{D-2}\nabla^2)\delta^D(x-x')$, with no $\ln a$, of the form of the counterterms (173) and (188); its pole and its finite part would be absorbed into α_2 and α_3. Whether the outer-operator reduction produces other local differences at coincidence has not been settled (Sec. V H); the point is only partially resolved. In particular, a finite local term of mass type, $H^4 a^D \delta^D(x-x')$, would not be of counterterm form; its consequences for the late-time analysis are discussed in Secs. VII G and VIII D.

E. Local terms: the coefficients β_i

This subsection corresponds to (I.160)–(I.165). The regulated self-mass-squared is the sum of the 4-point contribution and the local, logarithm and normal 3-point contributions. Our Table 28 shows where each piece goes. The terms $3L\,a^D\Box$, $-4L\,a^{D-2}\nabla^2$ and $9H^2a^D$ (times $i\delta^D$) of (I.44) are not of counterterm form. They cancel against the local part of (I.67) for general D, up to an $O(D-4)$ remainder (Sec. IV); both carry the reversed sign in Paper I, so the cancellation holds in either convention.

What remains has counterterm form (paper convention; cf. (I.160)):

$$-iM^2_{\rm reg}(x;x') = i\kappa^2 a^D\Big(\beta_1\Box^2 + \beta_2 H^2\Box + \beta_3 H^2\frac{\nabla^2}{a^2}\Big)\delta^D(x-x') + \text{Table X} + O(D-4). \tag{203}$$

TABLE 29. The counterterm coefficients β_i of (I.161)–(I.165), in units of $H^{D-4}/(4\pi)^{D/2}$, up to $O(D-4)$. The shifts of β_2 and β_3 are B3 and B4.

	Printed (paper convention)	Corrected (paper convention)	Corrected (definition sign)
β_1	0	0	0
β_2	$-\frac{61}{3}$	$-\frac{49}{3}$	$+\frac{49}{3}$
β_3	$-\frac{4}{D-4}+\frac{58}{3}+2\gamma_{\rm E}$	$-\frac{4}{D-4}+\frac{52}{3}+2\gamma_{\rm E}$	$+\frac{4}{D-4}-\frac{52}{3}-2\gamma_{\rm E}$

Paper I prints $\beta_2\Box+\beta_3\nabla^2/a^2$ without the factors H^2 (C17).[6]

β_1. The structure $\Box^2$ occurs only in the first column of Table VIII, whose total vanishes. Hence (cf. (I.161))

$$\beta_1 = 0. \tag{204}$$

β_2 and β_3 from the printed inputs. The coefficient of $i\kappa^2H^2a^D\Box\delta^D$ receives $k[\frac{1}{4}D(D-1)\hat\pi-2]$ from (I.44), $k[-\frac{D-1}{D-3}\hat\pi+2]$ from (I.59), and the second column of Table VIII. The coefficient of $i\kappa^2H^2a^{D-2}\nabla^2\delta^D$ receives $k[-D\hat\pi+2]$ from (I.44), $k[\frac{D-1}{D-3}\hat\pi-\frac{7}{2}]$ from (I.59), and the third column of Table VIII. In both cases the $\hat\pi$ terms combine, and the constants give 0 and $-\frac{3}{2}k$ respectively; the latter is written by its $D=4$ value, which is legitimate because it multiplies no pole. With $(D-1)\Gamma(D-1)=\Gamma(D)$ and $\Gamma(\frac{D}{2})/(D-2)=\frac{1}{2}\Gamma(\frac{D}{2}-1)$ we obtain (cf. (I.162)–(I.165))

$$\beta_2 = \frac{H^{D-4}}{(4\pi)^{D/2}}\Big\{\frac{(-D^3+D-4)\,\Gamma(\frac{D}{2}-1)}{4(D-1)(D-3)}-\frac{(D+1)(D-4)\,\Gamma(D)\,\pi\cot(\frac{\pi D}{2})}{4(D-3)\,\Gamma(\frac{D}{2})}\Big\}=\frac{H^{D-4}}{(4\pi)^{D/2}}\Big\{-\frac{61}{3}+O(D-4)\Big\}, \tag{205}$$

$$\begin{aligned}\beta_3 &= \frac{H^{D-4}}{(4\pi)^{D/2}}\Big\{\frac{(D^3-16D^2+28D-16)\,\Gamma(\frac{D}{2})}{2(D-1)(D-2)(D-3)(D-4)}+\frac{(D^2-4D+1)\,\Gamma(D-1)\,\pi\cot(\frac{\pi D}{2})}{(D-3)\,\Gamma(\frac{D}{2})}-3\Big\}\\ &= \frac{H^{D-4}}{(4\pi)^{D/2}}\Big\{-\frac{4}{D-4}+\frac{58}{3}+2\gamma_{\rm E}+O(D-4)\Big\},\end{aligned} \tag{206}$$

where $(D-4)\pi\cot(\frac{\pi D}{2})\to 2$ and $\gamma_{\rm E}$ enters through $\psi(2)=1-\gamma_{\rm E}$ and $\psi(3)=\frac{3}{2}-\gamma_{\rm E}$. These reproduce the printed (I.161)–(I.165) exactly (`check_160_167.py` [17]), so Paper I did use the printed (I.44) and (I.59). Within Paper I's treatment of the outer operators of the normal terms, the coefficient β_2 is finite and the only divergence is in the de Sitter-breaking coefficient β_3, consistent with Sec. VI C. A double counting of the kind discussed at the end of Sec. VI D, if present, would shift poles and finite parts of β_2 and β_3 by counterterm-form amounts, absorbed into α_2 and α_3.

Corrected β_2 and β_3. Two finite errors in these inputs were found earlier (B1 and B2 of our Table 39). With the correct (I.40) (Sec. III), the constants ∓2 are absent from (I.44); this adds $+2\bar k\to+4$ to β_2 and $-2\bar k\to-4$ to β_3. The constant of the ∇^2 term of (I.59) is $-\frac{5}{2}$, the sum of the paper's own (I.57) and (I.58), not $-\frac{7}{2}$ (Sec. IV); this adds $+\bar k\to+2$ to β_3. Equivalently, the bracket of Eq. (205) acquires $+2\bar k$, and the constant -3 in Eq. (206) becomes -5. The corrected values are compared with the printed ones in our Table 29. Only the finite parts change; they are absorbed into the arbitrary finite counterterms, and the pole is unchanged by these two corrections. The same shifts apply to the counterterm coefficients α_2 and α_3 quoted in (II.8)–(II.9) (B5, B6).

F. Table X

This subsection corresponds to Table X and the text before (I.160). Table X collects every finite nonlocal term: Table III (the nonlocal logarithm term of (I.67), Sec. V B), Table V (the finite normal terms, as printed after the audit of Sec. V E), Table VII and Eq. (202). The redundant operators are eliminated with $\gamma_3=2\gamma_1+\gamma_2$ and $\epsilon_3=2\epsilon_1+\epsilon_2$; the row $(aa')^2(a+a')^2H^2\nabla^2$ of Table III is ϵ_3. Combined operators of Table VII are read at $D=4$, for example

[6] The β_i of (I.161)–(I.165) all have the dimension $H^{D-4}/(4\pi)^{D/2}$, the counterterm vertices (I.131) and (I.145) carry H^2, and (I.166) requires $\alpha_i=-\beta_i+\Delta\alpha_i$. The factors H^2 are therefore required; the same misprint occurs in (I.167), (I.170), (II.10) and (II.34) (C18, C22, C20, C21).

TABLE 30. Sources of the rows of Table X of Paper I (paper convention, coefficients of $\kappa^2 H^4/(4\pi)^4$). "VII α" is the α row of Table VII, "V α" the α row of Table V, and so on. The last column compares the total of the listed sources with the printed Table X.

Row	Sources	vs. printed
$(aa')^4\Box^3/H^2$	VII α	equal
$(aa')^4\Box^2$	VII α; VII γ_1 ($\Box^2/H^4$ part); V α	equal
$(aa')^4H^2\Box$	VII γ_1 ($\Box/H^2$ part); Eq. (202): $\frac{2}{x}-4\ln x$	equal
$(aa')^4H^4$	VII γ_1; V γ_1; 2×V γ_3; Eq. (202): -12	equal
$(aa')^3(a^2+a'^2)\Box^3/H^2$	VII β	equal
$(aa')^3(a^2+a'^2)\Box^2$	VII β	equal
$(aa')^3(a^2+a'^2)H^2\Box$	VII γ_2 ($\Box/H^2$ part); V β	equal
$(aa')^3(a^2+a'^2)H^4$	VII γ_2; V γ_2; V γ_3	equal to printed -54; see text
$(aa')^3H^2\nabla^2$	VII ϵ_1; V ϵ_1; 2×V ϵ_3; III: $-\frac{12}{x}+24\ln x$; 2×III ϵ_3: $\frac{8}{x}$	equal
$(aa')^3\nabla^2\Box$	VII ϵ_1 ($\Box/H^2$ part)	equal
$(aa')^2(a^2+a'^2)H^2\nabla^2$	VII ϵ_2; V ϵ_2; V ϵ_3; III ϵ_3: $\frac{4}{x}$	equal
$(aa')^2(a^2+a'^2)\nabla^2\Box$	VII ϵ_2 ($\Box/H^2$ part); V δ	equal (printed twice)
$(aa')^2\nabla^4$	V ζ; III ζ: $8\ln x-16x\ln x$	III omitted in print

$\beta\Box/H^2=(aa')^3(a^2+a'^2)\Box^2$ and $\gamma_1\Box^2/H^4=(aa')^4\Box^2$. In all operators the scale factors stand to the left of the derivatives, which act on the coefficient functions. Our Table 30 lists the sources of each row.

Twelve of the thirteen rows follow exactly from the inputs; the row $(aa')^2(a^2+a'^2)\nabla^2\Box$ is printed twice in Table X, a misprint (C19; Table 1 of Paper II lists it once). Paper I added the first two rows of Table III to the ϵ_1 and ϵ_2 rows; for example $84\ln x=60\ln x+24\ln x$ and $-\frac{16}{x}=(\frac{1}{3}-\frac{26}{3}-\frac{11}{3}-12+8)\frac{1}{x}$. It omitted the third row, which belongs to ζ, although the text states that Table X includes Table III (S2). Paper I prints

$$(aa')^2\nabla^4:\qquad \begin{aligned}&\tfrac{10}{3}\ln x-24x\ln x+24x^2\ln x-36x^2 \quad \text{(printed)},\\ &\tfrac{34}{3}\ln x-40x\ln x+24x^2\ln x-36x^2 \quad \text{(corrected)}.\end{aligned} \tag{207}$$

The difference, $8\ln x-16x\ln x$, is non-analytic.

The row $(aa')^3(a^2+a'^2)H^4$ agrees with the print when Table V is used as printed, but Table V lacks a finite term $72\gamma_2$ (paper convention; S3), found globally in the audit of Tables IV–V (Sec. V G). In Table X this changes the constant of that row from -54 to $+18$. Equivalently, one may add $-12x$ to the δ row of Table V, since $\nabla^2\Box x=-6aa'H^4$ gives $\delta[-12x]=72\gamma_2[1]$, or make the γ_2 row of Table V $-\frac{5}{x}+54$. The pointwise test below fixes only the sum, so which row the term belongs to is a matter of interpretation. The corrected Table X is given in our Table 31, in both conventions; we denote its content by

$$\mathcal{T}(x;x')\equiv\frac{\kappa^2H^4}{(4\pi)^4}\sum_{\mathcal{O}}\mathcal{O}\big[c_{\mathcal{O}}(y/4)\big],\qquad y\equiv y(x;x'), \tag{208}$$

where the sum runs over the thirteen operators $\mathcal{O}$ of our Table 31 and $c_{\mathcal{O}}$ are the paper-convention coefficients, written in the table as functions of $x\equiv y/4$.

Global pointwise test. The two corrections were established by a test that does not depend on how Table X was assembled. At $D=4$ and $x\neq x'$ it compares two quantities, both in the paper convention. The left-hand side is the direct sum $\sum_k P_k$ of the normal 3-point contributions plus the nonlocal term $-\frac{\kappa^2H^2}{8\pi^2}\,2H(aa')^2(a\partial_0+a'\partial_0')\nabla^2A$ of (I.67), evaluated with naive (pointwise) inner derivatives and the actual $D=4$ propagator functions $A=h\{4/y-2\ln(y/4)-1\}$, $B=C=4h/y$, $h\equiv H^2/16\pi^2$; it does not use the outer-operator reduction of the appendix of Paper I. The right-hand side is the thirteen operators of Table X applied pointwise to their coefficient functions. Equation (202) does not enter, since its two terms cancel pointwise ($\mathcal{L}_y[8/y-4\ln(y/4)]=12$), and at $D=4$ the nonlocal parts of Eqs. (194)–(195) reproduce $(4/y)^2$ and $(4/y)^3$ pointwise. The evaluation uses three random rational spacelike pairs of points (seed 29) and 30-digit arithmetic (`check_tableX_global.py` [17]). Our Table 32 shows the residuals. With both corrections the agreement is exact to 30 digits; neither correction alone is enough. The test is numerical, at sample points, and checks only the sum of all rows; it is not a symbolic identity.

TABLE 31. Corrected Table X of Paper I: all finite nonlocal contributions, coefficients of $\kappa^2 H^4/(4\pi)^4$, $x = y/4$. The rows are numbered from 0 as in the derivation notes and the code; these row numbers are used in Sec. VII. The paper-convention column is to be compared with the printed Table X and with Table 1 of Paper II (which lists the same rows in the same order); the definition-sign column is -1 times it. The two changed entries (S3 and S2) are marked with an asterisk: printed $(aa')^3(a^2+a'^2)H^4$ constant -54, printed $(aa')^2\nabla^4$ row $\frac{10}{3}\ln x - 24x\ln x + 24x^2\ln x - 36x^2$. The row $(aa')^2(a^2+a'^2)\nabla^2\Box$ appears once.

#	Operator $\mathcal{O}$	Paper convention, $c_{\mathcal{O}}$	Definition sign, $-c_{\mathcal{O}}$
0	$(aa')^4\Box^3/H^2$	$-\frac{\ln x}{3x}$	$\frac{\ln x}{3x}$
1	$(aa')^4\Box^2$	$\frac{26\ln x}{3x} + \frac{38}{3x} - 6\ln^2 x - 18\ln x$	$-\frac{26\ln x}{3x} - \frac{38}{3x} + 6\ln^2 x + 18\ln x$
2	$(aa')^4H^2\Box$	$-\frac{6\ln x}{x} + \frac{4}{x} - 4\ln x$	$\frac{6\ln x}{x} - \frac{4}{x} + 4\ln x$
3	$(aa')^4H^4$	$\frac{4\ln x}{x} + \frac{18}{x} - 120 - 108\ln x$	$-\frac{4\ln x}{x} - \frac{18}{x} + 120 + 108\ln x$
4	$(aa')^3(a^2+a'^2)\Box^3/H^2$	$\frac{\ln x}{6x}$	$-\frac{\ln x}{6x}$
5	$(aa')^3(a^2+a'^2)\Box^2$	$-\frac{\ln x}{3x} + \frac{1}{6x}$	$\frac{\ln x}{3x} - \frac{1}{6x}$
6	$(aa')^3(a^2+a'^2)H^2\Box$	$-\frac{2\ln x}{3x} + \frac{5}{x} - 18\ln x$	$\frac{2\ln x}{3x} - \frac{5}{x} + 18\ln x$
7	$(aa')^3(a^2+a'^2)H^4$	$\frac{4\ln x}{3x} - \frac{32}{3x} + 18^*$	$-\frac{4\ln x}{3x} + \frac{32}{3x} - 18^*$
8	$(aa')^3H^2\nabla^2$	$-\frac{2\ln x}{3x} - \frac{16}{x} + 84\ln x - 48x\ln x + 96x$	$\frac{2\ln x}{3x} + \frac{16}{x} - 84\ln x + 48x\ln x - 96x$
9	$(aa')^3\nabla^2\Box$	$\frac{\ln x}{3x}$	$-\frac{\ln x}{3x}$
10	$(aa')^2(a^2+a'^2)H^2\nabla^2$	$-\frac{7\ln x}{3x} + \frac{11}{2x} - 12\ln x + 48x\ln x + 12x$	$\frac{7\ln x}{3x} - \frac{11}{2x} + 12\ln x - 48x\ln x - 12x$
11	$(aa')^2(a^2+a'^2)\nabla^2\Box$	$-\frac{17\ln x}{6x} - \frac{49}{6x} + 4\ln^2 x + 10\ln x + 12x\ln x$	$\frac{17\ln x}{6x} + \frac{49}{6x} - 4\ln^2 x - 10\ln x - 12x\ln x$
12	$(aa')^2\nabla^4$	$\left(\frac{34}{3}\ln x - 40x\ln x\right)^* + 24x^2\ln x - 36x^2$	$\left(-\frac{34}{3}\ln x + 40x\ln x\right)^* - 24x^2\ln x + 36x^2$

TABLE 32. Global pointwise test of Table X of Paper I: left-hand side minus the operator sum at three random spacelike pairs. For scale, |LHS| = 7.10, 0.107 and 8.60.

Version of Table X	Point 1	Point 2	Point 3
As printed	-11.52	64.14	-3.526
With the $(aa')^2\nabla^4$ correction only	1.182	65.28	0.0161
With the $(aa')^2\nabla^4$ and $72\gamma_2$ corrections	4×10^{-31}	0	-3×10^{-31}

Physical significance of the two corrections. The $72\gamma_2$ term is independent of y (analytic). In the Schwinger–Keldysh formalism it is identical in the $++$ and $+-$ kernels, so it drops out of the retarded self-mass-squared identically and has no effect on any retarded quantity. The term $(aa')^2\nabla^4(8\ln x - 16x\ln x)$ is non-analytic through $\ln x$; it enters the retarded kernel and changes the physical nonlocal self-mass-squared. Its effect on the late-time mode function is analyzed in Sec. VII: it contributes only terms that fall off at late times (it removes the $\hat{k}^4/a$, $\hat{k}^6/a, \ldots$ tails that the printed row leaves in the source).

G. The renormalized self-mass-squared

This subsection corresponds to (I.166)–(I.167). The counterterm vertices (173) and (188) add $i\kappa^2 a^D[\alpha_1\Box^2 + \alpha_2 H^2\Box + \alpha_3 H^2\nabla^2/a^2]\delta^D$ to $-iM^2_{\rm reg}$. The choice (cf. (I.166))

$$\alpha_i = -\beta_i + \Delta\alpha_i \tag{209}$$

removes the divergence and leaves arbitrary finite constants $\Delta\alpha_i$; this relation holds in either convention. Because $\alpha_i a^D$ cancels $\beta_i a^D$ exactly, the expansion $a^D = a^4[1+(D-4)\ln a + \ldots]$ leaves no $\ln a$ behind, and we may set $D = 4$. In the paper convention, with the factors H^2 restored and the corrected Table X, this is (cf. (I.167))

$$-iM^2_{\rm ren}(x;x')\Big|_{\text{paper conv.}} = i\kappa^2 a^4\Big(\Delta\alpha_1\Box^2 + \Delta\alpha_2 H^2\Box + \Delta\alpha_3 H^2\frac{\nabla^2}{a^2}\Big)\delta^4(x-x') + \mathcal{T}(x;x'). \tag{210}$$

As established in Secs. III and IV, the printed self-mass-squared is -1 times the one defined by Paper I's own (I.168). Both the 4-point part and the 3-point part are reversed, so their relative sign, and the local cancellations used above, are consistent. In the definition sign the regulated result is Eq. (203) with $\beta_i \to -\beta_i$ and Table X $\to -$Table X; with the corrections of Sec. VI E,

$$\beta_1 = 0, \qquad \beta_2 = \frac{H^{D-4}}{(4\pi)^{D/2}}\Big\{\frac{49}{3} + O(D-4)\Big\}, \qquad \beta_3 = \frac{H^{D-4}}{(4\pi)^{D/2}}\Big\{\frac{4}{D-4} - \frac{52}{3} - 2\gamma_{\rm E} + O(D-4)\Big\} \qquad \text{(definition sign)}, \tag{211}$$

and Eq. (209) again removes the divergence. The final result of this section is

$$\boxed{\begin{aligned} -iM^2_{\rm ren}(x;x') = i\kappa^2 a^4\Big(\Delta\alpha_1\Box^2 + \Delta\alpha_2 H^2\Box + \Delta\alpha_3 H^2\frac{\nabla^2}{a^2}\Big)\delta^4(x-x') \\ - \frac{\kappa^2 H^4}{(4\pi)^4}\sum_{\mathcal{O}}\mathcal{O}\big[c_{\mathcal{O}}(y/4)\big] \qquad \text{(definition sign)} \end{aligned}} \tag{212}$$

where the sum runs over the thirteen operators of our Table 31, $c_{\mathcal{O}}$ are its paper-convention coefficients (so the nonlocal part is $-\mathcal{T}$, the definition-sign column), $y \equiv y(x;x')$, and the $\Delta\alpha_i$ are arbitrary finite constants.

Since the $\Delta\alpha_i$ are arbitrary, the sign of the local part has no physical meaning. The nonlocal part has no counterterm freedom, and its sign is physical: it is opposite to the printed one. In the late-time analysis of Paper II this flips every entry of Table 5 (whose total remains zero) and the sign of the particular solution Φ_1. Section VII shows that the non-falling total of the source is zero at late times, so only terms that fall off change sign, and the late-time conclusion of Paper II is unchanged.

Comparison with Sec. IV of Paper I. Equations (I.126)–(I.159), Tables VI–IX and the structure of (I.160) and (I.167) are confirmed, with the misprints noted above: the omission of two (shift-forbidden) terms from the list (I.128) (C14), the sign of the γ_2 entry of Table IV (C15), the factor i in the header of Table VII (C16), the missing factors H^2 in (I.160) and (I.167) (C17, C18), and the duplicated row of Table X (C19). We also noted a notational remark without effect: the dilatation factor (k in (I.134)) must be positive. The coefficients (I.161)–(I.165) are reproduced exactly from the printed inputs; with the corrected (I.40) and (I.59) (B1, B2), the finite parts of (I.163) and (I.165) change (B3, B4), $-\frac{61}{3} \to -\frac{49}{3}$ and $\frac{58}{3} \to \frac{52}{3}$ (paper convention), as in our Table 29, with the pole unchanged; these shifts are absorbable. The relation (I.166) is confirmed. Table X needs the two corrections of Sec. VI F (S2, S3), and the overall sign of the self-mass-squared is reversed with respect to the definition (I.168) (S1).

Limitations. The coefficient functions of Table V enter as printed; the appendix of Paper I from which they come was audited through worked examples and the global pointwise test, not redone row by row (Sec. V E). The possible extra local term of Sec. V H is only partially resolved; any double counting of the kind identified has counterterm form. The global tests are numerical, at sample points, with 30-digit precision; the completeness of the counterterm basis was checked at $D = 4, 5, 6$, and the breaking of the gauge symmetry by class (iv) at the level of the action in $D = 3$. These points, and the assumptions shared with Paper I, are collected in Sec. VIII D.

VII. SCHWINGER–KELDYSH EFFECTIVE FIELD EQUATION AND LATE-TIME MODE FUNCTIONS

In this section the renormalized self-mass-squared obtained above, i.e. the corrected Table X with the overall sign fixed by Paper I's own definition (I.168), is used as the input of the Schwinger–Keldysh (SK) analysis of Paper II [12]. We follow the order of Paper II, Secs. 2–3: the input table and the counterterms (cf. (II.7)–(II.12) and Table 1 of Paper II), the four SK polarities and the effective field equation (cf. (II.15)–(II.21)), the perturbative mode equation and its local part (cf. (II.22)–(II.34)), the SK differences and the integrals of Tables 2–4 (cf. (II.35)–(II.43)), and the Table 5 cancellation within the simplification (II.44). We then drop (II.44), keep the full dependence on k, and solve for the late-time one-loop mode function. Tables 2–5 of Paper II were not used as input; they were compared with our results only at the end. The full derivation is Ch. 9 of the derivation notes, and the scripts quoted below are in `code/09_schwinger_keldysh/` of Ref. [17].

Conventions. Two signs appear. The *paper convention*, which in this section we call the *Table-X convention*, is the sign in which Paper I prints Table X and Paper II prints Table 1, Eq. (II.10) and Table 5. The *definition sign* is the sign fixed by (I.168) [Eq. (2); Sec. II F]; Paper II restates the same equation as (II.1). The definition sign is -1 times the Table-X convention, and physical statements are made in the definition sign. As in Paper II, everything refers to the renormalized self-mass-squared in the unregulated limit $D = 4$; general D appears only in the counterterm

coefficients. We use

$$x \equiv \frac{y}{4}, \qquad \Delta\eta \equiv \eta - \eta', \qquad r \equiv \|\vec{x}\,' - \vec{x}\|, \qquad X_0 \equiv \tfrac{1}{4} a a' H^2 \Delta\eta^2, \qquad \hat{k} \equiv \frac{k}{H}, \tag{213}$$

set $H = 1$ in intermediate steps and restore it by dimensional analysis. We count κ^2 once: $\Phi = u + \kappa^2 \Phi_1$ (cf. (II.23)), and Φ_1 is sourced by the one-loop kernel with κ^2 stripped off, Eq. (221) below. (The original working report of this step wrote an explicit κ^2 also in the source and in the particular solution, i.e. it counted κ^2 twice. Ch. 9 of the derivation notes [17] and the present paper count it once. No conclusion depends on this.) Table X/Table 1 entries are coefficients of $\kappa^2 H^4/(4\pi)^4$; the source $\mathcal{R}_1$ and the entries of Table 5 are quoted in units of $H^4 u(0,k)/16\pi^2$.

A. Input: the renormalized self-mass-squared

Paper II prints in its Table 1 the nonlocal contributions to $-iM^2(x;x')$ of Paper I Table X, with the row $(aa')^2(a^2 + a'^2)\nabla^2\Box$ listed once (Paper I prints it twice, erratum C19). The corrected table is our Table 31 (paper-convention column); its rows are listed in the same order as in Table 1 of Paper II, and we use its row numbers 0–12 throughout this section. It differs from the printed one in two entries, both established by the global pointwise test of Table X in Sec. VI F: in row 12 (the ζ row $(aa')^2\nabla^4$) the Table III contribution $8\ln x - 16x\ln x$ is restored (S2, non-analytic), and in row 7 the constant is $+18$ instead of -54 (S3, the $72\gamma_2$ correction, analytic).

The length function is $y(x;x') = aa'H^2\{\|\vec{x} - \vec{x}\,'\|^2 - (|\eta - \eta'| - i\delta)^2\}$; Eq. (II.12) is its $H = 1$ form. We denote by $\mathcal{T}(y)$ the sum of all rows of our Table 31 (Table-X convention). Scale factors stand to the left of the operators, and $\Box$, ∇^2 act only on $f(y)$. This ordering is an assumption; it is the ordering under which the pointwise test of Table X agrees to 30 digits, and the factor a^{4-K} in (II.45) implies the same ordering.

Counterterms (cf. (II.7)–(II.9)). Paper II prints $\alpha_1 = 0 + \Delta\alpha_1$, $\alpha_2 = \frac{H^{D-4}}{(4\pi)^{D/2}}\{\frac{61}{3}\} + \Delta\alpha_2$ and $\alpha_3 = \frac{H^{D-4}}{(4\pi)^{D/2}}\{\frac{4}{D-4} - \frac{58}{3} - 2\gamma_{\rm E}\} + \Delta\alpha_3$. With $\alpha_i = -\beta_i + \Delta\alpha_i$, (I.166), and the corrected $\beta_{2,3}$ of Sec. VI E we obtain

$$\alpha_1 = 0 + \Delta\alpha_1, \qquad \alpha_2 = \frac{H^{D-4}}{(4\pi)^{D/2}}\Big\{ \pm \frac{49}{3}\Big\} + \Delta\alpha_2, \qquad \alpha_3 = \frac{H^{D-4}}{(4\pi)^{D/2}}\Big\{ \pm \Big[\frac{4}{D-4} - \frac{52}{3} - 2\gamma_{\rm E}\Big]\Big\} + \Delta\alpha_3, \tag{214}$$

with the upper sign in the Table-X convention and the lower sign in the definition sign. The pole is unchanged; only the finite parts $\frac{61}{3} \to \frac{49}{3}$ and $-\frac{58}{3} \to -\frac{52}{3}$ (Table-X convention) move, and they are absorbed into the arbitrary $\Delta\alpha_{2,3}$ (B5, B6). They enter none of the results below.

Renormalized self-mass (cf. (II.10)). Paper II writes $-iM^2 = i\kappa^2 a^4(\Delta\alpha_1\Box^2 + \Delta\alpha_2\Box + \Delta\alpha_3\nabla^2/a^2)\delta^4(x-x') +$ (Table 1), without the factors H^2 next to $\Delta\alpha_2$ and $\Delta\alpha_3$ that dimensions require (erratum C20; the same omission as in (I.160), (I.167) and (I.170), errata C17, C18 and C22; it has no effect). In the definition sign, with these factors shown,

$$-iM^2_{++}(x;x') = i\kappa^2 a^4\Big(\Delta\alpha_1\Box^2 + \Delta\alpha_2 H^2\Box + \Delta\alpha_3 H^2\frac{\nabla^2}{a^2}\Big)\delta^4(x-x') - \mathcal{T}(y_{++}). \tag{215}$$

The minus sign in front of $\mathcal{T}$ is the overall sign correction: the physical nonlocal part is $-$(corrected Table X). The sign of the local part is immaterial, since the $\Delta\alpha_i$ are arbitrary.

B. Schwinger–Keldysh polarities and the effective field equation

In the SK formalism [13, 14] each propagator has four polarities, obtained by replacing y in the Feynman propagator by (cf. (II.15)–(II.18))

$$y_{\pm\pm'} = aa'H^2\big[r^2 - (\ldots)^2\big], \qquad (\ldots) = |\Delta\eta| - i\delta\ (++), \quad \Delta\eta + i\delta\ (+-), \quad \Delta\eta - i\delta\ (-+), \quad |\Delta\eta| + i\delta\ (--). \tag{216}$$

The $+$ vertex is the in-out vertex and the $-$ vertex is its conjugate; all lines at a vertex carry its polarity. At one loop $-iM^2_{++}$ is the in-out self-mass-squared, Eq. (215). The $+-$ self-mass follows in three steps: (i) the vertex at x' has $-$ polarity, and since the vertex functionals are real its factor is $(i\,\delta^n S)^* = -i\,\delta^n S$, an overall factor -1; (ii) every line from x to x' becomes $i\Delta_{+-}$, i.e. $y \to y_{+-}$, while the external operators (derivatives and scale factors) are unchanged; (iii) there are no $\delta^4(x-x')$ terms, because the 4-point vertex needs coincident legs and the $\partial_0|\Delta\eta|$ terms of the time-ordered propagator do not occur in y_{+-}. Hence

$$-iM^2_{+-}(x;x') = +\mathcal{T}(y_{+-}), \qquad -iM^2_{--} = \big(-iM^2_{++}\big)^*, \qquad -iM^2_{-+} = \big(-iM^2_{+-}\big)^*, \tag{217}$$

where the last two relations are (II.20), evaluated with $\mathcal{T}(y)^* = \mathcal{T}(y^*)$ (real coefficients and scale factors), $y^*_{++} = y_{--}$ and $y^*_{+-} = y_{-+}$. This is the rule stated in Paper II below (II.19) (drop the δ-function terms, reverse the sign, replace y by y_{+-}), written in the definition sign. Since the operators of our Table 31 are symmetric under $x \leftrightarrow x'$ and $y_{-+}(x;x') = y_{+-}(x';x)$, one has $M^2_{-+}(x';x) = M^2_{+-}(x;x')$.

Varying $\Gamma[\varphi_+,\varphi_-]$ of (II.19) with respect to $\varphi_+(x)$, using the symmetry of M^2_{++} and the last relation, and setting $\varphi_\pm = \Phi$ gives (II.21),

$$a^4\Box\Phi(x) - \int_{\eta_i}^0 d\eta' \int d^3x' \left\{M^2_{++}(x;x') + M^2_{+-}(x;x')\right\}\Phi(x') = 0, \qquad \eta_i = -\frac{1}{H}, \tag{218}$$

where $\delta S/\delta\Phi = \partial_\mu(a^2\partial^\mu\Phi) = a^4\Box\Phi$. The variation with respect to φ_- gives the complex conjugate equation. From Eqs. (215) and (217), in the definition sign,

$$M^2_{++} + M^2_{+-} = -\kappa^2 a^4\Big(\Delta\alpha_1\Box^2 + \Delta\alpha_2 H^2\Box + \Delta\alpha_3 H^2\frac{\nabla^2}{a^2}\Big)\delta^4(x-x') + i\big[\mathcal{T}(y_{+-}) - \mathcal{T}(y_{++})\big]. \tag{219}$$

For $\eta' > \eta$, $(\Delta\eta + i\delta)^2 = (|\Delta\eta| - i\delta)^2$, so $y_{+-} = y_{++}$ and the nonlocal part vanishes. For $\eta' < \eta$, $y_{+-} = y^*_{++}$ and the nonlocal kernel is $2\,\mathrm{Im}\,\mathcal{T}(y_{++})$, which is real. For spacelike separation y is real and positive for both polarities and the difference vanishes. The kernel is therefore real and causal, with support on and inside the past light cone, as stated in Paper II below (II.21).

C. Perturbative mode equation and the local term

With $M^2_{++} + M^2_{+-} = \sum_{\ell\geq 1}\kappa^{2\ell}M^2_\ell$ and $\Phi(x;\vec{k}) = \sum_\ell \kappa^{2\ell}\Phi_\ell(\eta,k)e^{i\vec{k}\cdot\vec{x}}$ (cf. (II.22)–(II.23)), the tree-order solution is (cf. (II.24), (II.28))

$$\Phi_0 = u(\eta,k) = \frac{H}{\sqrt{2k^3}}\Big(1 - \frac{ik}{aH}\Big)e^{ik/(aH)} = \frac{H}{\sqrt{2k^3}}\Big\{1 + \frac{\hat{k}^2}{2a^2} + \frac{i\hat{k}^3}{3a^3} + O\Big(\frac{\hat{k}^4}{a^4}\Big)\Big\}. \tag{220}$$

On a plane wave $a^4\Box = -a^4[\partial_t^2 + 3H\partial_t + k^2/a^2]$, with t comoving time. The first correction, $\Phi = u + \kappa^2\Phi_1$, therefore obeys (cf. (II.29))

$$\Big[\partial_t^2 + 3H\partial_t + \frac{k^2}{a^2}\Big]\Phi_1 = \mathcal{R}_1(\eta,k) \equiv -\frac{1}{a^4}\int_{\eta_i}^0 d\eta'\, u(\eta',k)\int d^3x'\, M^2_1(x;x')\, e^{i\vec{k}\cdot(\vec{x}\,'-\vec{x})}, \tag{221}$$

where M^2_1 is the one-loop part of Eq. (219) with κ^2 stripped off.

Late-time inversion (cf. (II.30)–(II.33)). With $\theta \equiv a\partial_a$, $\partial_t = H\theta$ and $P' \equiv dP/d\ln a$,

$$(\theta^2 + 3\theta)\big[a^n P(\ln a)\big] = a^n\big[(n^2+3n)P + (2n+3)P' + P''\big]. \tag{222}$$

For $n = 0$, $3P' + P'' = R$ gives $P = \frac{1}{6}\ln^2 a - \frac{1}{9}\ln a$ for $R = \ln a$ and $P = \frac{1}{3}\ln a$ for $R = 1$ (the leading terms of (II.30), (II.31)). For $n = -1$, $-2P + P' + P'' = R$ gives $P = -\frac{1}{2}\ln a - \frac{1}{4}$ for $R = \ln a$ and $P = -\frac{1}{2}$ for $R = 1$ ((II.32), (II.33), which keep only the leading terms and the arbitrary constant). For $n = -2$, $-2P - P' + P'' = \alpha\ln a + \beta$ gives $P = -\frac{\alpha}{2}\ln a + \frac{\alpha}{4} - \frac{\beta}{2}$. The case $n = -3$ is resonant with the homogeneous solution: $-3P' + P'' = R$ gives $P = -\frac{R}{3}\ln a$ for constant R. As argued in Paper II, Sec. 3.2, a constant in Φ_1 is a field-strength renormalization, and terms that fall off relative to u cannot be separated from corrections to the free Bunch–Davies initial state. Only contributions to $\mathcal{R}_1$ that grow or tend to a nonzero constant can produce a significant late-time effect.

Local part (cf. (II.34)). Inserting the local part of Eq. (219) into Eq. (221) gives $[\Delta\alpha_1\Box^2 + \Delta\alpha_2 H^2\Box + \Delta\alpha_3 H^2\nabla^2/a^2]$ acting on $u(\eta,k)e^{i\vec{k}\cdot\vec{x}}$. The d'Alembertian annihilates the tree-order solution, so

$$\mathcal{R}_{1,\mathrm{local}} = -\Delta\alpha_3\frac{H^2k^2}{a^2}\,u(\eta,k) \;\longrightarrow\; 0. \tag{223}$$

The corrected finite parts of $\beta_{2,3}$ only redefine $\Delta\alpha_{2,3}$ and leave no trace in the mode function.

D. SK differences and the integrals of Tables 2–4

Identities (cf. (II.35)–(II.36)). At $D=4$, $\Box f(y)=H^2[(4y-y^2)f''+4(2-y)f']$. For $f=\ln(y/4)$ this gives $\Box f/H^2=4/y-3$, and for $f=\frac12 L^2-L$ with $L=\ln(y/4)$ it gives $\Box f/H^2=\frac{4}{y}L-3L+2$. Hence

$$\frac{4}{y}=\frac{\Box}{H^2}\ln\frac{y}{4}+3,\qquad \frac{4}{y}\ln\frac{y}{4}=\frac{\Box}{H^2}\Big\{\frac12\ln^2\frac{y}{4}-\ln\frac{y}{4}\Big\}+3\ln\frac{y}{4}-2, \tag{224}$$

in agreement with Paper II. Since $\ln y$ and $\ln^2 y$ are integrable, no δ^4 remainder arises, and $\Box_x$ commutes with $\int d^4x'$ over the fixed region $[\eta_i,0]\times\mathbb{R}^3$.

SK differences (cf. (II.39)–(II.41)). Inside the past light cone ($\Delta\eta>r$), $y_{++}=-|y|+i0$ and $y_{+-}=-|y|-i0$, so on the principal branch $\ln y_{+\pm}=\ln|y|\pm i\pi$. For $\Delta f\equiv f(y_{++}/4)-f(y_{+-}/4)$ and $y_r\equiv aa'H^2(r^2-\Delta\eta^2)$,

$$\begin{aligned}\Delta\ln x&=2\pi i\,\theta(\Delta\eta-r),&\qquad\Delta\ln^2 x&=4\pi i\,\theta(\Delta\eta-r)\ln|x|,\\ \Delta\big(x^j\ln x\big)&=2\pi i\,x^j\,\theta(\Delta\eta-r),&\qquad\Delta\Big(\frac1x\Big)&=4\Big[\frac{1}{y_r+i0}-\frac{1}{y_r-i0}\Big]=-8\pi i\,\delta(y_r),\end{aligned} \tag{225}$$

with $x\to-\frac14 aa'H^2(\Delta\eta^2-r^2)$ in the polynomial factors, (II.41), and $|x|=X_0(1-z^2)$, $z\equiv r/\Delta\eta$ (in this section z denotes $r/\Delta\eta$, not the variable $4/y$ of Secs. V and VI). Analytic functions (constants, x, x^2) are identical for both polarities, so $\Delta=0$ for them. Consequently the $72\gamma_2$ correction ($-54\to+18$ in row 7), and the terms -120, $96x$, $12x$ and $-36x^2$, drop out of the retarded kernel identically. The equivalent form of the $72\gamma_2$ correction as an extra $-12x$ in the δ row of Paper I Table V drops out for the same reason.

Master formulas. Paper II lists in Tables 2–4 the integrals

$$S_K[f]\equiv-\frac{i}{16\pi^2}\int d^4x'\,a'^K\,\Delta f\qquad(H=1),\qquad K=3,4,5,\quad f=\frac1x,\ \frac{\ln x}{x},\ \ln x,\ \ln^2 x. \tag{226}$$

The entries for $K=3,4$ were taken there from Ref. [26]. We derived all of them independently. With $d\eta'=da'/a'^2$, $\Delta\eta=1/a'-1/a$, $a(\eta_i)=1$, $\int d^3r\,\theta(\Delta\eta-r)=\frac{4\pi}{3}\Delta\eta^3$ and $B_j\equiv\int_0^1 z^2(1-z^2)^j dz$ ($B_0=\frac13$, $B_1=\frac{2}{15}$, $B_2=\frac{8}{105}$), Eq. (225) gives

$$\begin{aligned}S_K[\ln x]&=\frac16\int_1^a da'\,a'^{K-2}\Delta\eta^3,\qquad S_K[x^j\ln x]=\frac{(-1)^jB_j}{2}\int_1^a da'\,a'^{K-2}X_0^j\,\Delta\eta^3,\\ S_K[\ln^2 x]&=\int_1^a da'\,a'^{K-2}\Delta\eta^3\Big[\frac13\ln X_0+C_0\Big],\qquad C_0\equiv\int_0^1 z^2\ln(1-z^2)\,dz=\frac23\ln2-\frac89,\\ S_K\Big[\frac1x\Big]&=-\frac1a\int_1^a da'\,a'^{K-3}\Delta\eta.\end{aligned} \tag{227}$$

The last line uses $\int d^3r\,\delta\big(aa'(r^2-\Delta\eta^2)\big)=2\pi\Delta\eta/(aa')$. It evaluates the $1/x$ terms directly on the light cone, independently of (II.35); the results agree term by term with $\mathcal{D}\,S_K[\ln x]$, where

$$\mathcal{D}\equiv-\big(a^2\partial_a^2+4a\,\partial_a\big),\qquad \mathcal{D}_k\equiv\mathcal{D}-\frac{\hat k^2}{a^2},\qquad \mathcal{D}a^n=-n(n+3)a^n, \tag{228}$$

is $\Box/H^2$ acting on a function of η alone; $\mathcal{D}_k$ is the operator of (II.37).

With $a'=as$ one has $\Delta\eta=(1-s)/(as)$ and $X_0=(1-s)^2/(4s)$, independent of a, so that

$$\int_1^a da'\,a'^{Q-2}\Delta\eta^N X_0^jG(X_0)=a^{Q-1-N}\int_{1/a}^1 ds\,s^{Q-2-N}(1-s)^N X_0^jG(X_0),\qquad \ln X_0=2\ln(1-s)-\ln s-2\ln2. \tag{229}$$

Everything reduces to $\int_\epsilon^1 s^p ds$, $\int_\epsilon^1 s^p\ln s\,ds$ and $\int_\epsilon^1 s^p\ln(1-s)\,ds$ ($\epsilon=1/a$). In the last one the terms s^n/n of $-\ln(1-s)$ with $n\le q\equiv\max(0,-p-1)$ are kept as divergent or logarithmic integrals, and the remainder leaves the constant

$$C_p=-\frac{H_{p+1}}{p+1}\ (p\ge0),\qquad C_{-1}=-\frac{\pi^2}{6},\qquad C_p=-\frac{H_q}{q}\ (p\le-2), \tag{230}$$

with H_n the harmonic numbers. The only source of π^2 in Tables 2–5 is $C_{-1} = -\pi^2/6$. Two examples:

$$\begin{aligned} S_4[\ln x] &= \frac{1}{6}\Big[\ln a - 3\frac{a-1}{a} + 3\frac{a^2-1}{2a^2} - \frac{a^3-1}{3a^3}\Big] = \frac{1}{6}\ln a - \frac{11}{36} + \frac{1}{2a} - \frac{1}{4a^2} + \frac{1}{18a^3}, \\ S_4[\ln^2 x] &= \frac{2}{3}\Big(-\frac{\pi^2}{6} + \frac{49}{36}\Big) - \frac{1}{3}\Big(-\frac{\ln^2 a}{2} + \frac{85}{36}\Big) - \frac{8}{9}\Big(\ln a - \frac{11}{6}\Big) + O\Big(\frac{\ln a}{a}\Big) \\ &= \frac{1}{6}\ln^2 a - \frac{8}{9}\ln a + \frac{7}{4} - \frac{\pi^2}{9} + O\Big(\frac{\ln a}{a}\Big), \end{aligned} \tag{231}$$

where in the second line $\frac{1}{3}\ln X_0 + C_0 = \frac{2}{3}\ln(1-s) - \frac{1}{3}\ln s - \frac{8}{9}$ (the $\ln 2$ cancels) and the three brackets are the $\ln(1-s)$, $\ln s$ and constant integrals of $(1-s)^3/s$.

TABLE 33. Tables 2–4 of Paper II: $S_K[f]$ of Eq. (226) ($H = 1$). The third column is both the printed entry and our result, which agree exactly through the order kept in Paper II; the last column is the next order, which Paper II leaves as $O(\cdot)$.

K	f	Printed = derived	Next order (this work)
3	$\frac{1}{x}$	$-\frac{\ln a}{a} + \frac{1}{a}$	$-\frac{1}{a^2}$
3	$\frac{\ln x}{x}$	$-\frac{\ln^2 a}{2a} + \frac{2\ln a}{a} - \frac{3}{a} + \frac{\pi^2}{3a}$	$-\frac{\ln a+1}{a^2}$
3	$\ln x$	$\frac{1}{6} - \frac{\ln a}{2a} + \frac{1}{4a}$	$-\frac{1}{2a^2}$
3	$\ln^2 x$	$\frac{1}{3}\ln a - \frac{11}{9} - \frac{\ln^2 a}{2a} + \frac{2\ln a}{a} - \frac{9}{4a} + \frac{\pi^2}{3a}$	$-\frac{\ln a}{a^2}$
4	$\frac{1}{x}$	$-\frac{1}{2}$	$\frac{1}{a}$
4	$\frac{\ln x}{x}$	$\frac{3}{4}$	$\frac{\ln a-1}{a}$
4	$\ln x$	$\frac{1}{6}\ln a - \frac{11}{36}$	$\frac{1}{2a}$
4	$\ln^2 x$	$\frac{1}{6}\ln^2 a - \frac{8}{9}\ln a + \frac{7}{4} - \frac{\pi^2}{9}$	$\frac{\ln a-1}{a}$
5	$\frac{1}{x}$	$-\frac{1}{6}a$	$\frac{1}{2a}$
5	$\frac{\ln x}{x}$	$\frac{17}{36}a$	$\frac{2\ln a-3}{4a}$
5	$\ln x$	$\frac{1}{24}a - \frac{1}{6}$	$\frac{1}{4a}$
5	$\ln^2 x$	$-\frac{13}{144}a - \frac{1}{3}\ln a + \frac{5}{9}$	$\frac{2\ln a-3}{4a}$

Result. Equations (227)–(230) were implemented in exact rational arithmetic (`engine.py`, `tables234.py` [17]). All 12 entries of Tables 2–4 agree exactly with Paper II through the order kept there (our Table 33). As an independent test, seven integrals of the type (227) (including $\ln^2 x$, $x\ln x$, $x^2\ln x$, an r^2 weight and a shifted power of a') were evaluated by direct 30-digit quadrature at $a = 50$ and $a = 400$ (`numcheck.py`); the largest difference from the exact expansion is 1.0×10^{-19}.

E. Table 5 within the simplification (II.44), definition sign

Paper II replaces $u(\eta', k) \to u(0, k)$ and $e^{i\vec{k}\cdot\Delta\vec{x}} \to 1$ in Eq. (221), which is the simplification (II.44). Rows containing ∇^2 then vanish, because $\int d^3x'\, F(\|\vec{x} - \vec{x}\,'\|)$ does not depend on $\vec{x}$. For a row term $c\,a^p a'^q(\Box/H^2)^n f$ of our Table 31, using $(4\pi)^4 = (16\pi^2)^2$ and Eq. (219) in the Table-X convention,

$$\mathcal{R}_1 \ni -\frac{u(0,k)}{a^4}\,\frac{ic}{(4\pi)^4}\,a^p\Box^n\int d^4x'\,a'^q\,\Delta f = \frac{H^4 u(0,k)}{16\pi^2}\times\Big\{c\,a^{p-4}\,\mathcal{D}^n S_q[f]\Big\} \qquad \text{(Table-X convention)}, \tag{232}$$

and in the definition sign the entry is $-c\,a^{p-4}\mathcal{D}^n S_q[f]$. The factor $(aa')^3(a^2 + a'^2)$ is the sum of $(p,q) = (5,3)$ and $(3,5)$; since $p + q = 8$, $a^{p-4} = a^{4-q}$, which is the factor a^{4-K} of (II.45). Three examples (Table-X convention), using $\mathcal{D}\ln^2 a = -2 - 6\ln a$, $\mathcal{D}\ln a = -3$ and Eq. (231):

$$\begin{aligned} (aa')^4\Box^2 \times (-6\ln^2 x): &\quad -6\,\mathcal{D}^2 S_4[\ln^2 x] = -6\,\mathcal{D}\Big(\frac{7}{3} - \ln a\Big) = -18, \\ (aa')^4 H^4 \times (-108\ln x): &\quad -108\Big(\frac{\ln a}{6} - \frac{11}{36}\Big) = 33 - 18\ln a, \\ (aa')^3(a^2+a'^2)H^4 \times \Big(-\frac{32}{3x}\Big): &\quad -\frac{32}{3}\Big[a\,S_3\Big[\frac{1}{x}\Big] + \frac{1}{a}S_5\Big[\frac{1}{x}\Big]\Big] = -\frac{80}{9} + \frac{32}{3}\ln a. \end{aligned} \tag{233}$$

TABLE 34. Table 5 of Paper II (Table-X convention, as printed) and the same quantity in the definition sign (this work), in units of $H^4u(0,k)/16\pi^2$; non-falling part, within (II.44). Rows of our Table 31 not shown contribute zero.

External operator $\times f(x)$	Paper II (Table-X convention)	This work (definition sign)
$(aa')^4\Box^3/H^2 \times \big(-\frac{\ln x}{3x}\big)$	0	0
$(aa')^4\Box^2 \times \frac{26\ln x}{3x}$	0	0
$(aa')^4\Box^2 \times \frac{38}{3x}$	0	0
$(aa')^4\Box^2 \times (-6\ln^2 x)$	-18	18
$(aa')^4\Box^2 \times (-18\ln x)$	0	0
$(aa')^4H^2\Box \times \big(-\frac{6\ln x}{x}\big)$	0	0
$(aa')^4H^2\Box \times \frac{4}{x}$	0	0
$(aa')^4H^2\Box \times (-4\ln x)$	2	-2
$(aa')^4H^4 \times \frac{4\ln x}{x}$	3	-3
$(aa')^4H^4 \times \frac{18}{x}$	-9	9
$(aa')^4H^4 \times (-108\ln x)$	$33-18\ln a$	$-33+18\ln a$
$(aa')^3(a^2{+}a'^2)\Box^3/H^2 \times \frac{\ln x}{6x}$	$\frac{-325+12\pi^2}{27}+\frac{14}{3}\ln a-\frac{2}{3}\ln^2 a$	$\frac{325-12\pi^2}{27}-\frac{14}{3}\ln a+\frac{2}{3}\ln^2 a$
$(aa')^3(a^2{+}a'^2)\Box^2 \times \big(-\frac{\ln x}{3x}\big)$	$\frac{85-12\pi^2}{27}-4\ln a+\frac{2}{3}\ln^2 a$	$\frac{-85+12\pi^2}{27}+4\ln a-\frac{2}{3}\ln^2 a$
$(aa')^3(a^2{+}a'^2)\Box^2 \times \frac{1}{6x}$	$\frac{8}{9}-\frac{2}{3}\ln a$	$-\frac{8}{9}+\frac{2}{3}\ln a$
$(aa')^3(a^2{+}a'^2)H^2\Box \times \big(-\frac{2\ln x}{3x}\big)$	$\frac{160-12\pi^2}{27}-\frac{10}{3}\ln a+\frac{2}{3}\ln^2 a$	$\frac{-160+12\pi^2}{27}+\frac{10}{3}\ln a-\frac{2}{3}\ln^2 a$
$(aa')^3(a^2{+}a'^2)H^2\Box \times \frac{5}{x}$	$\frac{55}{3}-10\ln a$	$-\frac{55}{3}+10\ln a$
$(aa')^3(a^2{+}a'^2)H^2\Box \times (-18\ln x)$	$-15+18\ln a$	$15-18\ln a$
$(aa')^3(a^2{+}a'^2)H^4 \times \frac{4\ln x}{3x}$	$\frac{-91+12\pi^2}{27}+\frac{8}{3}\ln a-\frac{2}{3}\ln^2 a$	$\frac{91-12\pi^2}{27}-\frac{8}{3}\ln a+\frac{2}{3}\ln^2 a$
$(aa')^3(a^2{+}a'^2)H^4 \times \big(-\frac{32}{3x}\big)$	$-\frac{80}{9}+\frac{32}{3}\ln a$	$\frac{80}{9}-\frac{32}{3}\ln a$
$(aa')^3(a^2{+}a'^2)H^4 \times 18$ (corrected constant)	(not listed)	0 (analytic)
rows 8–12 (with ∇^2, including ζ)	(not listed)	0 (under (II.44))
Total	0	0

Since $\mathcal{D}$ preserves powers of a, truncating before or after acting with $\mathcal{D}$ is consistent.

The script `table5.py` [17] reproduces all 19 printed entries exactly in the Table-X convention. In the definition sign every entry is -1 times the printed one (our Table 34), and the total is again zero. Individual entries grow as fast as $\frac{2}{3}\ln^2 a$; the $\ln^2 a$, π^2, $\ln a$ and constant parts cancel separately:

$$\begin{aligned} \ln^2 a: &\quad \tfrac{2}{3}(1-1-1+1)=0, \qquad \pi^2: \quad \tfrac{12}{27}(-1+1+1-1)=0,\\ \ln a: &\quad 18-\tfrac{14}{3}+4+\tfrac{2}{3}+\tfrac{10}{3}+10-18-\tfrac{8}{3}-\tfrac{32}{3}=0,\\ 1: &\quad 18-2-3+9-33+\tfrac{325-85-24-160-495+405+91+240}{27}=-11+11=0 \end{aligned} \tag{234}$$

(definition sign). Neither the ζ correction (a ∇^4 row) nor the $72\gamma_2$ correction (analytic) enters our Table 34. A heuristic reason to expect the $k=0$ cancellation, which we state as an assumption, is that the scalar couples to gravity only through derivatives, so that diagrammatically $M^2(x;x')=\partial_\mu\partial'_\nu K^{\mu\nu}(x;x')$ and $\int d^4x'\,M^2_{\rm ret}(x;x')\cdot 1$ leaves only terms from the initial surface $\eta'=\eta_i$, which decay. The vanishing total is then a consistency test of the rows without ∇^2; it does not test the ∇^2 rows, hence not the ζ row.

F. Beyond (II.44): the full k dependence

The justification of (II.44) fails term by term (S4). Paper II argues that the deviations $u(\eta',k)-u(0,k)$ and $e^{i\vec{k}\cdot\Delta\vec{x}}-1$ bring at least two factors of $1/a'$ or $1/a$, so that they, and with them all ∇^2 rows, only produce falling terms. Two effects spoil this argument for individual rows: positive powers $x^j\propto(aa')^j$ and the prefactors a^p multiply the contribution of the early-time region by powers of a; and at early η', $\ln|x|\simeq\ln[a/(4a')]$ contains $\ln a$. For example, the $(p,q)=(4,2)$ parts of rows 10 and 11 contain $x\ln x$, and from Eq. (227) with $X_0=\frac{1}{4}aa'\Delta\eta^2$,

$$S_2[x\ln x]=-\frac{1}{2}\cdot\frac{2}{15}\cdot\frac{a}{4}\int_1^a da'\,a'\Big(\frac{1}{a'}-\frac{1}{a}\Big)^5=-\frac{a}{60}\Big(\frac{1}{3}-\frac{5}{2a}+\dots\Big)=-\frac{a}{180}+\frac{1}{24}+\dots \tag{235}$$

With $\nabla^2 \to -\hat{k}^2$, $u \to u(0,k)$ and $\mathcal{D}a = -4a$, in the Table-X convention,

$$\text{row 10: } 48(-\hat{k}^2)\Big(-\frac{a}{180}\Big) = +\frac{4}{15}\hat{k}^2 a, \qquad \text{row 11: } 12(-\hat{k}^2)\,\mathcal{D}\Big(-\frac{a}{180}\Big) = -\frac{4}{15}\hat{k}^2 a. \tag{236}$$

In the definition sign rows 10 and 11 thus grow like $\mp\frac{4}{15}\hat{k}^2 a$. Either one alone would give $\Phi_1 \propto a$; they cancel only against each other. Likewise the k^2 corrections of the rows without ∇^2 give $O(1)$ constants. A complete expansion in k is therefore required.

Systematic expansion in k. We keep the exact $u(\eta',k)$, the phase, the k^2/a^2 part of $\Box$ and all ∇^2 rows. With $w \equiv \hat{k}/a'$,

$$\frac{u(\eta',k)}{u(0,k)} = (1-iw)e^{iw} = 1 + \frac{w^2}{2} + \frac{iw^3}{3} - \frac{w^4}{8} + \dots, \qquad \big\langle e^{i\vec{k}\cdot\vec{r}}\big\rangle_{\text{angles}} = \frac{\sin kr}{kr} = \sum_{m\ge 0}\frac{(-1)^m (kr)^{2m}}{(2m+1)!}, \tag{237}$$

together with $\Box/H^2 \to \mathcal{D}_k$, $\nabla^2 \to -k^2$, $1/x \to \mathcal{D}_k \ln x$ and $\ln x/x \to \mathcal{D}_k(\frac{1}{2}\ln^2 x - \ln x) + 3\ln x$ (Eq. (224); the constants drop out of Δ). Each term reduces to integrals of the type (227), with the power of a' shifted by $-n$ for the w^n term and an extra weight r^{2m} (integrals of z^{2+2m}). The integrand is an entire function of k and the region is compact ($\eta' \in [\eta_i,\eta]$, $r \le \Delta\eta$), so the expansion converges at every fixed a and for every k. Convergence at fixed a does not, however, control the behavior of the remainder as $a \to \infty$: individual rows contain terms growing like a at every even order computed (for example $\pm\frac{8a}{441}\hat{k}^6$ in rows 10 and 11, see below), and for $\hat{k} \gtrsim 1$, i.e. for modes inside the horizon at η_i, a truncated series need not approximate the source. The script `kexp.py` [17] expands every row exactly through k^{10}, and the statements below hold order by order through k^{10}.

As a hand check, consider row 3 at order $\hat{k}^2$ (Table-X convention). The $\hat{k}^2$ part of S_4 comes from $\frac{w^2}{2}$ and from $-\frac{k^2 r^2}{6}$ (radial weight $\frac{1}{5}$ instead of $\frac{1}{3}$). At leading order $\int_1^a da'\,\Delta\eta^3 \to \frac{1}{2}$ and $\int_1^a da'\,a'^2\Delta\eta^5 \to \frac{1}{2}$, so

$$G_4^{(2)}[\ln x] = \frac{1}{12}\int da'\,\Delta\eta^3 - \frac{1}{60}\int da'\,a'^2\Delta\eta^5 \to \frac{1}{24} - \frac{1}{120} = \frac{1}{30}, \qquad G_4^{(2)}[\ln^2 x] \ni \frac{\ln a}{15}, \tag{238}$$

where the $\ln a$ comes from $\ln X_0 \simeq \ln a$ at early times. Then $-108\ln x \to -\frac{18}{5}$, $\frac{4\ln x}{x} \to 4[\mathcal{D}(\frac{\ln a}{30}) + \frac{3}{30}] = 0$ and $\frac{18}{x} \to 0$: row 3 gives $-\frac{18}{5}$ at order $\hat{k}^2$ ($+\frac{18}{5}$ in the definition sign), as does the code. The early-time $\ln a$ in $\ln|x|$ is essential here.

TABLE 35. Non-falling part of $\mathcal{R}_1$ beyond (II.44), row by row (rows of our Table 31), in the definition sign and in units of $H^4 u(0,k)/16\pi^2$. The $\hat{k}^0$ column collects, row by row, the last column of our Table 34; the order $\hat{k}^1$ vanishes in every row. Rows 0 and 9 give zero at these orders.

#	$\hat{k}^0$	$\hat{k}^2$	$\hat{k}^3$	$\hat{k}^4$
1	18	0	0	0
2	-2	0	0	0
3	$18\ln a - 27$	$\frac{18}{5}$	$2i$	$-\frac{27}{35}$
4	$\frac{2}{3}\ln^2 a - \frac{14}{3}\ln a + \frac{325-12\pi^2}{27}$	$\frac{2}{9}\ln a - \frac{22}{27}$	$\frac{2i(6\ln a-23)}{81}$	$\frac{239}{900} - \frac{\ln a}{15}$
5	$-\frac{2}{3}\ln^2 a + \frac{14}{3}\ln a + \frac{12\pi^2-109}{27}$	$\frac{22}{27} - \frac{2}{9}\ln a$	$\frac{2i(23-6\ln a)}{81}$	$\frac{\ln a}{15} - \frac{239}{900}$
6	$-\frac{2}{3}\ln^2 a - \frac{14}{3}\ln a + \frac{12\pi^2-250}{27}$	$-\frac{2}{9}\ln a - \frac{20}{27}$	$-\frac{2i(6\ln a+19)}{81}$	$\frac{\ln a}{15} + \frac{181}{900}$
7	$\frac{2}{3}\ln^2 a - \frac{40}{3}\ln a + \frac{331-12\pi^2}{27}$	$\frac{2}{9}\ln a - \frac{61}{27}$	$\frac{4i(3\ln a-31)}{81}$	$\frac{629}{900} - \frac{\ln a}{15}$
8	0	$\frac{2}{5}$	0	$\frac{3}{35}$
10	0	$1 - \frac{4a}{15}$	0	$\frac{1}{5} - \frac{12a}{175}$
11	0	$\frac{4a}{15} - 2$	0	$\frac{12a}{175} - \frac{2}{5}$
12	0	0	0	$-\frac{1}{70}$
Total	0	0	0	0

Cancellation order by order. Our Table 35 lists the non-falling part of each row through $\hat{k}^4$ (`kexp.py`, `extract.py`). At order $\hat{k}^2$ the terms $\propto a$ cancel between rows 10 and 11, the $\ln a$ terms among rows 4–7, and the constants add to $\frac{18}{5} + \frac{2}{5} - \frac{81}{27} + 1 - 2 = 0$. At order $\hat{k}^4$ rows 3 and 8 give $-\frac{24}{35}$, rows 4–7 give $+\frac{9}{10}$, rows 10–11 give $-\frac{1}{5}$ and row 12 gives

$-\frac{1}{70}$, with total $\frac{-48+63-14-1}{70}=0$. Without the ζ row the $\hat{k}^4$ column would not cancel. Its $-\frac{1}{70}$ comes from the term $24x^2\ln x$, which is the same in the printed and in the corrected row, so the $\hat{k}^4$ cancellation does not depend on the ζ correction: with $S_2[x^2\ln x]=\frac{1}{2}\cdot\frac{8}{105}\cdot\frac{a^2}{16}\int da'\,a'^2\Delta\eta^7$ and $\int_1^a da'\,a'^2\Delta\eta^7\to\frac{1}{4}$, one finds $24\,a^{-2}\hat{k}^4\,S_2[x^2\ln x]\to\frac{\hat{k}^4}{70}$ in the Table-X convention. The code finds that the non-falling total vanishes separately at every order $k^0,k^1,\dots,k^{10}$, for the corrected table and for the printed one. At orders k^6 to k^{10} the terms $\propto a$ of rows 10 and 11 (e.g. $\pm\frac{8a}{441}\hat{k}^6$) again cancel each other. The series truncated at k^8 was compared with direct quadrature using the exact $u(\eta',k)$ and $\sin kr/kr$ at $a=200$ (`numcheck_k.py`): the differences are 10^{-10}–2×10^{-8} at $\hat{k}=0.3$ and 5×10^{-6}–9.3×10^{-4} at $\hat{k}=1$, the size of the expected k^9 truncation error. The conclusion of (II.44) therefore holds through k^{10}, although its justification does not hold row by row.

Contribution of the ζ correction. The correction is $\delta\mathcal{T}=(aa')^2\nabla^4(8\ln x-16x\ln x)$. By hand, in the Table-X convention, with $S_2[\ln x]=\frac{1}{6}(\frac{1}{2}-\frac{3}{a}+\dots)$ and Eq. (235), $8a^{-2}\hat{k}^4S_2[\ln x]=\frac{2}{3}\hat{k}^4/a^2+\dots$ and $-16a^{-2}\hat{k}^4S_2[x\ln x]=\frac{4}{45}\hat{k}^4/a-\frac{2}{3}\hat{k}^4/a^2+\dots$; the exact result (`analysis2.py`, `analysis3.py`) is

$$\delta\mathcal{R}_1=\frac{H^4u(0,k)}{16\pi^2}\Big[\hat{k}^4\Big(\frac{4}{45a}-\frac{4}{3a^3}\Big)+\hat{k}^6\Big(\frac{4}{175a}-\frac{2}{15a^3}\Big)+\dots\Big]\qquad\text{(Table-X convention)}.\tag{239}$$

Only falling terms appear. This also holds for every k, without the k expansion: since $|u(\eta',k)/u(0,k)|=\sqrt{1+k^2\eta'^2}\le\sqrt{1+\hat{k}^2}$, $|e^{i\vec{k}\cdot\vec{r}}|=1$ and $|x|\le\frac{1}{4}aa'\Delta\eta^2$,

$$\big|\delta\mathcal{R}_{1,\ln x}\big|\le C\frac{\hat{k}^4}{a^2}\int_1^a da'\Big(\frac{1}{a'}-\frac{1}{a}\Big)^3\le C'\frac{\hat{k}^4}{a^2},\qquad \big|\delta\mathcal{R}_{1,x\ln x}\big|\le C\frac{\hat{k}^4}{a}\int_1^a da'\,a'\Big(\frac{1}{a'}-\frac{1}{a}\Big)^5\le C'\frac{\hat{k}^4}{a},\tag{240}$$

where the constants contain the factor $\sqrt{1+\hat{k}^2}$ and the unit $H^4u(0,k)/16\pi^2$. The ζ correction therefore falls off at late times for every mode, independently of the k expansion.

G. Late-time source and the one-loop mode function

Summing all rows of the corrected table (definition sign, through k^{10}; `extract.py`) gives

$$\mathcal{R}_1=\frac{H^4u(0,k)}{16\pi^2}\Big\{\frac{1}{a^2}\Big[12-\Big(8\ln a+\frac{103}{3}\Big)\hat{k}^2-8i\hat{k}^3+\frac{8}{3}\hat{k}^4+\frac{32i}{45}\hat{k}^5-\frac{2}{15}\hat{k}^6+\dots\Big]-\frac{12i\hat{k}^3}{a^3}+O\Big(\frac{\ln a}{a^4}\Big)\Big\}+\mathcal{R}_{1,\rm local}.\tag{241}$$

There is no a^{-1} term. The next a^{-2} coefficients are $-\frac{16i}{1575}\hat{k}^7-\frac{4}{945}\hat{k}^8-\frac{16i}{6615}\hat{k}^9+\frac{11}{14175}\hat{k}^{10}$, and there is no other a^{-3} term through k^{10}. With the printed ζ row (definition sign) $\mathcal{R}_1$ would contain the additional terms

$$\frac{H^4u(0,k)}{16\pi^2}\,\frac{1}{a}\Big[\frac{4}{45}\hat{k}^4+\frac{4}{175}\hat{k}^6+\frac{2i}{135}\hat{k}^7-\frac{8}{1323}\hat{k}^8-\frac{i}{525}\hat{k}^9+\frac{4}{8019}\hat{k}^{10}+\dots\Big],\tag{242}$$

plus a^{-3} terms. The correction thus changes the fall-off of the right-hand side from $\hat{k}^4/a$ to $\ln a/a^2$; we record this as an observation, not as an independent proof of the correction. Both behaviors are consistent with the statement of Paper II after Table 5 that the right-hand side falls at least like $1/a$ times powers of $\ln a$. The a^{-2} coefficients depend on the initial time and on $\hat{k}$ (e.g. the oscillatory $\hat{k}^3,\hat{k}^5$ terms), i.e. they are sensitive to the initial state and, in the sense of Paper II, Sec. 3.2, cannot be separated from initial-state corrections.

Solving for Φ_1. In Eq. (222) the a^{-2} terms of Eq. (241) have $\alpha=-8\hat{k}^2$ and $\beta=12-\frac{103}{3}\hat{k}^2-8i\hat{k}^3+\dots$, so $P=4\hat{k}^2\ln a-6+\frac{91}{6}\hat{k}^2+4i\hat{k}^3+\dots$; the resonant a^{-3} term gives $P=4i\hat{k}^3\ln a$. The k^2/a^2 term on the left-hand side changes the a^{-2} solution only at order a^{-4}. Dividing by H^2 (from $\partial_t^2+3H\partial_t=H^2(\theta^2+3\theta)$) we obtain

$$\Phi_1=C\,u(\eta,k)+\frac{H^2u(0,k)}{16\pi^2}\Big\{\frac{1}{a^2}\Big[4\hat{k}^2\ln a-6+\frac{91}{6}\hat{k}^2+4i\hat{k}^3+\dots\Big]+\frac{4i\hat{k}^3\ln a}{a^3}+\dots\Big\},\tag{243}$$

where the local source (223), itself of order $\hat{k}^2/a^2$, only adds a $\Delta\alpha_3$-dependent $\hat{k}^2/a^2$ term inside the braces. Here $C\,u(\eta,k)$ stands for the homogeneous part, which the late-time analysis does not fix; we write $C\equiv\frac{H^2}{16\pi^2}\tilde{C}$. In general the initial conditions at η_i fix a combination $c_1u+c_2u^*$ of both homogeneous solutions; since $u(0,k)=H/\sqrt{2k^3}$ is real, this combination tends to $(c_1+c_2)\,u(0,k)[1+O(a^{-2})]$ and is absorbed into $\tilde{C}$. Therefore

$$\boxed{\Phi(\eta,k)=u(\eta,k)+\kappa^2\Phi_1(\eta,k)\ \xrightarrow{a\to\infty}\ u(0,k)\Big\{1+\frac{\kappa^2H^2}{16\pi^2}\Big[\tilde{C}+O\Big(\frac{\ln a}{a^2}\Big)\Big]\Big\}}\tag{244}$$

where $\tilde{C}$ is a field-strength renormalization constant, and the tree-order approach of $u(\eta, k)$ to $u(0, k)$, Eq. (220), is not displayed. Through order k^{10}, and given the kernel and the local terms as derived in Secs. III–VI, there is no growing (a, $\ln a$, $\ln^2 a$) or drifting term: one-loop quantum gravity gives no significant correction to the mode function of the massless minimally coupled scalar at late times. This confirms the conclusion of Paper II with the corrected self-mass, in the definition sign, and without the simplification (II.44). Two qualifications are stated at the end of this section: the cancellation is not proven beyond k^{10}, and a possible finite local term from the outer-operator reduction of Paper I has not been excluded.

TABLE 36. Effect of the corrections of Paper I on the SK kernel and on the late-time mode function.

Correction	SK kernel / Table 5	Late-time mode function
S1: overall sign (definition $= -$printed)	every Table 5 entry flips; total still 0	the particular solution Φ_1 changes sign; since the non-falling total is zero, only falling terms change sign; no late-time effect
S2: ζ row, $+8\ln x - 16x\ln x$ (∇^4, non-analytic)	absent under (II.44); in the full calculation only falling terms	none ($\leq C\hat{k}^4/a$ for every k); removes the $\hat{k}^4, \hat{k}^6, \ldots$ tails $\propto 1/a$
S3: $72\gamma_2$, $-54 \to +18$ (analytic)	$\Delta f \equiv 0$, drops out of the retarded kernel	identically none
B1–B6: finite parts of β_2, β_3 and α_2, α_3	absorbed into $\Delta\alpha_{2,3}$	none; $\mathcal{R}_{1,\mathrm{local}} = -\Delta\alpha_3 H^2 k^2 u/a^2 \to 0$
C19: row printed twice in Table X	already single in Paper II Table 1	none

TABLE 37. Statements of Paper II compared with this work.

Paper II	This work
(II.7)–(II.9): $\alpha_2 \ni \frac{61}{3}$, $\alpha_3 \ni \frac{4}{D-4} - \frac{58}{3} - 2\gamma_{\mathrm{E}}$	$\frac{49}{3}$ and $\frac{4}{D-4} - \frac{52}{3} - 2\gamma_{\mathrm{E}}$ (Table-X convention; B5, B6); pole unchanged; finite parts absorbed
Table 1	two entries corrected (rows 7 and 12, our Table 31; S3 and S2)
(II.15)–(II.21): SK rules, real and causal kernel	derived; Eqs. (217)–(219) in the definition sign
(II.34): only $\Delta\alpha_3$ contributes locally	confirmed (H^2 restored, C21)
(II.35)–(II.41)	confirmed; $1/x$ also evaluated directly on the light cone
Tables 2–4 (12 entries)	identical through the order kept; next order obtained
Table 5 (19 entries), total 0	identical in the Table-X convention; -1 times each entry in the definition sign; total 0
(II.44) and its justification	S4: justification fails term by term (rows 10, 11 grow like $\mp\frac{4}{15}\hat{k}^2 a$, definition sign); the conclusion holds, verified through k^{10}
Right-hand side falls at least like $\ln^n a/a$	corrected table: $\ln a/a^2$; printed table: $\hat{k}^4/a$
No significant one-loop late-time correction	confirmed, Eq. (244)

Status and limitations. Our Tables 36 and 37 summarize the effect of the corrections of Paper I and the comparison with Paper II. The cancellation of the non-falling part beyond (II.44) is exact through k^{10} but has not been proven to all orders in k. The convergence of the k series at fixed a gives no control over the late-time behavior of the terms beyond k^{10}, since individual rows carry growing terms $\propto a$ at every even order computed; in particular, for $\hat{k} \gtrsim 1$ the truncated series does not establish the late-time form of the source. The a^{-2} coefficients of $\mathcal{R}_1$ and Φ_1 are sensitive to the initial state. The explanation of the $k = 0$ cancellation by the derivative structure of M^2 is a heuristic. The operator ordering of our Table 31 is assumed; it is the one verified by the pointwise test of Table X. The direct light-cone evaluation of $S_K[1/x]$ was done by hand only; the code uses (II.35) and agrees with it. Finally, this section takes the renormalized self-mass of Secs. III–VI as given, together with its stated limitations: in particular, the possible extra local term from the operator reduction in the appendix of Paper I is only partially resolved (a double counting would have counterterm form, but other local differences at coincidence have not been excluded by a full distributional analysis), and the global checks of the input table are numerical, at sample points. A finite local term of mass type, such as $H^4 a^4 \delta^4(x - x')$ in $-iM^2$, would give a source $\mathcal{R}_1$ that tends to a constant, and by (II.31)

a secular $\Phi_1 \propto \frac{1}{3}\ln a$; this is exactly the kind of effect excluded above, so the late-time conclusion holds given the local terms as derived. Heuristically, such a term would conflict with the shift symmetry that underlies the $k = 0$ cancellation, but this has not been demonstrated for the outer-operator reduction.

VIII. SUMMARY OF DIFFERENCES AND CONCLUSIONS

A. Main differences with the printed papers

Our Table 38 summarizes, item by item, how our results compare with Papers I and II and whether each difference matters physically. The first row is the physical question that motivated both papers; its answer does not change.

The finite errors that shift β_2 and β_3, and their consequences B3–B6, are listed in our Table 39. The printed (I.161)–(I.165) are reproduced exactly from the printed (I.44) and (I.59); the shifts come only from correcting these two inputs. In the definition sign all β_i change sign, while $\alpha_i = -\beta_i + \Delta\alpha_i$, (I.166), holds in either convention.

B. Errata

Our Table 40 lists the 23 typographical errors C1–C23. None of them affects any result: in each case the printed calculation used, or is consistent with, the correct expression, and the two terms missing from (I.128) (C14) are excluded by the shift symmetry. The labels C1–C23, S1–S4 and B1–B6 are those of the summary of findings in the public repository [17]. Three of the typographical errors (C2, C20 and C21) were first noticed while the English derivation notes were prepared. Two more, the missing factors H^2 in (I.170) (C22, the same omission as in C17, C18 and C20) and the slip "P_1 and P_6" for "P_1 and P_4" before (I.A31) (C23), were noticed while the write-up was refereed and were confirmed on the printed pages of Paper I; they are not discussed in the derivation-note chapters. The list of misprints is therefore not claimed to be exhaustive.

The derivation notes also record notational remarks, which are not counted as errors. In (I.6) the indices of $h^\nu{}_\rho$ are not staggered, which is unambiguous because $h_{\mu\nu}$ is symmetric. (I.71) requires $\Delta x_\sigma = \eta_{\sigma\nu}\Delta x^\nu$, so that $\Delta x_0 = -\Delta\eta$. The statement at (I.69) that no delta function arises holds for naive (pointwise) derivatives; the distributional derivative adds exactly the local term counted separately in (I.54)–(I.58), so the inner derivatives must be naive, as the footnote to (I.75) says. The $\Box$ in (I.92) is distributional, while (I.93) is pointwise. The dilatation factor in (I.134) must be positive, and in (I.135) $x \cdot x$ is the Minkowski product $\|\vec{x}\|^2 - \eta^2$. Finally, (I.125) and the polynomial and $\ln 2$ terms of Table V depend on the choice of integration constants, which the external operators annihilate or which cancel globally, except for S3.

C. Conclusions

We have recomputed, step by step and independently, the one-loop quantum-gravitational self-mass-squared of a massless, minimally coupled scalar on de Sitter background in the gauge of Paper I, and the late-time analysis of the effective mode equation of Paper II. The structure of the calculation is confirmed: the Feynman rules, the coincidence limits (apart from the finite part of (I.40)), the decomposition of the 3-point contribution, the extraction of derivatives (audited through worked examples and a global pointwise test), the separation of divergences, the counterterm analysis and the integral tables of Paper II. The differences we found fall into three classes.

First, an overall sign: the printed self-mass-squared is -1 times the one defined by the effective field equation (I.168) that both papers use. The error arises separately in the 4-point and in the 3-point part, so the two parts have the correct relative sign and the local cancellations between them hold. The sign of the local part is irrelevant, because the finite counterterms are arbitrary, but the nonlocal part has no counterterm freedom, so its sign is physical. Second, two corrections to the final table of nonlocal terms: a missing non-analytic contribution in the $(aa')^2\nabla^4$ row, which changes the physical nonlocal self-mass, and an analytic constant in the $(aa')^3(a^2 + a'^2)H^4$ row, which cannot affect any retarded quantity. Third, finite errors in two local terms, which shift the finite parts of β_2 and β_3 and are absorbed into the arbitrary finite counterterms. Within Paper I's treatment of the outer operators of the normal terms, the divergence structure, a single de Sitter-breaking pole in β_3, is unchanged (see point 2 of Sec. VIII D). In the labels of our Table 38, the first two classes are the substantive findings S1 and S2–S3, and the third consists of the absorbable errors B1–B6; the fourth substantive finding, S4, concerns the argument for (II.44) and is discussed below. In addition there are 23 typographical errors, C1–C23.

None of these corrections changes the late-time physics, within the limitations listed below. With the corrected self-mass-squared in the definition sign, and without the simplification (II.44), the non-falling part of the source of

TABLE 38. Main results of this work compared with the printed Papers I [8] and II [12]. Signs of table entries and of β_i are in the paper convention unless stated otherwise; the definition sign is -1 times the paper convention, Eq. (4). Table X entries are coefficients of $\kappa^2 H^4/(4\pi)^4$, β_i are in units of $H^{D-4}/(4\pi)^{D/2}$, and $x = y/4$. The class of each difference is given in parentheses in the first column: substantive findings S1–S4, absorbable finite errors B1–B6, and typographical errors C1–C23.

Item	Papers I/II (printed)	This work	Does it matter physically?
Late-time scalar mode function	**No significant one-loop correction** at late times; the right-hand side of (II.29) falls at least like $\ln^n a/a$ (Paper II, Secs. 3.2 and 3.4).	**Confirmed** with the corrected self-mass, in the definition sign and without the simplification (II.44), through order k^{10}: $\Phi \to u(0,k)\Big\{1 + \frac{\kappa^2 H^2}{16\pi^2}\Big[\tilde{C} + O\Big(\frac{\ln a}{a^2}\Big)\Big]\Big\}$ (tree-order $O(\hat{k}^2/a^2)$ terms of $u(\eta,k)$ not displayed); limitations in Sec. VIII D.	**The main conclusion stands: no significant one-loop correction at late times.**
Overall sign of $-iM^2$ (S1, substantive)	$-iM^2_{\rm ren}$ = local + Table X, (I.167).	The printed self-mass is -1 times the one defined by (I.168). The 4-point and 3-point parts are both reversed, so their relative sign is consistent. Physical nonlocal part: $-$(corrected Table X).	Every Table 5 entry flips and the particular solution Φ_1 changes sign; since the non-falling total is zero, only decaying terms change sign. No late-time effect.
Table X, row $(aa')^2\nabla^4$ (S2, substantive)	$\frac{10}{3}\ln x - 24x\ln x + 24x^2\ln x - 36x^2$	$\frac{34}{3}\ln x - 40x\ln x + 24x^2\ln x - 36x^2$: the Table III term $8\ln x - 16x\ln x$ was left out. Corrected Table X: our Table 31.	Non-analytic: changes the nonlocal self-mass. In the mode equation it adds only falling terms ($\le C\hat{k}^4/a$); it removes the $\hat{k}^4/a$, $\hat{k}^6/a, \ldots$ tails, so the source falls like $\ln a/a^2$. The k^4 cancellation beyond (II.44) uses the row's $24x^2\ln x$ term, which the correction leaves unchanged.
Table X, row $(aa')^3(a^2+a'^2)H^4$ (S3, substantive, analytic)	constant -54	constant $+18$ (equivalently, a term $-12x$ in the δ row of Table V). Only with this and the previous correction does Table X pass the global pointwise test (30 digits).	Analytic: drops out of the retarded kernel identically. No effect.
Finite local terms, β_2, β_3 (B1–B6, absorbable)	$\beta_2 = -\frac{61}{3}$, $\beta_3 = -\frac{4}{D-4} + \frac{58}{3} + 2\gamma_{\rm E}$	$\beta_2 = -\frac{49}{3}$, $\beta_3 = -\frac{4}{D-4} + \frac{52}{3} + 2\gamma_{\rm E}$, from finite errors in (I.40) and (I.59) (our Table 39). Pole unchanged; $\beta_1 = 0$ confirmed.	No: absorbed into the arbitrary finite counterterms $\Delta\alpha_2$, $\Delta\alpha_3$.
Divergences and counterterms	Two invariant counterterms and one noninvariant counterterm (I.145); only β_3 diverges.	Confirmed, including the de Sitter group analysis and the completeness of the noninvariant counterterms (checked at $D = 4, 5, 6$); the divergence structure holds within Paper I's treatment of the outer operators (Sec. V H).	—
Paper II, Tables 2–4	12 integrals	All 12 reproduced exactly in closed form; next order obtained.	—
Paper II, Table 5	19 entries, total 0	Reproduced exactly in the paper convention; in the definition sign every entry flips; total 0.	Confirms the cancellation at $k = 0$.
Simplification (II.44) (S4, argument)	Dropping $u(\eta') - u(0)$, $e^{i\vec{k}\cdot\Delta\vec{x}} - 1$ and the ∇^2 rows is justified by a fall-off argument.	The justification fails term by term: two rows grow like $\mp\frac{4}{15}\hat{k}^2 a$. The full non-falling part cancels at each order $k^0, \ldots, k^{10}$, for the corrected and for the printed table.	No change of the conclusion.
Misprints (typographical)	—	23 typographical errors C1–C23 (our Table 40).	None.

TABLE 39. Finite errors of Paper I that are absorbed into the counterterms (paper convention; $h \equiv H^2/16\pi^2$; β_i and α_i in units of $H^{D-4}/(4\pi)^{D/2}$). The corresponding counterterm coefficients of Paper II, (II.8)–(II.9), change accordingly.

	Location	Printed	Corrected	Effect
B1	(I.40), finite part	$+2\,\delta^0_\alpha\delta^\sigma_0\,h$	$-2\,\delta^0_\alpha\delta^\sigma_0\,h$	the printed value contradicts (I.38) through a trace identity; $\beta_2 \to \beta_2 + 4$, $\beta_3 \to \beta_3 - 4$
B2	(I.59), constant of the $a^{D-2}\nabla^2$ term	$-\frac{7}{2}$	$-\frac{5}{2}$, the sum of the paper's own (I.57) and (I.58)	$\beta_3 \to \beta_3 + 2$
B3	β_2, (I.163)	$-\frac{61}{3}$	$-\frac{49}{3}$	finite; absorbed into $\Delta\alpha_2$
B4	β_3, (I.165)	$-\frac{4}{D-4} + \frac{58}{3} + 2\gamma_{\rm E}$	$-\frac{4}{D-4} + \frac{52}{3} + 2\gamma_{\rm E}$	pole unchanged; finite part absorbed into $\Delta\alpha_3$
B5	α_2, (II.8)	$\frac{61}{3} + \Delta\alpha_2$	$\frac{49}{3} + \Delta\alpha_2$	enters no result
B6	α_3, (II.9)	$\frac{4}{D-4} - \frac{58}{3} - 2\gamma_{\rm E} + \Delta\alpha_3$	$\frac{4}{D-4} - \frac{52}{3} - 2\gamma_{\rm E} + \Delta\alpha_3$	enters no result (the local source $-\Delta\alpha_3 H^2k^2u/a^2$ depends only on the arbitrary $\Delta\alpha_3$ and falls off)

the one-loop mode equation cancels separately at each order in k through k^{10}; this is verified exactly through k^{10}, not proven to all orders. The overall sign reverses the particular solution Φ_1, but since the non-falling total vanishes only decaying terms change sign. The correction of the $(aa')^2\nabla^4$ row contributes only decaying terms; it removes the $\hat{k}^4/a$, $\hat{k}^6/a, \ldots$ tails that the printed row would leave, so that the source falls like $\ln a/a^2$. We record this faster fall-off as an observation, not as an independent confirmation of the correction. The mode function approaches its tree-order late-time value times a constant, Eq. (7), the constant being a field-strength renormalization. Through k^{10}, and given the kernel and local terms as derived, there is no term that grows like a, $\ln a$ or $\ln^2 a$. One-loop quantum gravity therefore gives no significant correction to the mode functions of a massless, minimally coupled scalar at late times, as concluded in Paper II. The expectation expressed in Paper II, that the same holds for the inflaton of scalar-driven inflation, is not affected by our corrections.

The analysis beyond (II.44) also clarifies how the cancellation works. The argument given for (II.44) does not hold row by row: the rows $(aa')^2(a^2+a'^2)H^2\nabla^2$ and $(aa')^2(a^2+a'^2)\nabla^2\Box$ each grow like a and cancel only against each other, and at order k^4 the $(aa')^2\nabla^4$ row is needed for the cancellation. Its contribution at that order comes from the term $24x^2\ln x$, which is the same in the printed and in the corrected row; the correction $8\ln x - 16x\ln x$ changes only falling terms. A heuristic explanation of the cancellation at $k = 0$ is that the scalar couples to gravity only through its derivatives, so that the self-mass-squared has the form $\partial_\mu\partial'_\nu K^{\mu\nu}(x;x')$ and annihilates a constant mode up to initial-surface terms that decay. This is consistent with the observation of Paper II that the tree-order mode function approaches a constant whose derivatives vanish.

Finally, a remark on method. The recomputation was carried out with an AI model as the working tool, under the discipline described in Sec. I D: apart from the exceptions stated there, every step was derived from its inputs without using the target, and every analytic result was checked by independent code where possible. Every tabulated or numerical result is backed by a script in the public repository [17] that reproduces the logged output (Appendix A); the few steps done by hand only are listed in Sec. VIII D (points 1 and 5). The global pointwise tests proved particularly effective: they test sums of many separately derived pieces against a reference that does not depend on the intermediate manipulations. The test of Tables IV–V located the $72\gamma_2$ inconsistency, and the test of Table X showed that both corrections of Table X are needed and that together they are sufficient at the sample points tested.

D. Limitations and open points

We list explicitly what has not been established.

1. *The appendix of Paper I was audited, not redone.* For the appendix "Extracting derivatives," worked examples (for instance the P_{3c} row of the γ_3 operator and the totals of the β and γ_1 tables) were checked by hand, and the totals of all ten tables were verified by the global pointwise test, with generic power functions at $D = 4$ and $D = 5$ and with the actual $D = 4$ propagators. The tables were not rederived row by row from scratch. Tables IV and V were computed from the printed appendix totals after these had been checked, and the polynomial terms

TABLE 40. Typographical errors in Papers I and II. None of them affects any result. Paper I's Table X and Paper II's Table 1 are in the paper convention.

	Location	Printed	Correct
C1	(I.1), first term	$-\frac{1}{2}\partial_\mu\phi\,\partial_\mu\phi\,g^{\mu\nu}\sqrt{-g}$	$-\frac{1}{2}\partial_\mu\phi\,\partial_\nu\phi\,g^{\mu\nu}\sqrt{-g}$
C2	(I.6), left-hand side	$-\frac{1}{2}\partial_\mu\phi\,\partial_\nu g^{\mu\nu}\sqrt{-g}$	$-\frac{1}{2}\partial_\mu\phi\,\partial_\nu\phi\,g^{\mu\nu}\sqrt{-g}$
C3	(I.7), left-hand side	$-\frac{1}{2}\kappa a^{D-2}\partial_\alpha\phi\,\partial_\beta\phi\,h^{\alpha\beta}$	$+\frac{1}{2}\kappa a^{D-2}\partial_\alpha\phi\,\partial_\beta\phi\,h^{\alpha\beta}$; the vertex $V_1 = i\kappa a^{D-2}\partial_1^\alpha\partial_2^\beta$ is correct
C4	(I.14), first coefficient of $A(y)$	$\Gamma(\frac{D}{2}-1)/(\frac{D}{2}-1)$	$\Gamma(\frac{D}{2})/(\frac{D}{2}-1) = \Gamma(\frac{D}{2}-1)$, as used in (I.52)
C5	Table II, U_2	$-\frac{i}{4}\kappa^2 a^{D-2}\eta^{\alpha\rho}\eta^{\beta\sigma}\partial_1\cdot\partial_2$	$+\frac{i}{4}\kappa^2 a^{D-2}\eta^{\alpha\rho}\eta^{\beta\sigma}\partial_1\cdot\partial_2$, as used in (I.33)
C6	(I.39), finite part	$[2\delta_0^\rho\delta_0^\sigma + 2\bar\eta_{\rho\sigma}]H^2/16\pi^2$	$[2\delta_0^\rho\delta_0^\sigma + 2\bar\eta^{\rho\sigma}]H^2/16\pi^2$ (index placement)
C7	text before (I.54)	"the 4th term of (47)"	"the 2nd term of (47)"
C8	(I.89), second term	$-4(aa')^2(a+a')^2\nabla^2(yA')$	$-4H^2(aa')^2(a+a')^2\nabla^2(yA')$, as in Table III
C9	Table XII, header	$\beta = (aa')^{D-2}(a^2+a'^2)H^2\Box$	$\beta = (aa')^{D-1}(a^2+a'^2)H^2\Box$, as in (I.105)
C10	Appendix, (I.A18)	$-2(D-1)aa'H^2\,I[f'A]$	$-2(D-1)aa'H^2\,I[f'A']$
C11	Table XIII, row P_{7c}	repeats part of the total	$-\frac{1}{2}(D-1)(D-2)\,y\,I[A'\Delta C'] - \frac{1}{2}(D-1)^2(D-2)\,I^2[A'\Delta C']$
C12	Table XV, row P_{4b}	$(D-1)\,y'(A'B')'$	$(D-1)\,y^2(A'B')'$
C13	Table XVIII, total	$\frac{1}{4}\,y\,I^2[A'^2\Delta C']$	$\frac{1}{4}\,y\,I^2[A'\Delta C']$
C14	(I.128), list of four-derivative invariants	five terms	the reduced list also contains $\phi^2 R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ and $\phi^2\Box R$; both are excluded by the shift symmetry, so (I.129) is unaffected
C15	Table IV, γ_2 row, second column	$-\frac{D^3-3D^2-4D+8}{4(D-1)}$	$+\frac{D^3-3D^2-4D+8}{4(D-1)}$, as used in Tables VI–IX and required by shift symmetry
C16	Table VII, header	coefficient of $i\kappa^2H^4/(4\pi)^4$	coefficient of $\kappa^2H^4/(4\pi)^4$
C17	(I.160)	$\beta_2\Box + \beta_3\nabla^2/a^2$	$\beta_2H^2\Box + \beta_3H^2\nabla^2/a^2$
C18	(I.167)	$\Delta\alpha_2\Box + \Delta\alpha_3\nabla^2/a^2$	$\Delta\alpha_2H^2\Box + \Delta\alpha_3H^2\nabla^2/a^2$
C19	Table X	row $(aa')^2(a^2+a'^2)\nabla^2\Box$ printed twice	printed once, as in Table 1 of Paper II
C20	(II.10)	$\Delta\alpha_2\Box + \Delta\alpha_3\nabla^2/a^2$	$\Delta\alpha_2H^2\Box + \Delta\alpha_3H^2\nabla^2/a^2$
C21	(II.34)	$\Delta\alpha_3$ term without H^2	with H^2
C22	(I.170)	$\Delta\alpha_2\Box + \Delta\alpha_3\nabla^2/a^2$; result $-\kappa^2a^4\Delta\alpha_3(k^2/a^2)\Phi_0$	$\Delta\alpha_2H^2\Box + \Delta\alpha_3H^2\nabla^2/a^2$; result $-\kappa^2a^4\Delta\alpha_3H^2(k^2/a^2)\Phi_0$
C23	Appendix, text before (I.A31)	"Applying the second identity (A18) to P_1 and P_6"	"… to P_1 and P_4" (the subparts that follow are $P_{1a}, P_{1b}, P_{4a}, \dots$)

of Table V enter Table X as printed. In the same spirit, the gauge-fixed graviton propagator (I.15)–(I.25) was taken from Paper I and Ref. [9] as input and was not re-derived.

2. *A possible additional local term is only partially resolved.* If the outer d'Alembertian in the operators α and δ acts as a distribution, the local term (I.54)+(I.58) could be counted twice. We showed that any such double counting has exactly the counterterm form $c(D)\,(a^D\Box - a^{D-2}\nabla^2)\delta^D(x-x')$, with a constant $c(D)$ that may contain a pole and without $\ln a$; it is absorbed into α_2 and α_3. The finiteness of β_2 and the pole of β_3 are therefore confirmed within Paper I's treatment of the outer operators; a double counting, if present, would shift poles and finite parts by counterterm-form amounts. We did not show that the generic reductions (I.A35)–(I.A38) of Paper I produce no other local difference at coincidence, for example a finite $H^4a^D\delta^D(x-x')$; this would require a full distributional analysis. The vanishing $a^D\Box^2$ total of Table VIII and the finiteness of Table IX give partial support. Such a mass-type term would give a source $\mathcal{R}_1$ that tends to a constant and, by (II.31), a secular $\Phi_1 \propto \frac{1}{3}\ln a$. Heuristically it would conflict with the shift symmetry that underlies the $k=0$ cancellation, but this has not been demonstrated.

3. *The global checks are numerical.* The tests of the appendix totals, of Tables IV–V and of Table X compare sums at two or three random spacelike points in 30-digit arithmetic. They are not symbolic identities, and they test only sums. Some further checks were done in specific dimensions: the gauge-breaking Euler–Lagrange check in $D=3$; the completeness of the noninvariant counterterms among fifteen candidate densities at $D=4,5,6$, with general-D formulas derived for eleven of them; and the pointwise check of the nonlocal part of (I.60)–(I.67) at $D=4$.

4. *The location of the $72\gamma_2$ correction is an interpretation.* Reading it as a term $-12x$ dropped from the δ row of

Table V, or as a change of the constant of the γ_2 row, gives the same sum; the tests fix only the sum.

5. *The late-time cancellation is exact through k^{10}, not proven to all orders.* The k expansion converges at fixed a (an entire integrand on a compact region), but this gives no control over the behavior of the terms beyond k^{10} as $a \to \infty$: individual rows carry terms growing like a at every even order computed (for example $\pm\frac{8a}{441}\hat{k}^6$ in the rows $(aa')^2(a^2+a'^2)H^2\nabla^2$ and $(aa')^2(a^2+a'^2)\nabla^2\Box$), and for $\hat{k} \gtrsim 1$ (modes inside the horizon at η_i) the truncated series need not approximate the source. The a^{-2} coefficients of $\mathcal{R}_1$ and Φ_1 depend on the initial time and on $\hat{k}$, i.e. they are sensitive to the initial state. The explanation of the $k = 0$ cancellation given above is a heuristic. The operator ordering (scale factors to the left of the derivative operators of Table X) is assumed; it was verified in the pointwise test of Table X. Steps done by hand only, without a code check, are the direct light-cone evaluation of the $1/x$ integrals (the code uses (II.35) and agrees with it), the analytic bound (240) on the $(aa')^2\nabla^4$ correction, and the heuristic argument for the $k = 0$ cancellation.

6. *Assumptions shared with Paper I.* Dimensional regularization preserves the shift symmetry and the residual symmetries of the gauge-fixed action; ultraviolet divergences are polynomial in H; coincidence limits are taken in dimensional regularization; and $D = 4$ is set only in terms not multiplied by a pole.

7. *The overall sign rests on the definition (I.168)* with $-iM^2 = \delta^2(i\Gamma_1)/\delta\Phi\,\delta\Phi$. Several independent arguments support it: a lemma on second functional derivatives of quadratic forms with derivatives (with two proofs), the field-equation route, the comparison with the paper's counterterm vertex (I.131), explicit test functions, a lattice Hessian, and the consistency of the 3-point steps. Once (I.168) is adopted, the sign of the nonlocal part is physical.

Of these, points 2 and 5 bear most directly on the late-time result: a finite local term of the type mentioned in point 2, if it were present, would enter the mode equation directly and could produce a secular $\ln a$, and point 5 limits the proven cancellation to the orders computed. The late-time conclusion is therefore established through k^{10}, given the kernel and the local terms as derived. The nonlocal kernel used in the late-time analysis was verified by the global pointwise test of Table X against a sum that does not depend on the outer-operator reduction of the appendix.

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Appendix A: Verification methods

This appendix describes the checking tools that carry the most weight in the conclusions: the exact engine for the late-time Schwinger–Keldysh integrals, the global pointwise tests of the self-mass tables, the momentum expansion beyond (II.44) with its numerical tests, and the reproducibility of all scripts. Throughout, "30 digits" means `mpmath` arithmetic with a working precision of 30 significant digits.

1. Exact engine for the late-time integrals

All late-time integrals of Paper II, with or without the simplification (II.44), are generalizations of the integrals $S_K[f] = -\frac{i}{16\pi^2}\int d^4x'\, a'^K\{f(\frac{y_{++}}{4}) - f(\frac{y_{+-}}{4})\}$ of Tables 2–4 of Paper II ($H = 1$), with an additional shift of the power of a' (from the expansion of $u(\eta', k)$) and an additional weight r^{2m}, $r \equiv \|\vec{x}\,' - \vec{x}\|$ (from the expansion of the phase). Their reduction to single integrals over a', including the direct light-cone evaluation of the $1/x$ terms, is given in Sec. VII (Eq. (227)). We normalize $a(\eta_i) = 1$ and write $\Delta\eta \equiv \eta - \eta' = 1/a' - 1/a$ and $X_0 \equiv \frac{1}{4}aa'\Delta\eta^2$.

Scaling variable. With $a' = as$ one has $\Delta\eta = (1-s)/(as)$ and $X_0 = (1-s)^2/(4s)$, which is independent of a, so that every integral that occurs reduces to the form (229), with $G = 1$ or $G = \ln X_0$. After binomial expansion only three elementary integrals remain, $\int_\epsilon^1 s^p\,ds$, $\int_\epsilon^1 s^p \ln s\,ds$ and

$$\int_\epsilon^1 ds\, s^p \ln(1-s) = C_p - \sum_{n=1}^{q} \frac{1}{n} \int_\epsilon^1 ds\, s^{p+n} + \sum_{n>q} \frac{\epsilon^{p+n+1}}{n(p+n+1)}, \qquad q \equiv \max(0, -p-1), \qquad \epsilon \equiv \frac{1}{a}, \tag{A1}$$

which follows from $\ln(1-s) = -\sum_n s^n/n$: the terms with $n \le q$ are kept as divergent or logarithmic integrals, and those with $n > q$ are integrated term by term. The constants $C_p = -\sum_{n>q} 1/[n(n+p+1)]$ are given in closed form in Eq. (230). The result of every integral is thus an exact expansion in $\epsilon = 1/a$ whose coefficients are polynomials in $\ln a$ with rational coefficients and the constants $\ln 2$ and π^2; the only source of π^2 is $C_{-1} = -\pi^2/6$. Worked examples are given in Eq. (231), and the next order of every entry of Tables 2–4 of Paper II in our Table 33.

Implementation. The helper module `engine.py` implements Eqs. (229), (A1) and (230) in exact rational arithmetic in `sympy`, including the shift of the power of a' and the weight r^{2m}, the operator $\mathcal{D} \equiv -(a^2\partial_a^2 + 4a\partial_a)$, which is $\Box/H^2$ acting on functions of η alone, and truncation in powers of a. A first attempt, which used generic symbolic integration followed by an expansion in $1/a$, failed because the resulting polylogarithms could not be expanded asymptotically; it was replaced by this closed-form engine. With it all 12 entries of Tables 2–4 and all 19 entries of Table 5 of Paper II are reproduced exactly (in the paper convention), and the next order of Tables 2–4 is obtained. Paper II's Tables 2–5 were never used as input.

2. Global pointwise tests of the self-mass tables

The final tables of Paper I are obtained after long chains of manipulations: extraction of derivatives, reduction to external operators, and separation of divergences. A global pointwise test compares the end product directly with a reference that does not depend on these manipulations. All quantities are in the paper convention, at $D = 4$ and for $x \neq x'$.

Reference side. The normal part of the 3-point contribution is the sum of the seven parts $P_1, \ldots, P_7$ defined in Eqs. (I.A1)–(I.A7) of the appendix of Paper I, each of the generic form $\partial\partial'[(aa')^{D-2} F\, \partial\partial' A]$ with $F \in \{A, B, \Delta C\}$. They are evaluated by brute-force differentiation in explicit coordinates, with naive (pointwise) inner derivatives. For the test of Table X the nonlocal logarithm term of (I.67), $-\frac{\kappa^2 H^2}{8\pi^2}\, 2H(aa')^2(a\partial_0 + a'\partial_0')\nabla^2 A$, is added. The actual $D = 4$ propagator functions are used: $A = h\{4/y - 2\ln(y/4) - 1\}$, $B = C = 4h/y$ and $\Delta C = 4(C - A)$, with $h \equiv H^2/16\pi^2$.

Table side. The external operators of the table act pointwise on their coefficient functions, with the scale factors standing to the left of the derivative operators.

Appendix totals and Tables IV–V. The ten external operators acting on the totals of the appendix Tables XI–XX reproduce the direct sum $\sum_k P_k$ exactly (to 30 digits), once the five appendix misprints C9–C13 of our Table 40 are corrected. This was tested with the generic powers $A = y^{-7/3}$, $B = y^{-5/4}$, $\Delta C = y^{2/7}$ at $D = 5$ and at $D = 4$ (two random spacelike points each), which checks the reduction algebra independently of the propagators, and with the actual $D = 4$ propagators (three random points). Evaluated at $D = 4$, the printed Tables IV and V miss the direct sum by residuals of order 10^{-2} to 10; the residual vanishes to 30 digits only when the sign of the γ_2 entry of Table IV is corrected (C15) and a term equivalent to $72\gamma_2$ acting on a constant is added (S3; our Table 19).

Table X. Three random rational spacelike pairs of points (seed 29) were used. The two terms of (I.159) cancel pointwise, so (I.159) does not enter the test, and at $D = 4$ the nonlocal parts of (I.152)–(I.153) reproduce $(4/y)^2$ and $(4/y)^3$ pointwise, so that the rows of Table X that come from Table VII can be compared pointwise with the reference side. The residuals are listed in our Table 32: for the printed Table X and for Table X with the $(aa')^2\nabla^4$ correction alone they range from about 10^{-2} to 10^2 (with |LHS| between 0.1 and 9), and they vanish to 30 digits ($|\text{residual}| \lesssim 4 \times 10^{-31}$) only with both the $(aa')^2\nabla^4$ (S2) and the $72\gamma_2$ ($-54 \to +18$, S3) corrections.

The tests are numerical, at sample points, and they test sums only (Sec. VIII D). Their strength is that the reference side is independent of the outer-operator reduction of the appendix and of the separation of divergences, and that random points make an accidental agreement to 30 digits implausible.

3. Momentum expansion and quadrature checks

Beyond (II.44) the full k dependence of the source $\mathcal{R}_1$ of Eq. (5) is kept: the exact $u(\eta', k)/u(0, k) = (1 - iw)e^{iw}$ with $w \equiv \hat{k}/a'$, the angular average $\sin kr/kr$ of the phase, the k^2/a^2 part of $\Box$, and all rows containing ∇^2 (Sec. VII,

TABLE 41. Code folders of the repository [17] and the main checks they contain.

Folder	Main checks
`01_feynman_rules`	expansion (I.6) against the exact metric; vertices of Tables I–II
`02_four_point`	derivative rule and overall sign of the 4-point part (test functions, lattice Hessian, calibration with (I.131)); coincidence limits (I.34)–(I.40); (I.41)–(I.44)
`03_three_point_setup`	3-point rule (I.45)–(I.46); the nine terms (I.47)–(I.49) and their sign; (I.52); first coefficient of (I.14)
`04_three_point_local_log`	local terms (I.54)–(I.59); logarithm terms (I.60)–(I.67) and their cancellation against (I.44) in general D
`05_normal_terms_tableIII`	derivative-extraction identities (I.68)–(I.80); (I.81)–(I.93) and Table III
`06_tables_IV_V`	(I.94)–(I.125); global test of the appendix totals; Table IV in general D; Table V and the $72\gamma_2$ hypothesis tests
`07_desitter_counterterms`	de Sitter isometries and the residual symmetries of the gauge; (I.136)–(I.145) and the completeness of the noninvariant counterterms
`08_renormalization_tableX`	(I.146)–(I.159) and Tables VI–IX; β_i of (I.161)–(I.165); assembly of Table X; global pointwise test of Table X
`09_schwinger_keldysh`	exact late-time engine; Tables 2–5 of Paper II; momentum expansion through k^{10}; quadrature checks; late-time coefficients of the source

Eq. (237)). Each term of the expansion in k is an integral of the type (229). The integrand is an entire function of k and the integration region is compact ($\eta' \in [\eta_i, \eta]$, $r \le \Delta\eta$), so the expansion converges for every k at fixed a. This convergence gives no control over the late-time behavior of the terms beyond the order computed, since individual rows contain terms growing like a at every even order computed (Sec. VII F). The script `kexp.py` expands every row of the corrected Table X exactly through k^N and prints the part of the source that does not fall off as $a \to \infty$. For $N = 2, 4, 6, 10$ this non-falling part vanishes at every order $k^0, \dots, k^N$. The same holds for the printed table (scripts `analysis2.py` and `extract.py`), whose difference from the corrected one consists of falling terms only. Hand computations reproduce selected entries: the growing terms $\mp\frac{4}{15}\hat{k}^2 a$ of the rows $(aa')^2(a^2 + a'^2)H^2\nabla^2$ and $(aa')^2(a^2+a'^2)\nabla^2\Box$, the order-$\hat{k}^2$ entry of the row $(aa')^4H^4$, and the order-$\hat{k}^4$ entry of the $(aa')^2\nabla^4$ row, which comes from its $24x^2 \ln x$ term (the same in the printed and in the corrected row). For the correction of the $(aa')^2\nabla^4$ row an analytic bound valid for every k shows that its contribution to $\mathcal{R}_1$ falls at least like $\hat{k}^4/a$, independently of the expansion.

Two numerical tests compare the engine and the expansion with direct quadrature in `mpmath`.

- `numcheck.py`: the exact expansion through a^{-6} versus direct quadrature over η' and $z \equiv r/\Delta\eta$, at $a = 50$ and $a = 400$, for seven integrals, including $\ln^2 x$, $x \ln x$, $x^2 \ln x$, an r^2 weight and a power of a' shifted as by the $u(\eta', k)$ correction. The largest difference is 1.0×10^{-19}, on values between 10^{-2} and 10^2.

- `numcheck_k.py`: the series truncated at k^8 versus quadrature with the exact $u(\eta', k)$ and $\sin kr/kr$, at $a = 200$. The differences are between 10^{-10} and 2×10^{-8} at $\hat{k} = 0.3$, and between 5×10^{-6} and 9.3×10^{-4} at $\hat{k} = 1$, consistent with the expected truncation error of order $\hat{k}^9$.

4. Reproducibility

The repository [17] contains the English derivation notes (nine chapters, `notes/chapters/`, and a summary chapter listing every finding with the labels S1–S4, B1–B6 and C1–C23) and, for each chapter, a folder `code/NN_…/` with standalone Python scripts (Python 3.10 or later, `sympy`, `mpmath`), a README describing what each script checks, and the expected output of every script. Each script runs from its own folder with `python3 <script>.py`, without network access, in at most a few minutes. All scripts were re-run for the public version and reproduce the outputs recorded in the original working logs; the only difference is the order in which one script prints two tied entries, now fixed by a sort key. Our Table 41 lists the folders and the main checks they contain.

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